Decidability in Borel Monadic Logic

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Abstract

Monadic second-order logic provides a rich framework for decidability results. The limits of decidability are indicated by undecidability of the monadic theory of $(\mathbb{R}, \leq)$. Restricting the monadic quantifier enables decidability. We prove decidability of the Borel monadic theory.

The thesis first surveys Shelah's undecidability proof of the unrestricted monadic theory of $(\mathbb{R}, \leq)$ and his reduction of the monadic theory to quantification over sufficiently uniform sets and Cantor sets. We then explore the connection of the Borel monadic theory of $(\mathbb{R}, \leq)$ with Baire category and, building on Shelah's reduction, deduce decidability. The Boolean combinations of F_σ sets form an elementary substructure of the Borel sets. Under determinacy hypotheses, the proof extends to larger classes of sets.

The Borel monadic theory of the binary tree is a common generalization of the Borel monadic theory of $(\mathbb{R}, \leq)$ and the monadic theory of two successors, S2S. Moreover, it expresses Borel determinacy as a sentence. We present joint work with Mikołaj Bojańczyk, Antonio Casares and Paweł Parys on a potential automata model to obtain decidability of this theory. A conjecture on finite memory determinacy would imply the missing complementation lemma for these automata. We provide equivalent formulations of this conjecture and solve particular cases, in particular the cases of Büchi automata and automata with a single state.

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0.1 Introduction

We begin by giving an introduction to the context of this thesis. After introducing notation, we discuss in detail some elementary results on the expressiveness of monadic logic. Finally, we state the results of the thesis and consider variants.

0.1.1 Background

First-order logic quantifies over elements of a universe. Second-order logic additionally quantifies over n-ary relations. Monadic (second-order) logic is intermediate, quantifying over unary relations.

A starting point of modern mathematical logic is Hilbert's "Entscheidungsproblem" [HA72]. Undecidability of first-order arithmetic $(\mathbb{N}, +, \cdot)$ - essentially Gödel's first incompleteness theorem - provides a negative solution. The study of decidability in model theory aims to identify the most expressive theories that remain decidable. Seminal results include decidability of Presburger arithmetic $(\mathbb{N}, +)$ [Pre91] and the real field $(\mathbb{R}, +, \cdot)$ [Tar48].

Decidability in higher-order logic can rarely be obtained. Second-order theories of the natural numbers are undecidable even for restricted languages like $(\mathbb{N}, +1)$ (Proposition 0.1.11). Monadic logic, however, often yields theories that are expressive, but still decidable. While the monadic theory of $(\mathbb{N}, +)$ is still undecidable (Corollary 0.1.16), S1S, the monadic theory of $(\mathbb{N}, +1)$, or equivalently of $(\mathbb{N}, \leq)$, is decidable. Even the weak monadic theory of $(\mathbb{N}, +1)$ interprets expansions of Presburger arithmetic [Büc60]. Decidability of S1S can be proven either using automata for infinite words [Büc62] or employing semigroups and Ramsey theory [She75, Theorem 3.4]. These approaches lead to different generalizations of S1S.

The automata for infinite words used in the decidability proofs of S1S were generalized by Rabin to automata for infinite trees [Rab69]. This yields decidability of S2S, the monadic theory of the nodes of the infinite binary tree $2^{<\omega}$ with the two successor operations $x \mapsto x0, x \mapsto x1$ or equivalently with the prefix relation and lexicographic order. The main part of the decidability proof, the complementation lemma, is deduced in modern accounts [GH82] from a generalization of the Büchi-Landweber theorem on finite memory determinacy [BL69].

The semigroup-theoretic approach (also called the model-theoretic approach) was employed by Shelah for more general total orders [She75]. This for example yields decidability of the monadic theory of countable total orders, or equivalently $(\mathbb{Q}, \leq)$ (although this result also can be deduced from decidability of S2S). Results for general orders enable further results, for example Shelah's proof that Aronszajn lines share the monadic theory of the rationals [She75, Theorem 6.2].

However, decidability of monadic theories of order is limited by the undecidability of the theory of $(\mathbb{R}, \leq)$ [She75, Section 7], which is discussed in chapter 1. A survey of the earlier developments is [Gur85]. Generalizations of the undecidability result from the reals to other spaces have been studied [Gur77b; Gur79b; Gur79a; GS79], see also Remark 1.3.35.

Shelah's undecidability proof employs sets of high complexity in the sense of descriptive set theory. Thus, restricting the monadic quantifier enables decidability results. In contrast, quantification only over F_σ sets, the second level of the Borel hierarchy, yields a decidable theory [Rab69, Theorem 2.9].

0.1.2 Notation and prerequisites

Let X be a set. Let $\#X$ denote the cardinality of X. Let 2^X denote the power set of X. The disjoint union or coproduct of sets A, B is denoted by $A \amalg B$, which has lower precedence than cartesian products (that is, $A \amalg B \times C = A \amalg (B \times C)$).

For an ordinal α, let $X^{<\alpha}$ denote the disjoint union of the sets of maps $\beta \to X$ over $\beta < \alpha$, and define $X^{\leq\alpha} = X^{<\alpha+1}$ and $X^{=\alpha} = X^{\leq\alpha} \setminus X^{<\alpha}$. The prefix relation on $X^{<\alpha}$ is denoted by $\preceq$. For $x \in X^{<\alpha}$, let $|x|$ denote the length of x, that is, the unique ordinal such that $x \in X^{=|x|}$. For $x \in X^{=\alpha}$ and $\beta \leq \alpha$, we denote by $x|_\beta$ the unique prefix of length β. If $|x|$ is a successor ordinal $\alpha + 1$, write $x^- = x|_\alpha$.

For us, $0 \in \mathbb{N}$. We abbreviate $\{0, \dots, k\}^{<\alpha}$ by $(k+1)^{<\alpha}$, and $k^{\leq\alpha}, k^{=\alpha}, k^{\mathbb{N}}$ are defined analogously. For instance, for $y \in k^{\mathbb{N}}$, we have $\limsup y \in \{0, \dots, k-1\}$.

A **labeling** of a set is a map to a finite set, whose elements are called labels. If P is a labeling, then $\#P$ denotes the cardinality of its codomain. A labeling $Q\colon X \to J$ **refines** a labeling $P\colon X \to I$ **along** a map $f\colon J \to I$ if $P = f \circ Q$. In this case we say that P **coarsens** Q. If (X, P) is a labeled set, that is, P is a labeling of X, and $Y \subseteq X$, then we abuse notation by writing (Y, P) for the restriction $(Y, P|_Y)$. If P is clear from the context, we often refer to properties of $(Y, P|_Y)$ or $P|_Y$ by saying that Y has this property.

We usually consider labeled total orders. Embeddings and isomorphisms of these are required to preserve the labels. We thus obtain a category total orders labeled by a fixed set as objects and monotone label-preserving maps as morphisms. An isomorphism class in this category is called **labeled order type**.

Let $()$ denote the empty tuple. If P is an n-tuple, we denote its components by $P_1, \dots, P_n$. If $P_1, \dots, P_n$ are elements, we denote the tuple of them by P. We often use notation for elements also for tuples, for example $\exists P$ for $\exists P_1 \exists P_2 \dots \exists P_n$.

Logic

Let $\mathscr{L}$ be a **multi-sorted relational first-order language**, that is, a set I of sorts, a set of relation symbols, and a map a from the relation symbols to tuples of sorts. An atomic $\mathscr{L}$-formula in variables x_k of sorts i_k is a pair $R(x_{k_1}, \dots, x_{k_n})$ of a relation symbol r and a tuple of variables such that $(i_{k_1}, \dots, i_{k_n}) = a(R)$. General

$\mathscr{L}$-formulas are built as usual from atomic ones using the Boolean connectives $\top, \wedge, \neg$ and quantifiers $\exists$ for each sort, from which the symbols $\bot, \vee, \Rightarrow, \Leftarrow, \Leftrightarrow, \forall$ can be defined. Satisfaction of formulas can be defined recursively as usual. We will always assume that there is an equality relation symbol for each sort that is interpreted correctly.

Definable without further qualification means definable without parameters. Here we only study decidability and definability results, and will thus regard any two representations of a class of formulas as equivalent if there is a computable interpretation. In particular, we assume that all first-order languages are relational by replacing an n-ary function symbol by an $(n+1)$-ary relation symbol for its graph.

We now define monadic formulas and the monadic theory of a structure. Higher-order logic can be defined as multi-sorted first-order logic. For instance, to obtain the **monadic language** for a one-sorted first-order language $\mathscr{L}$ with sort X, we add a sort 2^X for the power set and a binary relation symbol $\in$ on $X \times 2^X$. Thus, any first-order $\mathscr{L}$-structure induces a monadic $\mathscr{L}$-structure with the same universe. In this structure, it is easy to define the Boolean algebra structure on 2^X, for example the symbols $\subseteq, \emptyset, \cap, \cup, \setminus$. Moreover, being a singleton set is definable as being an atom in this Boolean algebra, and we introduce a predicate isAtom for this purpose.

However, it is also possible to reduce monadic logic to one-sorted first-order logic. In this case, the sort represents the power set, and we replace each relation symbol $R(x_1, \dots, x_m, Y_1, \dots, Y_n)$ with element variables x_i and set variables Y_j by the relation symbol $R'(X_1, \dots, X_m, Y_1, \dots, Y_n)$ interpreted as "there are elements $x_1, \dots, x_m$ such that $X_i = \{x_i\}$ for all i and $R(x_1, \dots, x_m, Y_1, \dots, Y_n)$ holds". Clearly, this language defines the same relations on 2^X as the original language and they interpret each other. Whenever we refer to usual first-order properties, like elementary embeddings, in the monadic setting, they are to be interpreted in this language.

Remark 0.1.1. There are alternative ways to express first-order relations R as monadic relations. For example, we could replace $R(x_1, \dots, x_m, Y_1, \dots, Y_n)$ by a relation symbol $R'(X_1, \dots, X_m, Y_1, \dots, Y_n)$ with the interpretation "for all $x_1 \in X_1, \dots, x_m \in X_m$, the relation $R(x_1, \dots, x_m, Y_1, \dots, Y_n)$ holds.

Restricted monadic logic can be handled in exactly the same way, replacing the power set 2^X by a subset $\mathcal{B}$ in this discussion. In our discussions, $\mathcal{B}$ always contains all singletons, enabling the reduction to first-order logic, but it will not always be a Boolean subalgebra.

We use Gale-Stewart games, usually only called **games**, and their determinacy for Borel sets [Mar75]. For standard notation, see also [Kec95, 20, 21]. They are also compared with other game models in chapter 3.

Topology

Let A, B be subsets of a topological space X. Let $\overline{A}, \mathring{A}, \partial A$ denote the closure, interior, and boundary of A. We call A **perfect** if it is closed, nonempty, and has no isolated points. The set A is **nowhere dense** if $\overline{\mathring{A}}$ is empty. It is **somewhere dense** else. We call A **dense in** B if $A \cap B$ is dense in the subspace B, and similarly for somewhere dense or (co-)meager sets. A subset of a topological space is **a Cantor set** if it is homeomorphic to $2^{\mathbb{N}}$.

A topological space is the **disjoint union** of subsets $(U_i)_{i \in I}$ if its underlying set is their disjoint union, but the U_i need not be clopen. A family $(V_j)_{j \in J}$ of sets is a **refinement** of $(U_i)_{i \in I}$ if $\bigcup_j V_j = \bigcup_i U_i$ and for all j there is an i with $V_j \subseteq U_i$.

We collect basic facts of descriptive set theory. A reference is [Kec95].

A subset of a Polish space has the **Baire property** if it is the symmetric difference of an open set and a meager set. Borel sets, and more generally σ-combinations of analytic sets, have the Baire property.

Fact 0.1.2 (Baire alternative). Let A have the Baire property. Then exactly one of the following conditions holds:

(i) A is meager,

(ii) there is a nonempty open U such that A is comeager in U.

Another useful immediate implication of Baire's theorem is the following fact.

Fact 0.1.3. For a Cantor set C in a Polish space, the following are equivalent:

(i) C has empty interior,

(ii) C is nowhere dense,

(iii) C is meager.

Proof. (i) $\iff$ (ii) holds as C is closed, (ii) $\implies$ (iii) by definition and (iii) $\implies$ (i) by Baire's theorem. □

Fact 0.1.4. For a topological space X, the following conditions are equivalent:

(i) $X \cong 2^{\mathbb{N}}$.

(ii) X is nonempty, compact, totally disconnected, metrizable, and has no isolated points.

(iii) X is nonempty, compact, totally disconnected, second countable, Hausdorff, and has no isolated points.

Proof. (ii) $\iff$ (iii) follows from Urysohn's metrizability theorem. (i) $\implies$ (iii) is clear. Thus it remains to show that any two spaces satisfying (iii) are homeomorphic. Under Stone duality, X corresponds to a nonzero countable atomless Boolean algebra. The result follows by back-and-forth. □

This implies the following fact.

Fact 0.1.5. (a) Let $A \subseteq 2^{\mathbb{N}}$. Then A is homeomorphic to $2^{\mathbb{N}}$ if and only if A is compact, nonempty and has no isolated points.

(b) Let $A \subseteq \mathbb{R}$. Then A is homeomorphic to $2^{\mathbb{N}}$ if and only if A is compact, nonempty, has no isolated points and empty interior.

The next fact follows since open subsets of spaces without isolated points have no isolated points.

Fact 0.1.6. If $U \subseteq 2^{\mathbb{N}}$ is nonempty clopen, then U is homeomorphic to $2^{\mathbb{N}}$.

Since any second countable topological space is the disjoint union of a closed subset without isolated points and a countable open subset, we obtain the following fact.

Fact 0.1.7. (a) If $A \subseteq 2^{\mathbb{N}}$ is compact, then A is either countable or a disjoint union of a Cantor set and a countable set.

(b) If $A \subseteq \mathbb{R}$ is compact with empty interior, then A is either countable or a disjoint union of a Cantor set and a countable set.

Order theory

For some spaces, like $\mathbb{R} \setminus \mathbb{Q}$ or $2^{\mathbb{N}}$, the monadic theory of order is more expressive than the theory of topology. Thus, we need refined versions of the previous results concerning order isomorphisms and not only homeomorphisms.

Fact 0.1.8. Let $A \subseteq \mathbb{R}$ or $A \subseteq 2^{\mathbb{N}}$ be a Cantor set. Then A is order isomorphic to $2^{\mathbb{N}}$.

Proof. We may assume $A \subseteq \mathbb{R}$ as $2^{\mathbb{N}}$ embeds into $\mathbb{R}$ simultaneously as order and as topological space. For a proof of this folklore result see [Das05, Section 1.2, Theorem, p. 3]. □

Fact 0.1.9. Let $A \subseteq \mathbb{R}$ be countable and dense. Then $\mathbb{R} \setminus A$ is order isomorphic to $\mathbb{R} \setminus \mathbb{Q}$.

Proof. By Cantor's theorem A is order isomorphic to $\mathbb{Q}$. Extending such an isomorphism to completions, we obtain an order automorphism $f \colon \mathbb{R} \to \mathbb{R}$ with $f(A) = \mathbb{Q}$. Thus $f|_{\mathbb{R} \setminus A}$ is the desired isomorphism. □

Let $(X, \leq)$ be a total order. We use interval notation $[a, b], (a, b), (a, b], [a, b)$ for intervals in X. A **jump** in X is a pair $(x, y) \in X \times X$ with $x < y$ such that there is no $z \in X$ with $x < z < y$. The jumps in X are totally ordered by $(x, y) < (x', y')$ if and only if $x < x'$ if and only if $y < y'$ if and only if $y \leq x'$. If X is complete and dense and $C \subseteq X$ is closed, we call the connected components of $X \setminus C$ **complementary interval**s of C in X. The complementary intervals are totally ordered by $I < J$ if and only if there are i, j with $I \ni i < j \in J$ if and only if all $i \in I, j \in J$ satisfy $i < j$.

Fact 0.1.10. The set $2^{\mathbb{N}}$ has a cocountable subset order isomorphic to $\mathbb{R}$ and first-order definable in $(2^{\mathbb{N}}, \leq)$.

Proof. Consider the set X of $y \in 2^{\mathbb{N}} \setminus \{0^\infty, 1^\infty\}$ such that there is no $x \in 2^{\mathbb{N}}$ such that (x, y) is a jump. Then X is cocountable and definable. It is folklore that X is order isomorphic to $\mathbb{R}$. □

0.1.3 Undecidability in second-order logic

To familiarize the reader with monadic theories and motivate the main result, we prove some folklore results. Our main example will be the natural numbers. Recall that in first-order logic, the structures $(\mathbb{N}), (\mathbb{N}, +1), (\mathbb{N}, \leq), (\mathbb{N}, +), (\mathbb{N}, +, \cdot)$ have strictly increasing strength and all but the last one are decidable.

We first observe that studying higher-order logic beyond monadic logic yields undecidability even for restricted languages.

Proposition 0.1.11. *The second-order theory of $(\mathbb{N}, +1)$ defines addition and multiplication.*

Proof. The usual recursive definitions are second-order. For example, the graph of addition can be defined as the least set G of triples such that $(n, 0, n) \in G$ for all n and $(k, m, n) \in G$ implies $(k, m + 1, n + 1) \in G$. □

The second-order theory in the empty language, although it defines no nontrivial subsets, has the same expressive power. More precisely, it can quantify over pairs of elements $z \in \mathbb{N}$ and maps $s\colon \mathbb{N} \to \mathbb{N}$ satisfying the Peano axioms: s is injective, $\mathrm{im}(f) = \mathbb{N} \setminus \{z\}$ and every subset that contains z and is stable under s equals $\mathbb{N}$. In particular, the Turing degree of $(\mathbb{N})$ is full second-order arithmetic.

In contrast, the monadic theory of $(\mathbb{N})$ is the theory of atomic Boolean algebras, thus decidable. Much less trivially, we will see that the monadic theory of $(\mathbb{N}, +1)$, S1S, is decidable. While it does not define addition (this will follow from Corollary 0.1.16), it still defines order.

Proposition 0.1.12. *The monadic theories of $(\mathbb{N}, \leq)$ and $(\mathbb{N}, +1)$ are interdefinable.*

Proof. Defining $+1$ from $\leq$ is possible in first-order logic. For the converse use that $x \leq y$ if and only if there is a set $A \subseteq \mathbb{N}$ with $x, y \in A$, such that $a \in A \setminus \{y\}$ implies $a + 1 \in A$, but $y + 1 \notin A$, and $a \in A \setminus \{x\}$ implies $a \neq 0$ and $a - 1 \in A$, but $x = 0$ or $x - 1 \notin A$. □

Lemma 0.1.13. *In monadic logic of a monoid $(X, \cdot)$, the map sending $A \subseteq X$ to the submonoid $\langle A \rangle$ generated by A is definable.*

Proof. Being a submonoid is definable, and $\langle A \rangle$ is the least submonoid including A. □

Corollary 0.1.14. *The monadic theory of $(\mathbb{N}, +)$ defines divisibility.*

Lemma 0.1.15. *The first-order theory of $(\mathbb{N}, +, |)$ defines multiplication.*

Proof. It suffices to define squaring since $2xy = (x + y)^2 - x^2 - y^2$. Now, $x^2 = \mathrm{lcm}(x, x + 1) - x$ as $x, x + 1$ are always coprime. □

Corollary 0.1.16. *The monadic theory of $(\mathbb{N}, +)$ defines multiplication.*

Furthermore, first-order arithmetic defines a pairing function (that is, an injection $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$), whence the monadic theory of $(\mathbb{N}, +)$ interprets second-order arithmetic.

Generally, monadic logic of algebraic structures tends to be infeasible:

Proposition 0.1.17. *Let G be a group that is not torsion. Then the monadic theory of $(G, \cdot)$ is undecidable.*

Proof. For g of infinite order, $(G, \cdot, g)$ is undecidable by Corollary 0.1.16 (by reductions independent of g). Moreover, whether g has infinite order is definable using Lemma 0.1.13 by whether the generated submonoid is a group. □

Monadic logic can become feasible for algebraic structures if the monadic quantifier is restricted to more algebraic classes, like convex subgroups of an ordered abelian group [Gur77a] or ideals of a countable Boolean algebra [Rab69, Theorem 2.10], but we will not study such theories.

Example 0.1.18. We investigate the monadic theory of $(\mathbb{R}, +)$, or more generally of $\mathbb{Q}$-vector spaces in the language of additive groups.

(a) Let $V \oplus W$ be a decomposition of a $\mathbb{Q}$-vector space. We use this decomposition to code nonempty sets of subsets of V of cardinality at most $\#W$. Namely, let $A \subseteq 2^V$ be such that there is a surjection $f\colon W \to A$. Then $X = \bigcup_{w \in W} f(w) \oplus \{w\}$ is a subset of $V \oplus W$ that allows to recover

$$A = \{\{v \in V \mid v \oplus w \in X\} \mid w \in W\}.$$

(b) For each $x \in \mathbb{R}$, third-order arithmetic on $\mathbb{Q} \cdot x$ can be expressed in $(\mathbb{R}, +)$ by our previous reduction of second-order arithmetic and (a) with $V = \mathbb{Q} \cdot x$ and W a linear complement of V. Thus, the Turing degree of the the monadic theory of $(\mathbb{R}, +)$ equals the Turing degree of third-order arithmetic.

(c) The monadic theory of a $\mathbb{Q}$-vector space V defines whether V is n-dimensional for given n, finite-dimensional, countable-dimensional, or continuum-dimensional. First, using Lemma 0.1.13 we can define generated subgroups, thus generated $\mathbb{Q}$-subspaces as the unions of all cyclic subgroups containing a generator. Then fixed finite dimensions can be easily recognized. To obtain countable dimensions, code countable increasing chains of subspaces by (a) with $\dim W = 1$, employing arithmetic on W. Similarly, binary trees enable to code Cantor sets of subspaces.

The Borel (or projective or closed) monadic theory of $(\mathbb{R}, +)$ only has the Turing degree of second-order arithmetic.

0.1.4 Results

Chapter 1: Survey of undecidability

In section 1.2 we state an error in the proof of [GS82, Claim 3.3], give an explicit counterexample and a slight change to a definition that fixes the problem.

There are several variants of the undecidability proof of the monadic theory of $(\mathbb{R}, \leq)$. They differ in several regards. Some of them require the continuum hypothesis (or weaker set-theoretical assumptions) [She75; GS84], while others do not [GS82; She88]. Moreover, they give different lower bounds on the Turing degree. The trivial upper bound is the Turing degree of third-order arithmetic. The best known lower bound for the Turing degree of the monadic theory of $(\mathbb{R}, \leq)$ in a model V of set theory is the Turing degree of third-order arithmetic in the forcing extension of V obtained by adjoining a Cohen real [GS84; She88]. In some models, this matches the upper bound, but in general the Turing degree of third-order arithmetic may decrease when passing to forcing extensions. The only proof of this lower bound without the continuum hypothesis is thus [She88], which the author found hard to read. In section 1.3 we give a slightly changed variant of this undecidability proof.

Chapter 2: Monadic theories of order

The main result extends the decidability of the F_σ monadic theory of $(\mathbb{R}, \leq)$ [Rab69, Theorem 2.9], answering a question of Shelah [She75, Conjecture 7B; She90, $\alpha(5)$, p. 95; She00, 2.22].

Theorem 0.1.19. *The Borel monadic theory of $(\mathbb{R}, \leq)$ is decidable. The Boolean combinations of F_σ sets form an elementary substructure of the Borel sets.*

Assuming determinacy, we obtain variants.

Theorem 0.1.20. *Assume* $\mathsf{ZF} + \mathsf{DependentChoices} + \mathsf{Determinacy}$. *The monadic theory of $(\mathbb{R}, \leq)$ is decidable. The Boolean combinations of F_σ sets form an elementary substructure of all sets.*

To provide a general framework, we give a technical definition. We throughout work in $\mathsf{ZF} + \mathsf{DependentChoices}$ to accommodate determinacy.

Definition 0.1.21. A family $(\mathcal{B}_X)_X$ of Boolean subalgebras of 2^X for X ranging over the G_δ subsets of $\mathbb{R}$ is **sufficiently stable** if there are $\mathcal{A}_X \subseteq 2^X$ such that

(a) $\mathcal{A}_X$ generates $\mathcal{B}_X$ as a Boolean algebra;

(b) if $f\colon X \to Y$ is a homeomorphism and $A \in \mathcal{A}_X$, then $f(A) \in \mathcal{A}_Y$;

(c) if $A \subseteq X$ is G_δ and $B \in \mathcal{A}_X$, then $A \cap B \in \mathcal{A}_A$;

(d) if $\mathbb{R}$ is partitioned into a G_δ subset Y and countably many locally closed subsets X_i, then $A \in \mathcal{B}_\mathbb{R}$ if $A \cap Y \in \mathcal{A}_Y$ and $A \cap X_i \in \mathcal{A}_{X_i}$ for all i.

A Boolean subalgebra of $2^{\mathbb{R}}$ is sufficiently stable if it is a component of a sufficiently stable family of Boolean algebras.

We extend the notations $\mathcal{A}_X, \mathcal{B}_X$ to all X that are homeomorphic to a G_δ subset of $\mathbb{R}$ by transporting along any homeomorphism and often abuse notation by writing $\mathcal{B}$ instead of $\mathcal{B}_X$.

The Borel sets are sufficiently stable with $\mathcal{A}$ comprising all Borel sets. The Boolean combinations of F_σ sets are sufficiently stable with $\mathcal{A}$ consisting of the G_δ sets. Similarly, the Boolean combinations of $\mathbf{\Sigma}^0_\alpha$ sets for $\alpha \geq 2$ or the $\mathbf{\Delta}^0_\alpha$ sets for $\alpha \geq 3$ form sufficiently stable Boolean algebras, as do the Boolean or σ-combinations of $\mathbf{\Sigma}^1_n$ sets for $n \geq 1$ or the $\mathbf{\Delta}^1_n$ sets.

Theorem 0.1.22. *Let $\mathcal{B}, \mathcal{C}$ be sufficiently stable Boolean algebras with $\mathcal{B} \subseteq \mathcal{C}$ such that every subset of $2^{\mathbb{N}}$ in $\mathcal{C}$ is determined.*

(a) The monadic theory of $(\mathbb{R}, \leq)$ with quantification restricted to $\mathcal{B}$ is decidable.

(b) The embedding $\mathcal{B} \to \mathcal{C}$ is elementary.

An earlier version of this work has been submitted and is available as [Man24].

Chapter 3: Monadic theories of the binary tree

This chapter is based on joint work with Mikołaj Bojańczyk, Antonio Casares and Paweł Parys. When Shelah asked for the decidability of the Borel monadic theory of $(\mathbb{R}, \leq)$, answered affirmatively by Theorem 0.1.19, he also asked for the Borel monadic theory of $2^{\leq\omega}$ with the prefix relation and lexicographic order [She75, Conjecture 7A; Gur85, Open Question, p. 503; She00, 2.22; Boj+14, Conclusions]. This theory is a natural generalization of both the Borel monadic theory of $(\mathbb{R}, \leq)$, which is equivalent to that of $2^{=\omega}$, and S2S, the monadic theory of $2^{<\omega}$. Moreover, in contrast to these theories, it expresses Borel determinacy as a sentence [Boj+14, Section 5.3], whence Baire category methods cannot suffice [Fri71]. The semigroup-theoretic approach seems inappropriate. Instead, trying to also extend the result that Boolean combinations of F_σ sets form an elementary substructure, we define automata operating on trees with labeled nodes and branches.

We attempt to generalize the modern decidability proof of S2S. The core of the proof is the complementation lemma, which depends on finite memory determinacy. Thus, we define prioritized Borel games and formulate Conjecture 3.2.13 stating the required determinacy. We provide equivalent formulations of the conjecture (Theorems 3.5.3 and 3.5.10) and establish particular cases. In particular, we can negate Büchi automata by Corollary 3.4.17 and automata with a single state and edge-based acceptance conditions by Proposition 3.4.20.

The main results of this work have been submitted, but the submitted version is not publicly available at the time of print.

0.1.5 Related theories

Topology

The monadic language of topology is given by adding to the monadic language of equality a predicate for open sets. Equivalently, one may add a predicate for closed sets, the interior or closure operator, or the neighborhood relation. Since the definition of the order topology is monadic, the monadic theory of any order defines these relations. For the reals, we obtain a converse (although order is undefinable as it is not preserved by the autohomeomorphism $x \mapsto -x$).

Proposition 0.1.23. *The Turing degree of the monadic theory of $(\mathbb{R}, \leq)$ equals the Turing degree of the monadic theory of topology of $\mathbb{R}$.*

Proof. The monadic theory of topology defines the class of connected sets. Since a subset of $\mathbb{R}$ is connected if and only if it is convex, it defines the relation $x \leq z \leq y \vee y \leq z \leq x$. Thus, quantifying over two distinct parameters, we can recover order or its reverse. □

The same works for restricted monadic theories for any class of sets containing the closed intervals. In contrast, the monadic theory of topology of $\mathbb{R}^n$ for $n \geq 2$ is undecidable as it interprets arithmetic [Grz51, Corollary 1]. The proof goes through even if we restrict to small classes of sets (like Boolean combinations of

cubes). The main idea is that we may interpret $n \in \mathbb{N}$ by a finite set of n points and express for example equality by connecting the points pairwise with paths. The monadic theory of topology of $\mathbb{Q}^n$ is decidable as it is homeomorphic to $\mathbb{Q}$.

Many topological notions are definable in the monadic language of topology, for example, dense, connected and perfect sets, or sets with exactly n isolated points for fixed $n \in \mathbb{N}$. The Borel monadic theory of topology defines more properties.

Proposition 0.1.24. *Let $\mathcal{B}$ be a sufficiently stable Boolean algebra such that every subset of $2^{\mathbb{N}}$ in $\mathcal{B}$ is determined.*

(a) The restricted monadic theory defines the class of countable sets.

(b) The restricted monadic theory defines the class of F_σ sets.

(c) The restricted monadic theory defines the class of meager sets.

Proof. (a) follows from the definability of perfect sets and the perfect set property. Similarly, (b) follows from Hurewicz' theorem [Kec95, Theorem 21.18]: A set A is G_δ if and only if there is no Cantor set C such that A is countable dense in C. (c) follows as a set is meager if and only if it is included in an F_σ set with empty interior. □

Corollary 0.1.25. *The Borel monadic theory of topology of the class of zero dimensional Polish spaces is decidable.*

Proof. A space is zero dimensional Polish if and only if it is homeomorphic to a G_δ subset of the Cantor set. Since the class of G_δ sets is definable, the claim follows from decidability of the Borel monadic theory of topology of the Cantor set. □

Bi-separable orders

We investigate the class of orders for which our results on the reals apply. The folklore Proposition 0.1.26 is also discussed in [Ges16].

A subset A of a partial order is **join-dense** if every element is the supremum of a subset of A. For total orders this is equivalent to $A \cap (a, b] \neq \emptyset$ for every nonempty - but possibly unbounded - left open interval. In particular, join-dense subsets are dense in the order topology. The notion of being **meet-dense** is defined dually. We call a subset **bi-dense** if it is both join-dense and meet-dense. For dense total orders, being dense in the order topology implies being bi-dense, but for general orders bi-dense sets are better behaved than sets dense in the order topology. For example, the Dedekind completion $\hat{A}$ of A is the unique complete order such that A is bi-dense in $\hat{A}$.

Proposition 0.1.26. *The following conditions are equivalent for a total order X, and we call such orders bi-separable:*

(i) The order topology on X is separable and there are only countably many jumps, that is, pairs (x, y) such that there is no $x < z < y$.

(ii) X has a countable join-dense subset.

(iii) X has a countable meet-dense subset.

(iv) There is an order embedding $X \to \mathbb{R}$.

(v) There is an order embedding $X \to 2^{\mathbb{N}}$.

(vi) There is an order embedding $X \to \mathbb{N}^{\mathbb{N}}$.

Proof. (i) $\implies$ (ii): Let D be dense in the order topology and E be the set of right endpoints of jumps. We claim that $D \cup E$ is join-dense. If $(a, b]$ is an interval, then either (a, b) is a jump, whence $b \in E$, or it is no jump, whence $D \cap (a, b) \neq \emptyset$.

(ii) $\Longrightarrow$ (iv): Let $A \subseteq X$ be countable join-dense and $i\colon A \to \mathbb{Q}$ be an order embedding. Extend i to X by mapping x to the supremum of $\{i(a) \mid A \ni a \leq x\}$. This is clearly monotone and coincides with i on A. For injectivity, suppose that $i(x) \leq i(y)$. Thus $i(a) \leq i(y)$ for all $A \ni a \leq x$. This yields $a \leq y$. Since A is join-dense, $x \leq y$ follows.

(iv) $\Longrightarrow$ (i): There are only countably many jumps since for any jump (x, y) there is a rational $i(x) < q < i(y)$. Finally, taking for each open interval (p, q) with $p, q \in \mathbb{Q}$ such that $i(X) \cap (p, q) \neq \emptyset$ an element of $i^{-1}((p, q))$ yields a dense subset.

(i) $\Longrightarrow$ (iii) $\Longrightarrow$ (iv) is dual to (i) $\Longrightarrow$ (ii) $\Longrightarrow$ (iv). Finally, (iv) $\Longleftrightarrow$ (v) $\Longleftrightarrow$ (vi) follows as there are mutual order embeddings between $\mathbb{R}, 2^{\mathbb{N}}, \mathbb{N}^{\mathbb{N}}$. □

Example 0.1.27. Although the order topology of $\{1, 2\} \times \mathbb{R}$ is separable, it is not bi-separable. The Dedekind completion of $\{1, 2\} \times \mathbb{Q}$ is not $\{1, 2\} \times \mathbb{R}$, but the Cantor set $2^{\mathbb{N}}$.

Adding or removing endpoints to $\mathbb{R}, \mathbb{R} \setminus \mathbb{Q}$ or $2^{\mathbb{N}}$ does not change the Turing degree of their monadic theories. This is straightforward. One implication is also a particular case of Corollary 2.3.7. By the proof of [She75, Lemma 7.12] we obtain:

Lemma 0.1.28. *Let $\mathcal{C}$ be a class of total orders such that $\mathbb{R} \in \mathcal{C}$. Then the monadic theory of $(\mathbb{R}, \leq)$ is computable from the theory of $\mathcal{C}$.*

Proof. We show that for any sentence φ we have $\mathbb{R} \models \varphi$ if and only if for all nonempty complete densely ordered $X \in \mathcal{C}$ without endpoints there is a nonempty complete densely ordered $Y \subseteq X$ without endpoints such that $Y \models \varphi$. For sufficiency set $X = \mathbb{R}$ and note $Y \cong \mathbb{R}$. For necessity let $A \subseteq X$ be nonempty countable densely ordered without endpoints and let Y be the Dedekind completion of A. □

Proposition 0.1.29. *The following theories have the same Turing degree:*

(i) The monadic theory of $(\mathbb{R}, \leq)$.

(ii) The common monadic theory of the class of bi-separable orders.

(iii) The monadic theory of $(2^{\mathbb{N}}, \leq)$.

(iv) The monadic theory of $(\mathbb{R} \setminus \mathbb{Q}, \leq)$.

Proof. (iv)≥(iii) follows as $2^{\mathbb{N}}$ embeds into $\mathbb{R} \setminus \mathbb{Q}$ and being a Cantor set is definable by Fact 0.1.5 as compactness of a subset of $\mathbb{R} \setminus \mathbb{Q}$ is definable by being equivalent to being a complete total order with endpoints. (iii)≥(ii) is easy. (ii)≥(i) follows from Lemma 0.1.28. (i)≥(iv) follows from Fact 0.1.9 by the argument of Lemma 0.1.28. Namely, $\mathbb{R} \setminus \mathbb{Q} \models \varphi$ if and only if for all $X \subseteq \mathbb{R}$ with empty interior there is $Y \supseteq X$ with empty interior such that $Y \models \varphi$. □

The mentioned results continue to hold when quantification is restricted to a sufficiently stable Boolean algebra $\mathcal{B}$ (although our definition of sufficiently stable needs to be changed to larger index sets if we want to accommodate bi-separable orders that are not G_δ in their Dedekind completion). In this case, the monadic theory of the class of bi-separable orders needs to be replaced by the class of such orders which are members of $\mathcal{B}$ in their Dedekind completion. (This replacement is also necessary by Remark 1.3.11.) The only part of the argument which requires modifications is the definability of the complete orders. It persists as an order is complete if every bounded convex subset has infimum and supremum, and convex subsets of bi-separable orders are locally closed, thus in $\mathcal{B}$.

A larger class is the class of ω_1-short orders, that is, the orders which do not include a copy of $(\omega_1, \leq)$ or its order dual $(\omega_1, \geq)$. For these orders, obtaining decidability results is problematic.

Proposition 0.1.30. *The Borel monadic theory of $\{1, \ldots, n\} \times \mathbb{R}$ is decidable if and only if $n \leq 2$.*

Proof. $n = 0$ is trivial and $n = 1$ is Theorem 0.1.19. The case $n = 2$ reduces to $n = 1$ as every Borel subset of $\{1, 2\} \times \mathbb{R}$ is the union of $\{1, 2\} \times B$ with B Borel and a countable set. For $n \geq 3$, note that $\{2\} \times \mathbb{R}$ is a definable, open, discrete subset. Thus every subset is Borel and $\{1, \ldots, n\} \times \mathbb{R}$ interprets the unrestricted monadic theory of the reals. □

Decidability may be obtainable by considering more restricted monadic quantifiers.

Conjecture 0.1.31. *The common monadic theory of the class of ω_1-short orders with quantification restricted to the σ-algebra generated by the intervals is decidable.*

For bi-separable orders, this σ-algebra coincides with the Borel σ-algebra.

The Baire property

The result for sufficiently stable Boolean algebras, Theorem 0.1.22, enables decidability of the monadic theory of $(\mathbb{R}, \leq)$ for large classes of sets. For example, assuming the corresponding determinacy, we can obtain decidability for projective sets or even sets constructible from reals.

Nevertheless, undecidability of the theory with quantification over projective sets is also consistent with ZFC. More precisely, undecidability holds if a projective well-order of the reals exists. The undecidability proof applies as noted in [She75, p. 410], see also Remark 1.3.36. This holds without the continuum hypothesis using the proofs in [GS82] or [She88]. A detailed explanation under the continuum hypothesis is [Boj+14, Theorem 3.1]. In fact, the axiom of constructibility yields a Δ^1_2-well-order.

It is thus natural to ask about decidability for classes between $\mathbf{\Delta}^1_1$ and $\mathbf{\Delta}^1_2$ in ZFC. For example, decidability for the σ-algebra $\sigma(\mathbf{\Sigma}^1_1)$ generated by the analytic sets would follow from:

Conjecture 0.1.32. *Theorem 0.1.22 continues to hold if determinacy is replaced by the Baire property.*

Since the Baire property for all projective sets is consistent with ZFC without requiring the large cardinal hypothesis of determinacy [She84, Theorem 7.16], under Conjecture 0.1.32 decidability of the theory for projective sets is consistent with ZFC. Similarly, since the consistency strength of determinacy is not necessary to obtain the Baire property for all sets [She84, Conclusion 7.17], under Conjecture 0.1.32 decidability of the unrestricted monadic theory is independent of ZF + DependentChoices.

The largest sufficiently stable Boolean algebra with the Baire property comprises the subsets $A \subseteq \mathbb{R}$ with the Baire property in the restricted sense. ($A \subseteq \mathbb{R}$ has the Baire property in the restricted sense if for all $C \subseteq \mathbb{R}$ the set $A \cap C$ has the Baire property in C. It suffices to verify this condition when C is $\mathbb{R}$ or a Cantor set [Kur66, §11 VI].) These sets form a σ-algebra containing all universally Baire sets.

We finally justify why Conjecture 0.1.32 is plausible. The only location in the proof where determinacy beyond the Baire property is used is section 2.5.2, considering a generalized Wadge game. The lemma as stated implies the perfect set property and thus fails for $\sigma(\mathbf{\Sigma}^1_1)$. More generally, if Conjecture 0.1.32 is true, then every set without a perfect subset has the monadic type of a countable set. (For instance, one can show that no Bernstein set has the Baire property and so the sets without a perfect subset form an ideal under the assumptions of Conjecture 0.1.32.) There is an analogous statement for F_σ sets, whence Proposition 0.1.24 (a) and (b) fail for $\sigma(\mathbf{\Sigma}^1_1)$. However, we provide an alternative of (c) that only uses the Baire property in section 2.2.3.

Chapter 1

Undecidability

1.1 Prerequisites

1.1.1 Topology

For $y \in A^{<\omega}$, we write $[y] = \{z \mid y \preceq z\}$ regarded as a subset of $A^{\mathbb{N}}$ or $A^{\leq\omega}$ depending on context.

Consider $\omega + 1$ with the order topology - equivalently, this is the one-point compactification of $\mathbb{N}$ with the discrete topology - and equip $(\omega+1)^{=\omega}$ with the product topology. We equip $\mathbb{N}^{\leq\omega}$ with the quotient topology induced by the surjection $f\colon (\omega+1)^{=\omega} \to \mathbb{N}^{\leq\omega}$ that is the identity on $\mathbb{N}^{=\omega}$ and maps $(x_1, \dots, x_n, \omega, x_{n+2}, \dots)$ with $x_1, \dots, x_n \in \mathbb{N}$ to $(x_1, \dots, x_n)$.

Proposition 1.1.1. *Let $x \in \mathbb{N}^{\leq\omega}$. If $x \in \mathbb{N}^{\mathbb{N}}$, then the $[y]$ for $y \prec x$ form a neighborhood base of x. Else the $\bigcup_{m\geq k} \{(x_1, \dots, x_n)\} \cup [(x_1, \dots, x_n, m)]$ for $k \in \mathbb{N}$ form a neighborhood base of $x = (x_1, \dots, x_n)$.*

Proof. Since $f^{-1}([y]) = [y]$ for $y \in \mathbb{N}^n$ and $f^{-1}(\{(x_1, \dots, x_n)\}) = [(x_1, \dots, x_n, \omega)]$, the stated neighborhoods are open. If A is any neighborhood of $x \in \mathbb{N}^{\mathbb{N}}$, then $f^{-1}(A) \supseteq [y]$ for some $y \prec x$ and so $A \supseteq f([y]) = [y]$. Finally, if A is any neighborhood of $(x_1, \dots, x_n)$, then $f^{-1}(A)$ is a neighborhood of $(x_1, \dots, x_n, \omega, p_1, p_2, \dots)$ for any $p \in (\omega+1)^{=\omega}$, thus includes some $\bigcup_{m\geq k}[(x_1, \dots, x_n, m, p_1, \dots, p_\ell)]$, whence $A \supseteq \bigcup_{m\geq k} f([(x_1, \dots, x_n, m, p_1, \dots, p_\ell)]) = \bigcup_{m\geq k}[(x_1, \dots, x_n, m, p_1, \dots, p_\ell)]$, and, as p was arbitrary, $A \supseteq \bigcup_{m\geq k}[(x_1, \dots, x_n, m)]$. □

Lemma 1.1.2. *The following spaces are homeomorphic:*

(i) $2^{\mathbb{N}}$.

(ii) $(\omega+1)^{=\omega}$.

(iii) $\mathbb{N}^{\leq\omega}$.

Proof. By Fact 0.1.4 it suffices to show that these spaces are nonempty, compact, totally disconnected, Hausdorff, second countable, and have no isolated points. They are clearly nonempty. For (i), (ii) use that products of totally disconnected (resp. compact, Hausdorff) spaces have the same property, that countable products of second countable spaces are second countable, and that no infinite product of spaces with at least two points has isolated points. It remains to consider (iii). Quotients of compact Hausdorff spaces X by closed (in $X \times X$) equivalence relations are compact Hausdorff. There are only countably many basic neighborhoods, and since they contain at least two points, there are no isolated points. For total disconnectedness use that each basic neighborhood is clopen as its preimage is so. □

Fact 1.1.3. Let I be countable, X, Y be dense total orders without endpoints, and $X_i \subseteq X, Y_i \subseteq Y$ for $i \in I$ be pairwise disjoint countable dense subsets with $\bigcup_i X_i = X, \bigcup_i Y_i = Y$. Then there is an order isomorphism $X \cong Y$ mapping X_i onto Y_i.

Proof. This follows from a back-and-forth argument. □

Fact 1.1.4. If X is an uncountable Polish space and $A \subseteq X$ includes no perfect subset, then $\#(X \setminus A) = 2^{\aleph_0}$.

Proof. There is an embedding f of $2^{\mathbb{N}} \times 2^{\mathbb{N}} \cong 2^{\mathbb{N}}$ into X. Each $f(\{i\} \times 2^{\mathbb{N}})$ is perfect in X, thus meets $X \setminus A$. Since they are pairwise disjoint as i varies, this yields $2^{\aleph_0}$ distinct elements. □

Lemma 1.1.5. *If E_i are countably many pairwise disjoint countable dense subsets of $2^{\mathbb{N}}$, and so are F_i, then there is an autohomeomorphism of $2^{\mathbb{N}}$ mapping E_i onto F_i for all i.*

Proof. By composing with a translation we may assume that no element of some E_i or F_i is of the form $x0^\infty$ or $x1^\infty$. Then, since both $\bigcup_i E_i \cup \{x0^\infty \mid x\}$ and $\bigcup_i F_i \cup \{x0^\infty \mid x\}$ are countable dense total orders without endpoints, there is an order isomorphism mapping each E_i onto F_i and $\{x0^\infty \mid x\}$ onto itself by Fact 1.1.3. This isomorphism extends to an order automorphism $2^{\mathbb{N}} \to 2^{\mathbb{N}}$. □

Lemma 1.1.6. *[She88, Claim 3.5] Let $E_0, \dots, E_{n-1}$ be dense in $2^{\mathbb{N}}$. There are $2^{\aleph_0}$ many Cantor sets $P_i \subseteq 2^{\mathbb{N}}$ with $E_0, \dots, E_{n-1}$ dense in P_i and $P_i \cap P_j \subseteq \bigcup_k E_k$ whenever $i \neq j$.*

Proof. If $n = 0$, the lemma follows from $2^{\mathbb{N}} \cong 2^{\mathbb{N}} \times 2^{\mathbb{N}}$, setting $P_i = \{i\} \times 2^{\mathbb{N}}$ in the latter space. Suppose $n > 0$.

We replace $2^{\mathbb{N}}$ with the space $\mathbb{N}^{\leq\omega}$ of Lemma 1.1.2(iii). For $k, k' < n$, either $E_k \cap E_{k'}$ or $E_k \setminus E_{k'}$ is somewhere dense. Thus, replacing the space by a subinterval and the E_k by subsets, we may assume the E_k pairwise disjoint. Thus, by Lemma 1.1.5 we may assume that $E_k = \bigcup_{i \in \mathbb{N}} \mathbb{N}^{in+k}$.

Now for $x \in \mathbb{N}^{\mathbb{N}}$ let P'_x be the set of $y \in \mathbb{N}^{\mathbb{N}}$ such that $y_{n^2} = x_n$ for all n, and $P_x = \overline{P'_x}$. Since $P_x \setminus P'_x \subseteq \mathbb{N}^{<\omega} = \bigcup_k E_k$ and the P'_x are pairwise disjoint, we have that $P_i \cap P_j \subseteq \bigcup_k E_k$. Since P'_x is nonempty without isolated points, P_x is a Cantor set. Finally, E_k is dense in P_x: Each $y \in P_x$ is the limit of $y|_{in+k} \in E_k$ as $i \to \infty$, and infinitely many i satisfy $y|_{in+k} \in P_x$ as it is the limit of $(y_0, \dots, y_{in+k-1}, m) \in P'_x$ as $m \to \infty$. □

Lemma 1.1.7. *If $D_1, \dots, D_n$ are dense subsets of a Polish space, then there is a Cantor set P such that all $P \cap D_i$ are dense in P.*

Proof. The usual construction of Cantor schemes [Kec95, Section 6.A] can be varied such that each $x10^\infty$ maps into D_i with $|x| \equiv i \mod n$. □

1.1.2 Regular open sets

We represent a few well-know facts about the relation between nowhere dense and regular open sets. A reference is [Kur66, Ch. 1, §8, in particular Theorems III.1, V.1–4, VII.2]. Their analogue when "nowhere dense" is replaced by "meager" is discussed in [Kec95, 8.G].

We use the usual logical notation $\wedge, \vee, \neg, \Rightarrow, \Leftrightarrow$ to denote operations in the Boolean algebra of regular open subsets (e.g., $\neg\neg U$ means $\overline{\overset{\circ}{U}}$).

Let X be a topological space. For $U \subseteq X$ open and $A \subseteq X$ arbitrary, we write $U \Vdash A$ if $U \setminus A$ is nowhere dense. Then there is a greatest set $|A|$ with $|A| \Vdash A$. Explicitly,

$$|A| = \overline{\overset{\circ}{A}}.$$

In particular, if U is open, then $|U|$ is the least regular open set with $|U| \supseteq U$. In any case, $|A|$ is a regular open set.

Example 1.1.8. We say that a set has the **finite Baire property** if it is the union of an open set and a nowhere dense set (this is a strengthening of the usual Baire property). The sets with the finite Baire property form a Boolean subalgebra of the power set. Moreover, $A \mapsto |A|$ restricts to an isomorphism of Boolean algebras from the sets with the finite Baire property modulo the ideal of nowhere dense sets onto the regular open sets.

1.1.3 Boolean-valued structures

We recall the standard definition of Boolean-valued $\mathscr{L}$-structures. For the sake of simplicity, we assume that $\mathscr{L}$ is a relational language without equality. Let B be a complete Boolean algebra.

A B-**valued structure** consists of an underlying set X and a map $\varphi \mapsto \|\varphi\|$ from atomic $\mathscr{L}$-formulas with variables in X to B. We extend this to all $\mathscr{L}$-sentences by

$$\begin{aligned} \|\varphi \wedge \psi\| &= \|\varphi\| \wedge \|\psi\|, \\ \|\neg\varphi\| &= \neg \|\varphi\|, \\ \|\exists x \varphi(x)\| &= \bigvee_{x \in X} \|\varphi(x)\|. \end{aligned}$$

We use usual model-theoretic notation, in particular $X \models \varphi(x)$ for $\|\varphi(x)\| = \top$.

X is **full** if for every formula φ there is $x \in X$ with $\|\exists x \varphi(x)\| = \|\varphi(x)\|$.

An **isomorphism** of B-valued structures X, Y is a partial map $f\colon X \to Y$ that induces a bijection modulo the equivalence relations $X, Y \models a = b$ and such that $\|\varphi(x)\| = \|\varphi(f(x))\|$ for all formulas. It suffices to check this for atomic formulas.

If $f\colon B \to C$ is a map, then X becomes a C-valued $\mathscr{L}$-structure by $\|\varphi\|_C = f(\|\varphi\|_B)$ for all atomic φ. If f is a morphism of Boolean algebras, this rule extends to all quantifier-free φ. It extends to all φ if f is a complete morphism or if X is full.

If $B = \{\bot, \top\}$, then B-valued $\mathscr{L}$-structures are the same as usual $\mathscr{L}$-structures. Thus, using the inclusion $\{\bot, \top\} \to B$, we can regard any usual $\mathscr{L}$-structure as B-valued $\mathscr{L}$-structure. If X is a full B-valued structure and U is a ultrafilter on B, then $B \to B/\mathcal{U} \cong \{\bot, \top\}$ yields a usual $\mathscr{L}$-structure.

Example 1.1.9. An ultraproduct of $\mathscr{L}$-structures X_i for $i \in I$ can also be described as the quotient of the Boolean-valued structure for the power set of I induced by the X_i by an ultrafilter.

Example 1.1.10. For any complete Boolean algebra B and $M \models \mathsf{ZFC}$ transitive, there is a B-valued model $M^{(B)}$ of ZFC consisting of the forcing names for B. If an ultrafilter U on B is M-generic, then the quotient by U is well-founded and its Mostowski collapse is the generic extension $M[U]$.

A **Boolean-valued structure with equality** is a Boolean-valued structure in a language with a symbol for equality that satisfies $X \models \forall x\colon x = x$ and $X \models \forall x \forall y\colon x = y \Rightarrow (\varphi(x) \Leftrightarrow \varphi(y))$ for all formulas φ (it suffices to check this for atomic formulas). This implies that the structure models that $=$ is an equivalence relation.

1.2 Fixing an error

In the proof of [GS82, Claim 3.3] it is claimed that $\mathrm{rp}(X_\alpha) \cap Y_\beta$ consists of at most one element. We show that this is false. Moreover, we specify how to change definitions in [GS82] to fix this claim. We also show that Claim 3.3 is true nonetheless without changing definitions. The proof of [GS82], including our correction, is also included in [Blu25, B VI 5].

1.2.1 The counterexample

[GS82] shows undecidability of the monadic theory of a non-modest topological space U by coding a subset of $\mathbb{N}^{\leq\omega}$ (with the topology specified before Lemma 1.1.2) inside U. Our counterexample works inside $\mathbb{N}^{\leq\omega}$, and we begin by explaining how the coding construction trivializes for $U = \mathbb{N}^{\leq\omega}$. Juxtaposition or $\hat{}$ denotes concatenation, and exponentiation is used to denote repeated concatenation of finite sequences (e.g., 2^3 denotes the word $(2, 2, 2)$, not the word (8)).

Since $U \cong 2^{\mathbb{N}}$, we conclude that $N^{<\omega}$ contains no perfect subset of U. Since it has no isolated points, U defines a vicinity space of degree 1 in the sense of [GS82, Section 2]. The natural number p in the paper is set to 1, and the guard $D^0 = X_0$ is defined as the set of $x \in \mathbb{N}^{<\omega}$ of even length. A set Y is guarded in the sense of the paper if and only if $Y \cap D^0$ is dense in Y, thus in particular if $Y \subseteq D^0$. Since U is Polish and zero-dimensional, it satisfies the assumptions of section 3 of the paper.

Now we choose some sets satisfying the assumptions of Theorem 3.1, using the same names as in the paper. We set $D = \mathbb{N}^{<\omega}$; indeed $D \setminus D^0$, the set of $x \in \mathbb{N}^{<\omega}$ of odd length, is dense. We set $Q_0 = 4\mathbb{N}, Q_1 = 4\mathbb{N}+2$ and let $Q_2, Q_3, \dots$ be an arbitrary partition of $2\mathbb{N}+1$ into infinite parts, and $P_i = \{x \in \mathbb{N}^{<\omega} \mid |x| \in Q_i\}$ for all i. Since each Q_i is infinite, each P_i is dense, and the condition that D^0 is the union of guarded parts is vacuous for $p = 1$. Finally, set $F = \{P_0, P_1\}$ (we could also choose e.g., $F = \{\mathbb{N}^{<\omega} \setminus P_1, \mathbb{N}^{<\omega} \setminus P_0\}$ and the counterexample would work, but stick to the above F for definiteness).

Let us repeat the construction on page 187, choosing sets in such a way that the tree structure obtained there coincides with the one defined above. First, the Q_i defined above will be the Q_i in the paper. This is valid as for $p = 1$ (i) reduces to $Q_i \subseteq 2\mathbb{N}$ for the guarded P_0, P_1, and (ii) to $Q_i \subseteq 2\mathbb{N}+1$ else. We need to choose a base $\{B_n \mid n\}$ of clopen sets; this can be done in such a way that $B_0 = U$ and $B_i = U_{a,j}$ for some a with $|a| < i$ (thus the later requirements on B will cause no trouble). We finally need to order D by the order type of the naturals; for this we use the lexicographic order on $\mathbb{N}^{<\omega}$.

Now it is easy to see that the recursive construction can be done in such a way that $e = \mathrm{id}_{\mathbb{N}^{<\omega}}$ and - setting $H_i = U_{a,i}$ for the given $a \in \mathbb{N}^{<\omega}$ - $[a]$ defined there is indeed the $[a]$ defined above.

For Claim 3.3, we are to choose a sequence of coherent, guarded, colorless $X_\alpha \subseteq D$.

We choose

$$X_0 = \left\{ i_1^4 i_2^4 \dots i_n^4 \mid i \in \mathbb{N}^{<\omega} \right\}.$$

This is indeed coherent ($i_1^4 \dots i_n^4$ is not isolated as $i_1^4 \dots i_n^4 m^4$ converges to it for $m \to \infty$), guarded (it is a subset of D^0), and colorless: If $a = i_1^4 \dots i_n^4$ and $b = j_1^4 \dots j_m^4$ are elements which are no prefixes of each other, then $i_k \neq j_k$ for some k, but then $\{a, b\}$ varies on positions $\{4k, \dots, 4k+3\}$ and thus is not of color P_0 or P_1.

Next we have to choose an element $x_0 \in \overline{X_0} \setminus D$ (note that repletion is closure for coherent subsets of topological spaces), and we take $x_0 = 0^\infty$.

Finally, for choosing X_1 we fix an injection $f\colon \mathbb{N}^{<\omega} \to \mathbb{N}$. Then set

$$X_1 = \left\{ i_1 0^3 i_2 0^3 \dots i_n 0^3 01^3 \hat{} f(i) \hat{} 0 \hat{} f(i) \hat{} 0 j_1^4 \dots j_m^4 \mid i \in (\mathbb{N} \setminus \{0\})^{<\omega}, j \in \mathbb{N}^{<\omega} \right\}.$$

As X_0, this is coherent and guarded. And X_1 is colorless: Suppose that $a = i_1 0^3 \dots i_n 0^3 01^3 \hat{} f(i) \hat{} 0 \hat{} f(i) \hat{} 0 j_1^4 \dots j_m^4$ and $b = i_1' 0^3 \dots i_{n'}' 0^3 01^3 \hat{} f(i') \hat{} 0 \hat{} f(i') \hat{} 0 {j_1'}^4 \dots {j_{m'}'}^4$ are no prefixes of each other; we have to show that $\{a, b\}$ is not mono. First suppose $n \neq n'$, say $n > n'$. Then some $i_j 0^3$ lies on the same position as 01^3; since $i_j \neq 0$, this shows the claim. Now suppose $n = n'$ and $i \neq i'$. Then $f(i)0f(i)0$ lies on the same position as $f(i')0f(i')0$; since they are different at positions $0, 2$, this shows the claim. Finally suppose $i = i'$. Then $j \neq j'$, thus as they are no prefixes some $j_k \neq j_k'$, and we are done.

Furthermore, $\overline{X_1} \supseteq \left\{ i_1 0^3 i_2 0^3 \cdots \mid i \in (\mathbb{N} \setminus \{0\})^{\mathbb{N}} \right\}$.

And Y_0 is the union over $i \in \{0, 1\}$ of the sets $Y_0^{P_i}$ of $y \notin D$ such that $\{x_0|_n, y|_n\}$ is of color P_i. But $Y_0^{P_0} = \left\{ i_1 0^3 i_2 0^3 \cdots \mid i \in \mathbb{N}^{\mathbb{N}} \right\}$ and so

$$\overline{X_1} \cap Y_0 \supseteq \overline{X_1} \cap Y_0^{P_0} \supseteq \left\{ i_1 0^3 i_2 0^3 \cdots \mid i \in (\mathbb{N} \setminus \{0\})^{\mathbb{N}} \right\}$$

has cardinality $2^{\aleph_0}$ (and not at most 1 as claimed).

1.2.2 A solution

A possible solution is obtained by adding to the definition of colorless (before [GS82, Claim 3.2]) the requirement that every element of $\mathrm{rp}(X)$ of infinite length has prefixes in X of unbounded length. Although Claim 3.3 is true without this change as discussed below, the alternative definition was presumably intended by the authors and does not affect any main result.

Changing the definition

The X constructed in Claim 3.2 satisfies our definition of colorless since it is the image of an isotone map from $\mathbb{N}^{<\omega}$. In particular, it satisfies our definition of colorless.

We now prove Claim 3.3 using our definition of colorless. The problematic step is showing that $\mathrm{rp}(X_\alpha) \cap Y_\beta$ has at most one element. In fact, it suffices to show that it has cardinality less than $2^{\aleph_0}$. We only show countability (the author did not check whether it has cardinality at most 1).

For $A \in F$, let $Y_\beta(A)$ be the set of $y \notin D$ such that $\{x_\beta|_n, y|_n\}$ is of color A for all n. Since Y_β is the countable union of all $Y_\beta(A)$, it suffices to show that $\mathrm{rp}(X_\alpha) \cap Y_\beta(A)$ has at most one element.

Let x, y be distinct elements, say $x_\ell \neq y_\ell$. Since $x \in \mathrm{rp}(X_\alpha)$, the neighborhood $[x|_{\ell+1}]$ of x meets X_α in some x'. Similarly, $[y|_{\ell+1}]$ meets X_α in some y'. Since x', y' are no prefixes of each other, by the original definition of colorless, $\{x', y'\}$ is not mono, in particular not of color A. By our definition of colorless, we may arrange that $x' \prec x, y' \prec y$. This contradicts $x, y \in Y_\beta(A)$.

Claim 3.3 holds

We show directly that Claim 3.3 holds without changing the definition. For this, by Claim 3.3 for the changed definition verified above, it suffices to show that every coherent, guarded set X that is colorless in the sense of the paper has such a subset X' that is colorless in our sense. We essentially repeat parts of the proof of Claim 3.2. By recursion, define an isotone injection $f\colon \mathbb{N}^{<\omega} \to X$. For the recursion step, since $f(x)$ is not isolated, there are distinct k_i for $i \in \mathbb{N}$ such that $[f(x)^\frown k_i]$ meets X. Since X is guarded, there is an appropriate $f(x^\frown i) \in [f(x)^\frown k_i] \cap X$. Finally, $X' = \mathrm{im}(f)$ does the job.

1.3 Undecidability

We follow [She88]. We slightly adapt some definitions, partially because the author didn't understand Shelah's constructions, and partially for convenience. Here, some of the differences are listed. These remarks can be skipped by readers not familiar with [She88].

For example, we replace $A \cap P \subseteq W \cup D$ in Notation 3.1(3) by $P \subseteq W$ for convenience. We do not discuss the simplifications under the continuum hypothesis, Claim 3.4. Condition (iv) of Claim 3.6 seems to fail when this claim is applied in Lemma 3.8, so we drop this requirement and instead require in Notation 3.1(3) that $P \cap D$ contains no perfect subset. We weaken Lemma 3.8 by strengthening the second part of the definition of L^b to "for no i both D_1, D_2 meet D_i^r". In Lemma 3.9, ensuring $\nu_0(k) = \cdots = \nu_{10}(k)$ seems incompatible with the required denseness of $D_i^r \supseteq D_1, D_j^r \supseteq D_2$ in P. (If this was achievable, contrasting the footnote on page 348, this author checked that "the minimal number needed" is two.) We again fix this by proving a weaker variant, strengthening the assumptions to $D_1 = u \cap D_i^r, D_2 = u \cap D_j^r$ and consequently using simpler K^+, K^-.

The proof of Claim 2.10A only works for sets of reals containing the empty set. Of course, this is unproblematic and we fix it by shifting reals. Finally, we prove a variant of Lemma 1.2(3) that does not require R to be symmetric (by reducing it to the restricted version) to replace the necessary reductions as part in later proofs, which for example led to the appearance of ordinal arithmetic in the proof of Claim 1.4 or Claim 2.10A.

1.3.1 Combinatorics

Lemma 1.3.1. *[She88, Construction 3.7] There is a closed embedding $\mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}, i \mapsto \eta_i$ such that for all i, j, the set $\{k \mid \eta_i(k) = \eta_j(k)\} \subseteq \mathbb{N}$ is an initial segment.*

Proof. Fix injections $f_n\colon \mathbb{N}^n \to \mathbb{N}$. For $i \in \mathbb{N}^\mathbb{N}$ set $\eta_i(k) = f_{k+1}(i(0), \dots, i(k))$. For $i \neq j$, let ℓ be minimal with $i(\ell) \neq j(\ell)$. Then $\{k \mid \eta_i(k) = \eta_j(k)\} = \{0, \dots, \ell - 1\}$. This defines a closed embedding as $x \in \{\eta_i \mid i\}$ if and only if for all n there is $i \in \mathbb{N}^{n+1}$ with $x|_n = (f_1(i(0)), f_2(i(0), i(1)), \dots)$. □

Choose η_i as in Lemma 1.3.1 and let D_i be the set of $\alpha \in \mathbb{N}^\mathbb{N}$ such that $\{k \mid \alpha(k) \neq \eta_i(k)\}$ is finite. For $J \subseteq \mathbb{N}^\mathbb{N}$, set $D_J = \bigcup_{i \in J} D_i$.

Definition 1.3.2. [She88, Notation 3.1, Definition 3.3]

(a) For a tuple $E = (E_0, \dots, E_{n-1})$ of subsets of D, let $\mathrm{PR}(E)$ be the set of Cantor sets P such that all E_i are dense in P.

(b) Let $\mathrm{Pr}^J(W, E)$ mean that there is $P \in \mathrm{PR}(E)$ such that $P \cap D_J$ includes no perfect subset and $P \subseteq W$.

(c) We say that $W \subseteq \mathbb{N}^\mathbb{N}$ **separates** $\mathcal{E}_1$ from $\mathcal{E}_2$ if $\mathrm{Pr}^J(W, E)$ for all $E \in \mathcal{E}_1$ but $\mathrm{Pr}^J(W, E)$ for no $E \in \mathcal{E}_2$.

By Lemma 1.1.7 changing "Cantor set" to "perfect set" in $\mathrm{PR}(E)$ leaves $\mathrm{Pr}^J(W,E)$ unchanged. Thus $\mathrm{Pr}^J(W,E)$ is expressible in monadic logic.

Remark 1.3.3. If J is Borel, then for each n the set of $\alpha \in \mathbb{N}^{\mathbb{N}}$ such that $\{k \,|\, \alpha(k) \neq \eta_i(k)\} \subseteq \{0,\dots,n\}$ for some $i \in J$ is Borel and so D_J is Borel. We will not use this fact.

For now, fix an arbitrary $J \subseteq \mathbb{N}^{\mathbb{N}}$. We will only use the following lemma to separate sets of tuples of subsets of D.

Lemma 1.3.4. *[She88, Claim 3.6] Suppose there are $K^+, K^- \subseteq \mathrm{PR}()$ such that:*

(a) For all $E \in \mathcal{E}_1$ there is $P \in K^+ \cap \mathrm{PR}(E)$ with $P \cap D_J$ countable.

(b) For each $Q \in \mathrm{PR}(E)$ with $E \in \mathcal{E}_2$ there is $Q' \subseteq Q$ with $Q' \in K^-$.

(c) If $P \in K^+, Q \in K^-$, then $P \cap Q$ is countable.

(d) $\#\mathcal{E}_1 \leq 2^{\aleph_0}$.

Then some W separates $\mathcal{E}_1$ from $\mathcal{E}_2$.

Proof. Write $\mathcal{E}_1 = \left\{E_\alpha \,\middle|\, \alpha < 2^{\aleph_0}\right\}$ and $K^- = \left\{Q_\alpha \,\middle|\, \alpha < 2^{\aleph_0}\right\}$. We construct by recursion on $\alpha < 2^{\aleph_0}$ sets $P_\alpha \in \mathrm{PR}(E_\alpha)$ and $P'_\alpha \in K^+$ with $P_\alpha \subseteq P'_\alpha$ such that for all $\beta < \alpha$ we have $Q_\beta \cap P_\alpha \subseteq D_J$.

Given P_β for $\beta < \alpha$, by (a) there is $P'_\alpha \in K^+ \cap \mathrm{PR}(E_\alpha)$ with $P'_\alpha \cap D_J$ countable. By Lemma 1.1.6 there are $P_{\alpha,\gamma} \in \mathrm{PR}(E_\alpha)$ with $P_{\alpha,\gamma} \subseteq P'_\alpha$ for $\gamma < 2^{\aleph_0}$ such that for $\gamma \neq \gamma'$ we have $P_{\alpha,\gamma} \cap P_{\alpha,\gamma'} \subseteq D_J$. By (c) $P'_\alpha \cap \bigcup_{\beta<\alpha} Q_\beta$ has cardinality at most $\#\alpha + \aleph_0 < 2^{\aleph_0}$. Thus, since the $P_{\alpha,\gamma} \setminus D_J$ are pairwise disjoint, there is γ with $P_{\alpha,\gamma} \cap \bigcup_{\beta<\alpha} Q_\beta \subseteq D_J$. Set $P_\alpha = P_{\alpha,\gamma}$.

Set $W = \bigcup_\alpha P_\alpha$. Then for any $E_\alpha \in \mathcal{E}_1$, the set P_α witnesses $\mathrm{Pr}^J(W, E_\alpha)$. Let $E \in \mathcal{E}_2$ and $Q \in \mathrm{PR}(E)$ be such that $Q \cap D_J$ includes no perfect subset. By (b) there is β with $Q_\beta \subseteq Q$. Since $Q_\beta \cap D_J$ includes no perfect subset, Fact 1.1.4 yields $\#(Q_\beta \setminus D_J) = 2^{\aleph_0}$. Since $Q_\beta \cap P_\alpha \subseteq D_J$ for $\alpha > \beta$, we have $Q_\beta \cap W \setminus D_J \subseteq \bigcup_{\alpha\leq\beta} P_\alpha \cap Q_\beta$. Thus, using (c)

$$\#(Q_\beta \cap W \setminus D_J) \leq \sum_{\alpha\leq\beta} \#(P_\alpha \cap Q_\beta) \leq \#\beta + \aleph_0 < 2^{\aleph_0} = \#(Q_\beta \setminus D_J)$$

yields $Q \supseteq Q_\beta \setminus D_J \nsubseteq W$. □

The following separations will be used to encode the domain of a Boolean-valued structure and equality.

Lemma 1.3.5. *[She88, Lemma 3.8] Let $\mathcal{E}_1$ be the set of (E_0, E_1) such that there are a nonempty open U and i such that E_0 and E_1 are dense subsets of $U \cap D_i$. Let $\mathcal{E}_2$ be the set of (E_0, E_1) such that there is a nonempty open U such that E_0 and E_1 are dense subsets of $U \cap D_J$, but for no i both E_0, E_1 meet D_i. Then some W separates $\mathcal{E}_1$ from $\mathcal{E}_2$.*

Proof. Let S_i be the set of $(x,y,z) \in \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ such that for all n with $x|_{n+1}, y|_{n+1}, z|_{n+1}$ pairwise distinct, at least two of $x(n), y(n), z(n)$ equal $\eta_i(n)$. Let K^+ be the set of Cantor sets P such that there is i with $P \times P \times P \subseteq S_i$. Let K^- be the set of Cantor sets Q such that for no i and distinct $x, y, z \in Q$ we have $(x,y,z) \in S_i$.

We check the conditions of Lemma 1.3.4. (d) holds as all E_0, E_1 with $(E_0, E_1) \in \mathcal{E}_1$ are countable. For (c) we show $\#(P \cap Q) \leq 2$. Indeed, if $x, y, z \in P \cap Q$ are distinct, then $x, y, z \in P$ yields i with $(x,y,z) \in S_i$, contradicting $x, y, z \in Q$.

For (a) let $(E_0, E_1) \in \mathcal{E}_1$, say U is nonempty open and E_0, E_1 are dense subsets of $U \cap D_i$.

We recursively construct $\ell_n \in \mathbb{N}$ and maps $f_n \colon \{0,1\}^n \to \mathbb{N}^{\ell_n}$. Let $f_0()$ be such that $[f_0()] \subseteq U$. Given f_n, construct an extension f_{n+1} such that:

- for $s \in \{0,1\}^n, a \in \{0,1\}$ with $s_n = a$ and $\ell_n \leq m < \ell_{n+1}$ we have $f_{n+1}(sa)(m) = \eta_i(m)$,
- $f_{n+1}(sa)$ concatenated with $(\eta_i(\ell_{n+1}), \eta_i(\ell_{n+1}+1), \dots)$ lies in E_a,
- for $s \in \{0,1\}^n, a \in \{0,1\}$ with $s_n \neq a$ there are $\ell_n < k^1_{sa} < k^2_{sa} < \ell_{n+1}$ such that:

– for $\ell_n \leq m < k^1_{sa}$ and for $k^2_{sa} \leq m < \ell_{n+1}$ we have $f_{n+1}(sa)(m) = \eta_i(m)$,
– there is $k^1_{sa} \leq m < k^2_{sa}$ with $f_{n+1}(sa)(m) \neq \eta_i(m)$,
– the intervals $\left\{k^1_{sa}, \ldots, k^2_{sa} - 1\right\}$ are pairwise disjoint as sa vary.

Mapping $a \in 2^{\mathbb{N}}$ to the unique element of $\bigcap_{n\in\mathbb{N}}[f_n(a|_n)]$ defines a Cantor set P. Since $s0^\infty$ is mapped into E_0 and $s1^\infty$ into E_1, we have $P \in \mathrm{PR}(E_0, E_1)$. And since each element of P coincides with η_i infinitely often, $P \cap D_J \subseteq D_i$, which is countable. Finally, $P \times P \times P \subseteq S_i$ by construction.

For (b) let $(E_0, E_1) \in \mathcal{E}_2$, say U is nonempty open such that E_0 and E_1 are dense subsets of $U \cap D_J$, but for no i both E_0, E_1 meet D_i. Let $Q \in \mathrm{PR}(E_0, E_1)$.

We may assume that one of the following holds:

(i) There is i_0 such that $E_0 \cap Q \subseteq D_{i_0}$.

(ii) For all i, $D_i \cap E_0 \cap Q$ is nowhere dense in Q.

Indeed, if (ii) fails, by shrinking Q to an open subset such that $D_i \cap E_0 \cap Q$ is dense in Q and then replacing E_0 by $E_0 \cap D_i$ on Q, we obtain (i). Do the same for E_1.

We recursively construct $\ell_n \in \mathbb{N}$ and maps $f_n \colon \{0,1\}^n \to \mathbb{N}^{\ell_n}$. Let $f_0()$ be such that $[f_0()] \subseteq U$. Given f_n, construct an extension f_{n+1} such that, where $j \equiv n \mod 2$:

- for $s \in \{0,1\}^n, a \in \{0,1\}$, the sequence $f_{n+1}(sa)$ has an extension in Q,
- if (i) holds for E_j, then for all s, a we have $f_{n+1}(s0) \neq f_{n+1}(s1)$ and $f_{n+1}(sa)(\ell_{n+1} - 1) = \eta_{i_j}(\ell_{n+1} - 1)$,
- if (ii) holds for E_j, then the $f_{n+1}(sa)(\ell_{n+1} - 1)$ are distinct as sa varies.

Indeed, if (i) holds, use that $U \cap D_{i_j} \cap Q \supseteq E_j \cap Q$ is dense in $U \cap Q$, and if (ii) holds, use that each $U \cap D_J \cap Q \setminus (D_{i_1} \cup \cdots \cup D_{i_m})$ is dense in $U \cap Q$ to find extensions in distinct D_{i_k}.

Mapping $a \in 2^{\mathbb{N}}$ to the unique element of $\bigcap_{n\in\mathbb{N}}[f_n(a|_n)]$ defines a Cantor set $Q' \subseteq Q$. By definition of U either E_0 or E_1 is disjoint from D_i, say E_0. Then the entries at $\ell_{2n+1} - 1$ for n large enough witness $(x, y, z) \notin S_i$. □

The following lemma will enable us to encode arbitrary relation symbols.

Lemma 1.3.6. *[She88, Lemma 3.9] Let R be a symmetric function from $\mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ to the regular open subsets of $\mathbb{N}^{\mathbb{N}}$. Let $\mathcal{E}_1$ be the set of (E_0, E_1) such that there are distinct i, j and a nonempty open $U \subseteq R(i, j)$ with $E_0 = U \cap D_i$ and $E_1 = U \cap D_j$. Let $\mathcal{E}_2$ be the set of (E_0, E_1) such that there are distinct i, j and a nonempty open $U \subseteq \neg R(i, j)$ with $E_0 = U \cap D_i$ and $E_1 = U \cap D_j$. Then some W separates $\mathcal{E}_1$ from $\mathcal{E}_2$.*

Proof. Let K^+ be the set of Cantor sets P such that there are distinct i, j with $P \subseteq R(i, j)$ and such that for all $x \in P$ all but finitely many n satisfy $x(n) \in \{\eta_i(n), \eta_j(n)\}$. Let K^- be the set of Cantor sets Q such that there are distinct i, j with $Q \subseteq \neg R(i, j)$ and such that for all but countably many $x \in Q$ both $\{n \,|\, x(n) = \eta_i(n)\}, \{n \,|\, x(n) = \eta_j(n)\}$ are infinite.

We check the conditions of Lemma 1.3.4. (d) holds as all E_0, E_1 with $(E_0, E_1) \in \mathcal{E}_1$ are countable.

For (c) let $P \in K^+, Q \in K^-$. Let i^+, j^+ be as in the definition of K^+ and i^-, j^- be as in the definition of K^-. If $\{i^+, j^+\} = \{i^-, j^-\}$, then $P \cap Q \subseteq R(i^+, j^+) \cap \neg R(i^-, j^-) = \emptyset$ by symmetry of R. Otherwise, we may assume $i^- \notin \{i^+, j^+\}$. It suffices to show that any $x \in P \cap Q$ is among the countably many exceptions in the definition of K^-. Since $x \in P \in K^+$, for all but finitely many n we have $x(n) \in \left\{\eta_{i^+}(n), \eta_{j^+}(n)\right\} \not\ni \eta_{i^-}(n)$. Thus, $\{n \,|\, x(n) = \eta_{i^-}(n)\}$ is finite.

For (a) let $(E_0, E_1) \in \mathcal{E}_1$, say U is nonempty open and i_0, i_1 are distinct with $U \subseteq R(i_0, i_1)$ and $E_j = U \cap D_{i_j}$. Choose α with $[\alpha] \subseteq U$ and define a Cantor set $P \subseteq U \subseteq R(i_0, i_1)$ as the image of the map $2^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ that sends x to y with $y(n) = \alpha(n)$ for $n < |\alpha|$ and $y(n) = \eta_{i_{x(n)}}(n)$ else. Since $s0^\infty$ is mapped into E_0 and $s1^\infty$ into E_1, we have $P \in \mathrm{PR}(E_0, E_1)$. And since each other element of P coincides with both η_{i_0}, η_{i_1} infinitely often, $P \cap D_J \subseteq D_{i_0} \cup D_{i_1}$, which is countable. Finally, $P \in K^+$ since for $n \geq |\alpha|$ we have $x(n) \in \{\eta_{i_0}(n), \eta_{i_1}(n)\}$.

For (b) let $(E_0, E_1) \in \mathcal{E}_2$, say U is nonempty open and i_0, i_1 are distinct with $U \subseteq \neg R(i_0, i_1)$ and $E_j = U \cap D_{i_j}$. Let $Q \in \mathrm{PR}(E_0, E_1)$. We recursively construct strictly increasing $\ell_n \in \mathbb{N}$ and maps $f_n \colon \{0,1\}^n \to \mathbb{N}^{\ell_n}$. Let $f_0()$ be such that $[f_0()] \subseteq U$ and $[f_0()]$ meets Q. Given f_n, construct an extension f_{n+1} such that:

- for $s \in \{0,1\}^n, a \in \{0,1\}$ with $n > 0$, $s_n = a$ and $\ell_n \leq m < \ell_{n+1}$ we have $f_{n+1}(sa)(m) = \eta_{i_a}(m)$,
- $f_{n+1}(sa)(\ell_{n+1} - 1) = \eta_{i_a}(\ell_{n+1} - 1)$,
- $f_{n+1}(sa)$ concatenated with $(\eta_{i_a}(\ell_{n+1}), \eta_{i_a}(\ell_{n+1} + 1), \dots)$ lies in Q.

Mapping $a \in 2^{\mathbb{N}}$ to the unique element of $\bigcap_{n\in\mathbb{N}}[f_n(a|_n)]$ defines a Cantor set Q'. Since the dense set of $s0^\infty$ is mapped into Q', we have $Q' \subseteq Q$. And for any $x \in Q'$ whose preimage is not ultimately constant, $\{n \mid x(n) = \eta_i(n)\}, \{n \mid x(n) = \eta_j(n)\}$ are infinite, whence $Q' \in K^-$. □

1.3.2 Regularizing subsets

Recall that $|A| = \overline{\mathring{A}}^{\circ}$ denotes the greatest set with $|A| \Vdash A$.

Theorem 1.3.7. *[She88, Lemma 1.2] There are pairwise disjoint countable dense D_i for $i \in \mathbb{N}^{\mathbb{N}}$ such that the following hold:*

(a) There is a monadic formula Substorey *such that for all $J \subseteq \mathbb{N}^{\mathbb{N}}$ there is a tuple W_J of subsets of $\mathbb{N}^{\mathbb{N}}$ such that for all $X \subseteq \mathbb{N}^{\mathbb{N}}$ we have $\mathbb{N}^{\mathbb{N}} \models \mathsf{Substorey}(U, X, W_J)$ if and only if $U \subseteq \bigvee_{i\in J} \left|\complement(X \setminus D_i)\right|$.*

(b) There is a monadic formula Storey *such that for all $J \subseteq \mathbb{N}^{\mathbb{N}}$ there is a tuple W_J of subsets of $\mathbb{N}^{\mathbb{N}}$ such that for all $X \subseteq \mathbb{N}^{\mathbb{N}}$ we have $\mathbb{N}^{\mathbb{N}} \models \mathsf{Storey}(U, X, W_J)$ if and only if $U \subseteq \bigvee_{i\in J} \left|\complement(X \triangle D_i)\right|$.*

(c) For all $n \in \mathbb{N}$, there is a monadic formula Rel *such that for all maps R from $(\mathbb{N}^{\mathbb{N}})^n$ to the regular open subsets of $\mathbb{N}^{\mathbb{N}}$ there is a tuple W_R of subsets of $\mathbb{N}^{\mathbb{N}}$ such that for all $X_1, \dots, X_n \subseteq \mathbb{N}^{\mathbb{N}}$ we have $\mathbb{N}^{\mathbb{N}} \models \mathsf{Rel}(U, X_1, \dots, X_n, W_R)$ if and only if $U \subseteq \bigvee_{i_1,\dots,i_n} \bigwedge_{j=1}^n \left|\complement(X_j \triangle D_{i_j})\right| \wedge R(i_1, \dots, i_n)$.*

Proof. First let D_i be as in section 1.3.1. For (a), we may assume J nonempty. Let W_J separate $\mathcal{E}_1$ from $\mathcal{E}_2$ of Lemma 1.3.5. It suffices to construct a formula φ with parameters D_J, W_J such that the union of all U with $\mathbb{N}^{\mathbb{N}} \models \varphi(U, X)$ is a dense subset of $\bigvee_{i\in J} \left|\complement(X \setminus D_i)\right|$. Let $\varphi(U, X)$ state that U is open, $U \Vdash \complement(X \setminus D_J)$, and whenever $V \subseteq U$ is nonempty open and $X_0, X_1 \subseteq X$ are dense subsets of $V \cap D_J$, then $\Pr^J(W_J, X_0, X_1)$ holds.

The following lemma implies $\left|\complement(X \setminus D_i)\right| \subseteq \bigcup_{U\colon \mathbb{N}^{\mathbb{N}} \models \varphi(U,X)} U$ and so that the union is dense in the join.

Claim A. *If $U \Vdash \complement(X \setminus D_i)$ for some $i \in J$, then $\mathbb{N}^{\mathbb{N}} \models \varphi(U, X)$.*

Proof. Clearly $U \Vdash \complement(X \setminus D_i) \subseteq \complement(X \setminus D_J)$. Let $V \subseteq U$ be nonempty open and $X_0, X_1 \subseteq X$ be dense subsets of $V \cap D_J$. Since $V \subseteq U \Vdash \complement(X \setminus D_i) \subseteq \complement(X_j \setminus D_i)$, the sets $E_j = X_j \cap D_i$ are still dense in $V \cap D_J$. Thus $(E_0, E_1) \in \mathcal{E}_1$, whence $\Pr^J(W_J, E_0, E_1)$ and so $\Pr^J(W_J, X_0, X_1)$ follow. □

The following lemma implies that if $\mathbb{N}^{\mathbb{N}} \models \varphi(U, X)$, then U has a dense open subset V with $V \subseteq \bigcup_{i\in J} \left|\complement(X \setminus D_i)\right|$, whence $U \subseteq \bigvee_{i\in J} \left|\complement(X \setminus D_i)\right|$.

Claim B. *If $\mathbb{N}^{\mathbb{N}} \models \varphi(U, X)$ for nonempty U, then there are $i \in J$ and a nonempty open $V \subseteq U$ with $V \Vdash \complement(X \setminus D_i)$.*

Proof. First suppose there is a finite $K \subseteq J$ such that $U \cap X \setminus D_K$ is not dense in U. Let K be a $\subseteq$-minimal such set. Since J is nonempty, we are done if K is empty. Otherwise fix $k \in K$. Since $U \cap X \setminus D_K$ is not dense in U, there is a nonempty open V with $V \cap X \subseteq D_K$. By minimality of K, $E_0 = V \cap X \setminus D_{K\setminus\{k\}}$ is dense in V. If $E_1 = V \cap X \setminus D_k$ is not dense in V, then we are done, so assume E_1 is dense. Since $V \cap X \subseteq D_K$, we have $E_0 \cap E_1 = \emptyset$, hence $(E_0, E_1) \in \mathcal{E}_2$. Thus $\Pr^J(W_J, E_0, E_1)$ fails, contradicting $\mathbb{N}^{\mathbb{N}} \models \varphi(U, X)$.

Now suppose all $U \cap X \setminus D_K$ for $K \subseteq J$ finite are dense in U. Let $(V_j)_{j\in\mathbb{N}}$ be a base of the topology of U. Recursively choose $x_{j0}, x_{j1} \in V_j \cap X \cap D_J$ such that $x_{j\ell} \in D_i$ implies $x_{k,1-\ell} \notin D_i$ for all $i \in J$ and all $k \geq j$. Then $E_0 = \{x_{j0} \mid j\}$ and $E_1 = \{x_{j1} \mid j\}$ satisfy $(E_0, E_1) \in \mathcal{E}_2$. Again $\Pr^J(W_J, E_0, E_1)$ fails, contradicting $\mathbb{N}^{\mathbb{N}} \models \varphi(U, X)$. □

For (b), let $\mathsf{Storey}(U, X, W_J)$ express that $\mathsf{Substorey}(U, X, W_J)$ holds and whenever $\mathsf{Substorey}(U, X \cup Y, W_J)$ holds, then $U \Vdash \complement(Y \setminus X)$.

We now consider (c). First suppose $n = 2$ and R is symmetric with $R(i, i) = \bot$ for all i. Let W_R separate $\mathcal{E}_1$ from $\mathcal{E}_2$ of Lemma 1.3.6. Again it suffices to construct a formula ψ such that the union of all U with $\mathbb{N}^\mathbb{N} \models \psi(U, X, Y)$ is a dense subset of $\bigvee_{i,j} \left|\complement((X \triangle D_i) \cup (Y \triangle D_j))\right| \wedge R(i, j)$. Let $\psi(U, X, Y)$ state that $\mathsf{Storey}(U, X, W_{\mathbb{N}^\mathbb{N}})$ and $\mathsf{Storey}(U, Y, W_{\mathbb{N}^\mathbb{N}})$ hold and whenever $V \subseteq U$ is nonempty open, then there is $W \subseteq V$ nonempty open such that $\Pr^{\mathbb{N}^\mathbb{N}}(W_R, W \cap X, W \cap Y)$ holds. Again the following lemmas correspond to either inclusion.

Claim C. *If $U \Vdash R(i, j) \setminus ((X \triangle D_i) \cup (Y \triangle D_j))$ for some $i \in \mathbb{N}^\mathbb{N}$, then $\mathbb{N}^\mathbb{N} \models \psi(U, X, Y)$.*

Proof. Since $U \Vdash \complement((X \triangle D_i) \cup (Y \triangle D_j))$, both $\mathsf{Storey}(U, X, W_{\mathbb{N}^\mathbb{N}})$ and $\mathsf{Storey}(U, Y, W_{\mathbb{N}^\mathbb{N}})$ hold. If $V \subseteq U$ is nonempty open, then since $V \Vdash \complement((X \triangle D_i) \cup (Y \triangle D_j))$, there is $W \subseteq V$ nonempty open with $W \cap X = W \cap D_i$ and $W \cap Y = W \cap D_j$. Since $W \subseteq U \subseteq R(i, j)$, we have $(W \cap X, W \cap Y) \in \mathcal{E}_1$, whence $\Pr^{\mathbb{N}^\mathbb{N}}(W_R, W \cap X, W \cap Y)$ follows. □

Claim D. *If $\mathbb{N}^\mathbb{N} \models \psi(U, X, Y)$, then there are $i, j \in \mathbb{N}^\mathbb{N}$ and a nonempty open $V \subseteq U$ with $V \Vdash R(i, j) \setminus ((X \triangle D_i) \cup (Y \triangle D_j))$.*

Proof. Since $\mathbb{N}^\mathbb{N} \models \mathsf{Storey}(U, X, W_{\mathbb{N}^\mathbb{N}}) \wedge \mathsf{Storey}(U, Y, W_{\mathbb{N}^\mathbb{N}})$, there are i, j and a nonempty open $U' \subseteq U$ with $U' \Vdash \complement((X \triangle D_i) \cup (Y \triangle D_j))$. If U' meets $R(i, j)$, then $V = U' \cap R(i, j)$ does the job. Else $U' \subseteq \neg R(i, j)$. Since $U' \Vdash \complement((X \triangle D_i) \cup (Y \triangle D_j))$, there is $V \subseteq U'$ nonempty open with $V \cap X = V \cap D_i$ and $V \cap Y = V \cap D_j$. Since $V \subseteq U' \subseteq \neg R(i, j)$, we have $(W \cap X, W \cap X) \in \mathcal{E}_2$ for $W \subseteq V$ nonempty open, whence $\Pr^{\mathbb{N}^\mathbb{N}}(W_R, W \cap X, W \cap Y)$ fails, contradicting $\mathbb{N}^\mathbb{N} \models \psi(U, X, Y)$. □

Finally, we reduce the general case to the previous ones. Thus, we have to find pairwise disjoint countable dense E_i satisfying (a) to (c). Let σ be a bijection $\mathbb{N}^{\leq\omega} \to \mathbb{N}^{=\omega}$. Set $E_i = D_{\sigma i}$ for $i \in \mathbb{N}^\mathbb{N}$. Then (a) and (b) for the E_i follow from (a) and (b) for the D_i. Given an n-ary map R on $\mathbb{N}^\mathbb{N}$ and $1 \leq m \leq n$, let S_m be the binary map on $\operatorname{dom} \sigma$ defined by:

$$\begin{aligned}
S_m(i, j) &= \bot, && i, j \in \mathbb{N}^\mathbb{N},\\
S_m(f, g) &= \bot, && f, g \in (\mathbb{N}^\mathbb{N})^n,\\
S_m(i, f) &= \bot, && i \in \mathbb{N}^\mathbb{N}, f \in (\mathbb{N}^\mathbb{N})^n, f(m) \neq i,\\
S_m(f, i) &= \bot, && i \in \mathbb{N}^\mathbb{N}, f \in (\mathbb{N}^\mathbb{N})^n, f(m) \neq i,\\
S_m(i, f) &= R(f(1), \ldots, f(n)), && i \in \mathbb{N}^\mathbb{N}, f \in (\mathbb{N}^\mathbb{N})^n, f(m) = i,\\
S_m(f, i) &= R(f(1), \ldots, f(n)), && i \in \mathbb{N}^\mathbb{N}, f \in (\mathbb{N}^\mathbb{N})^n, f(m) = i.
\end{aligned}$$

To reduce (c) for R and the E_i to (c) for the S_m and the D_i, use $R(i_1, \ldots, i_n) = \bigvee_{f \in \operatorname{dom} \sigma} \bigwedge_{m=1}^n S_m(i_m, f)$. □

Remark 1.3.8. The formulas of Theorem 1.3.7 are invariant under taking subsets and unions in U and adding nowhere dense sets. We state the precise version only for Substorey, but analogues hold for Storey and Rel.

(a) $\mathbb{N}^\mathbb{N} \models U \subseteq V \wedge \mathsf{Substorey}(V, X, W) \Rightarrow \mathsf{Substorey}(U, X, W)$.

(b) $\mathbb{N}^\mathbb{N} \models \mathsf{Substorey}(\bigcup_i U_i, X, W)$ if and only if $\mathbb{N}^\mathbb{N} \models \mathsf{Substorey}(U_i, X, W)$ for all i.

(c) If U is open, then $\mathbb{N}^\mathbb{N} \models \mathsf{Substorey}(U, X, W) \Leftrightarrow \mathsf{Substorey}(|U|, X, W)$.

(d) If $X \triangle Y$ is nowhere dense, then $\mathbb{N}^\mathbb{N} \models \mathsf{Substorey}(U, X, W) \Leftrightarrow \mathsf{Substorey}(U, Y, W)$.

The statement of Theorem 1.3.7 ensures these conditions only for W_J, not for all W. However, the case of arbitrary W can be regarded either as a consequence of the proof or achieved afterwards by modifying Substorey without affecting its defining property.

1.3.3 Boolean-valued structures

Generally, Theorem 1.3.7 allows us to express arbitrary structures using parameters W. The remaining difficulty stems from eliminating these parameters. For this, we build formulas expressing that a parameter W satisfies the conclusions of Theorem 1.3.7 for sufficiently complicated structures.

Since it is impossible to interpret complicated structures into the monadic theory of $(\mathbb{R}, <)$ [GS89], we will use Boolean-valued structures instead. We specialize section 1.1.3 to the case where B is the completion of the (up to isomorphism unique) countable atomless Boolean algebra. Thus, B is the set of regular open subsets of $\mathbb{N}^{\mathbb{N}}$. This Boolean algebra is used in Cohen forcing, i.e., adjoining a single Cohen real.

Namely, we will show that any set W yields a B-valued structure. Using a finitely axiomatized undecidable theory, this already yields undecidability of the monadic theory, Corollary 1.3.10. However, we also can express some higher-order properties. A definition of well-foundedness recovers true arithmetic, Proposition 1.3.21. Finally, we reduce third-order arithmetic in the forcing extension given by adding a Cohen real to the monadic theory of topology of $\mathbb{N}^{\mathbb{N}}$.

Proposition 1.3.9. *Let $\mathscr{L} = \{R_1, \dots, R_n\}$ be a finite relational language without equality. Let $W = (W_\in, W_{R_1}, \dots, W_{R_n})$ be a tuple of subsets of $\mathbb{N}^{\mathbb{N}}$.*

(a) We can regard $\mathcal{X}(W) = \left\{X \subseteq \mathbb{N}^{\mathbb{N}} \,\middle|\, \mathbb{N}^{\mathbb{N}} \models \mathsf{Storey}(\mathbb{N}^{\mathbb{N}}, X, W_\in)\right\}$ as a full Boolean-valued structure by $U \leq \|R_n(X)\|$ if and only if $\mathbb{N}^{\mathbb{N}} \models \mathsf{Rel}(U, X, W_{R_n})$.

(b) There is a computable map $\varphi \mapsto \mathsf{Form}_\varphi$ from $\mathscr{L}$-formulas to monadic formulas such that $U \leq \|\varphi(X)\|$ if and only if $\mathbb{N}^{\mathbb{N}} \models \mathsf{Form}_\varphi(U, X, W)$.

(c) Given $x \in \mathbb{N}^{\mathbb{N}}$, let $\mathcal{U}_x$ be the ultrafilter $\{U \mid U \ni x\}$ on the regular open sets. Then $\mathcal{X}(W)/\mathcal{U}_x \models \varphi$ if and only if $\mathbb{N}^{\mathbb{N}} \models \exists U \ni x \colon \mathsf{Form}_\varphi(U, X, W)$.

Proof. We first show (a). It is a Boolean-valued structure by Remark 1.3.8. For fullness let U_i be disjoint regular open such that $\|\exists X \varphi(X)\| = \bigvee_i U_i$ and there are X_i with $U_i \leq \|\varphi(X_i)\|$. Then $X = \bigcup_i (U_i \cap X_i)$ satisfies $\|\varphi(X)\| = \|\exists X \varphi(X)\|$.

The map of (b) is constructed by recursion. Define $\mathsf{Form}_{R_n(X)}(U, X, W)$ as $\mathsf{Rel}(U, X, W_{R_n})$, $\mathsf{Form}_{\neg\varphi}(U, X, W)$ as "$\mathsf{Form}_\varphi(V, X, W)$ holds for no nonempty open $V \subseteq U$", $\mathsf{Form}_{\varphi\wedge\psi}(U, X, W)$ as $\mathsf{Form}_\varphi(U, X, W) \wedge \mathsf{Form}_\psi(U, X, W)$ and $\mathsf{Form}_{\exists X \varphi(X,Y)}(U, Y, W)$ as $\exists X \colon \mathsf{Storey}(\mathbb{N}^{\mathbb{N}}, X, W_\in) \wedge \mathsf{Form}_\varphi(U, X, Y, W)$. The proof of correctness is a straightforward induction, using fullness for the quantifier case.

(c) follows from (b). □

Corollary 1.3.10. *The monadic theory of topology of $\mathbb{N}^{\mathbb{N}}$ is undecidable.*

Proof. We reduce Robinson arithmetic Q (formulated in a relational language without equality) to the monadic theory. Let χ be a single sentence axiomatizing Q. By Proposition 1.3.9 it suffices to show that $\mathsf{Q} \models \varphi$ if and only if for all W, x we have $\mathcal{X}(W)/\mathcal{U}_x \models \chi \Rightarrow \varphi$. Necessity is clear.

Suppose that $\mathsf{Q} \not\models \varphi$. Löwenheim-Skolem yields a countable M with $(M, +, \cdot) \models \mathsf{Q} \cup \{\neg\varphi\}$. We may assume $M \subseteq \mathbb{N}^{\mathbb{N}}$. Let $W_\in$ be W_M of Theorem 1.3.7 and analogously define $W_+, W_\cdot$ corresponding to the maps $R_+, R_\cdot$ from $\mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ to the regular open sets with $R_\star(X, Y, Z) = \top$ if $X, Y, Z \in M$ with $X \star Y = Z$ and $R_\star(X, Y, Z) = \bot$ else. Then by induction we have $\mathcal{X}(W) \models \varphi(D_{m_1}, \dots, D_{m_n})$ if and only if $M \models \varphi(m_1, \dots, m_n)$. In particular, $\mathcal{X}(W)/\mathcal{U}_x$ is elementarily equivalent to M, hence models $\chi \not\Rightarrow \varphi$. □

Remark 1.3.11. For this remark, the language of topology is defined to contain a function symbol for the closure.

Already the fragment of the monadic theory given by formulas of the form $\exists W_\in \exists W_+ \exists W_\cdot \varphi$ such that φ quantifies over subsets of a fixed countable dense subset - $\bigcup_{n\in\mathbb{N}} D_{m_n}$ in the above argument - instead of $\mathbb{N}^{\mathbb{N}}$ is undecidable. Indeed, quantification over closed subsets can be reduced to quantification over countable sets since every closed set is the closure of a countable set, and for regular closed sets we may restrict to subsets of a fixed countable dense set.

With slightly more care, the initial quantifier block can be reduced to a single quantifier $\exists W \varphi$ [GS82, Corollary 4.9]. Second-order arithmetic in the forcing extension can also be reduced to formulas of bounded quantifier rank, but third-order arithmetic requires arbitrary quantifier ranks (as otherwise $\Sigma^3_n = \Sigma^3_{n+1}$ for some n in the forcing extension by a Cohen real).

1.3.4 Relating structures

Definition 1.3.12. W **admits equality** if for all $X, Y \in \mathcal{X}(W)$, we have $|X \triangle Y| \vee \left|\complement(X \triangle Y)\right| = \top$.

Since $|A|$ and $\left|\complement A\right|$ are always disjoint, this is equivalent to $\neg\, |X \triangle Y| = \left|\complement(X \triangle Y)\right|$. It is definable from W whether $\mathcal{X}(W)$ admits equality.

Proposition 1.3.13. *If $W_\in$ is given by Theorem 1.3.7, then $\mathcal{X}(W)$ admits equality.*

Proof. Since $X, Y \in \mathcal{X}(W)$, there are partitions of unity, indexed by $i, j \in \mathbb{N}^\mathbb{N}$, by regular open U with $U \Vdash \complement(X \triangle D_i)$ respectively $U \Vdash \complement(Y \triangle D_j)$. Passing to a common refinement, we arrange $U \Vdash \complement((X \triangle D_i) \cup (Y \triangle D_j))$. If $i = j$, this yields $U \leq \left|\complement(X \triangle Y)\right|$, and else, as D_i and D_j are disjoint, $U \leq |X \triangle Y|$. □

Proposition 1.3.14. *Setting $\|X = Y\| = \left|\complement(X \triangle Y)\right|$ makes $\mathcal{X}(W)$ a Boolean-valued structure with equality. If W admits equality, then $\|X \neq Y\| = |X \triangle Y|$.*

Proof. The substitution rule for equality follows from Remark 1.3.8 (d). If W admits equality, then

$$|X \triangle Y| = \neg \left|\complement(X \triangle Y)\right| = \neg \|X = Y\| = \|X \neq Y\|\,.$$ □

Definition 1.3.15. Let W', W admit equality. We say that W' is **subordinate** to W if for all $X' \in \mathcal{X}(W')$ there is $X \in \mathcal{X}(W)$ with $X \supseteq X'$.

Being subordinate is definable in monadic logic. We show that X' in Definition 1.3.15 is essentially unique.

Lemma 1.3.16. *Let $W', W \subseteq \mathbb{N}^\mathbb{N}$ such that W admits equality. For any dense $X' \in \mathcal{X}(W')$ there is at most one $X \supseteq X'$ with $X \in \mathcal{X}(W)$ up to nowhere dense sets.*

Proof. Let $X_0, X_1 \supseteq X'$ lie in $\mathcal{X}(W)$. We show $\mathcal{X}(W) \models X_0 = X_1$. Otherwise let $U \leq \|X_0 \neq X_1\| = |X_0 \triangle X_1|$ be nonempty regular open. Thus $U \Vdash X_0 \triangle X_1 \subseteq \complement X$, contradicting that X is dense. □

1.3.5 First-order arithmetic

Definition 1.3.17. [She88, Definition 1.5, Definition 1.7] Let $W = (W_\in, W_+, W_\cdot)$ be a tuple of subsets of $\mathbb{N}^\mathbb{N}$. We say that W **witnesses Robinson arithmetic** if W admits equality and $\mathcal{X}(W)$ models Robinson arithmetic. We say that W **witnesses (true) arithmetic** if additionally $\mathcal{X}(W)$ is well-founded in the sense that there is no infinite sequence of $X_n \in \mathcal{X}(W)$ with $\bigwedge_n \|X_{n+1} < X_n\| \neq \bot$.

It is definable by Proposition 1.3.9 if W witnesses Robinson arithmetic. We next show the analogue for true arithmetic, Proposition 1.3.19.

Proposition 1.3.18. *[She88, Claim 1.6] If $\mathcal{X}(W)$ witnesses Robinson arithmetic, then for $n \in \mathbb{N}$ there is $D_n(W)$ such that $\mathcal{X}(W) \models \varphi_n(D_n(W))$, where $\varphi_n(x)$ is $x = n$. These $D_n(W)$ are unique up to nowhere dense sets. They may be chosen pairwise disjoint.*

Proof. Existence follows from $\mathsf{Q} \models \exists x \varphi_n(x)$ and fullness of $\mathcal{X}(W)$. Uniqueness follows as $\mathsf{Q} \models \varphi_n(x) \wedge \varphi_n(y) \Rightarrow x = y$. Since $\mathsf{Q} \models \varphi_m(x) \wedge \varphi_n(y) \Rightarrow x \neq y$ for $m \neq n$, the $D_m(W) \cap D_n(W)$ are nowhere dense. Thus we may replace $D_n(W)$ by $D_n(W) \setminus \bigcup_{i=0}^{n-1} D_i(W)$, which are disjoint. □

Proposition 1.3.19. *[She88, Claim 1.8] It is definable in the monadic theory whether W witnesses arithmetic.*

Proof. It remains to express well-foundedness. We show that $\mathcal{X}(W)$ is ill-founded if and only if there is a $W' = (W'_\in, W'_{x \mapsto x+1})$ such that

- W' admits equality and is subordinate to W,
- $\mathcal{X}(W')$ satisfies the theory of $(\mathbb{N}, x \mapsto x + 1)$ on some nonempty regular open U,
- for $X, Y \in \mathcal{X}(W'), X', Y' \in \mathcal{X}(W)$ with $U \cap X \subseteq X', U \cap Y \subseteq Y'$, we have $U \cap \|X + 1 = Y\| \leq \|Y' < X'\|$.

This is sufficient by finite axiomatizability of the theory of the successor function.

Let W' satisfy these properties. As in Proposition 1.3.18, we obtain disjoint $D_n(W')$. For each such $D_n(W')$, there is $X_n \supseteq D_n(W')$ with $X_n \in \mathcal{X}(W)$. Then $\|X_{n+1} < X_n\| \geq U \wedge \|D_n(W') + 1 = D_{n+1}(W')\| = U$.

Suppose that $\mathcal{X}(W)$ is ill-founded, say $\bigwedge_n \|X_{n+1} < X_n\| \geq U > \bot$. Thus the $U \cap X_m \cap X_n$ are nowhere dense, whence we may assume the X_n pairwise disjoint. Let $X'_n \subseteq U \cap X_n$ be countable dense subsets. By Lemma 1.1.5 there is an autohomeomorphism σ with $\sigma(X'_n) = D_{i_n}$ for some D_{i_n} of Theorem 1.3.7. Define a map from $\mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ to the regular open sets by $R(i_n, i_{n+1}) = U$ and $R(i,j) = \bot$ else. Let W_R be as in Theorem 1.3.7 and set $W'_{x \mapsto x+1} = \sigma^{-1}(W_R)$. This admits equality by Proposition 1.3.13 and satisfies the theory of $\mathbb{N}$ by construction. Since each element locally equals some $\sigma^{-1}(D_{i_n}) = X'_n \subseteq X_n$, it is subordinate to W. Finally, this X_n is unique on U by Lemma 1.3.16, whence $U \wedge \|X + 1 = Y\| \leq \|Y' < X'\|$. □

Proposition 1.3.20. *There is a set W that witnesses arithmetic.*

Proof. Use Theorem 1.3.7 as in Corollary 1.3.10. □

Fix some W that witnesses arithmetic. Let $V^{(B)}$ be the Boolean-valued model of set theory given by the complete Boolean algebra B of regular open subsets of $\mathbb{N}^{\mathbb{N}}$. As usual, we use $\dot{x}$ to denote forcing names and $\check{x}$ to denote the canonical forcing names for sets $x \in V$.

Proposition 1.3.21. *[She88, Claim 2.3, Claim 2.4, Claim 2.5] $\mathcal{X}(W)$ is isomorphic to the Boolean-valued substructure of $V^{(B)}$ given by $\dot{n}$ with $V^{(B)} \models \dot{n} \in \mathbb{N}$ and usual arithmetic.*

Proof. Set $f(\dot{n}) = \bigcup_{m \in \mathbb{N}} \|\check{m} = \dot{n}\| \cap D_m(W)$. Clearly $f(\dot{n}) \in \mathcal{X}(W)$. Injectivity follows from

$$\|f(\dot{p}) = f(\dot{q})\| = \bigcup_m \bigcup_n \|\check{m} = \dot{p}\| \cap \|\check{n} = \dot{q}\| \cap \|D_m(W) = D_n(W)\| = \bigvee_m \|\check{m} = \dot{p}\| \cap \|\check{m} = \dot{q}\| = \|\dot{p} = \dot{q}\|.$$

Similarly, f is a semiring homomorphism. It remains to show surjectivity. If X_0 does not lie in the image, then there is a nonempty regular open U with $U \leq \|X_0 > D_m(W)\|$ for all m. Then we can recursively construct X_n with $U \leq \|X_{n+1} + D_1(W) = X_n\|$, contradicting well-foundedness. □

Corollary 1.3.22. *True first-order arithmetic reduces to the monadic theory of topology of $\mathbb{N}^{\mathbb{N}}$.*

1.3.6 Second-order arithmetic

Using the countably many slices for arithmetic, we can also encode reals, regarded as sets of naturals.

Definition 1.3.23. [She88, Definition 2.6, Definition 2.7] Let $X \subseteq \mathbb{N}^{\mathbb{N}}$.

(a) X **W-represents a real** if for all $Y \in \mathcal{X}(W)$ we have $\left|\complement(Y \setminus X)\right| \vee \left|\complement(Y \cap X)\right| = \top$.

(b) X **W-represents** $\dot{x}$ with $V^{(B)} \models \dot{x} \subseteq \mathbb{N}$ if for all $\dot{n}$ represented by $Y \in \mathcal{X}(W)$ we have $\|\dot{n} \in \dot{x}\| = \left|\complement(Y \setminus X)\right|$ and $\|\dot{n} \notin \dot{x}\| = \left|\complement(Y \cap X)\right|$.

Lemma 1.3.24. *[She88, Claim 2.9] Suppose that X satisfies $\|n \in \dot{x}\| = \left|\complement(D_n(W) \setminus X)\right|$ and $\|n \notin \dot{x}\| = \left|\complement(D_n(W) \cap X)\right|$ for $n \in \mathbb{N}$. Then X W-represents $\dot{x}$.*

Proof. We only show $\|\dot{n} \in \dot{x}\| = \left|\complement(Y \setminus X)\right|$ since $\|\dot{n} \notin \dot{x}\| = \left|\complement(Y \cap X)\right|$ is analogous. Since $\bigvee_{m \in \mathbb{N}} \|\dot{n} = m\| = \top$, we may work locally on $\|\dot{n} = m\|$. Then, for $Y = \bigcup_m \|\dot{n} = m\| \cap D_m(W)$ representing $\dot{n}$,

$$\|\dot{n} \in \dot{x}\| \wedge \|\dot{n} = m\| = \|m \in \dot{x}\| \wedge \|\dot{n} = m\| = \left|\complement(D_m(W) \setminus X)\right| \wedge \|\dot{n} = m\| = \left|\complement(Y \setminus X)\right| \wedge \|\dot{n} = m\|. \quad \square$$

Proposition 1.3.25. *[She88, Claim 2.8] X W-represents a real if and only if there is some name $\dot{x}$ of a real such that X W-represents $\dot{x}$. This $\dot{x}$ is unique up to equality in $V^{(B)}$.*

Proof. If X represents $\dot{x}$, then

$$\left|\complement(Y \setminus X)\right| \vee \left|\complement(Y \cap X)\right| = \|\dot{n} \in \dot{x}\| \vee \|\dot{n} \notin \dot{x}\| = \top.$$

Conversely, let X represent a real. Define $\dot{x}$ by $\|n \in \dot{x}\| = \left|\complement(X \setminus D_n(W))\right|$ for all n. Since $\left|\complement(Y \setminus X)\right| \wedge \left|\complement(Y \cap X)\right| = \left|\complement Y\right| = \bot$ for $Y \in \mathcal{X}(W)$ and X represents a real, $\neg \left|\complement(D_n(W) \setminus X)\right| = \left|\complement(D_n(W) \cap X)\right|$. Thus $\|n \notin \dot{x}\| = \left|\complement(D_n(W) \cap X)\right|$. By Lemma 1.3.24 X represents $\dot{x}$.

Finally, if X represents both $\dot{x}, \dot{y}$, then $\|\dot{n} \in \dot{x}\| = \|\dot{n} \in \dot{y}\|$ for all $\dot{n}$, whence $V^{(B)} \models \dot{x} = \dot{y}$. □

Proposition 1.3.26. *For each $\dot{x}$ with $V^{(B)} \models \dot{x} \subseteq \mathbb{N}$ there is an X that W-represents $\dot{x}$.*

Proof. Using Lemma 1.3.24, $\dot{x}$ is represented by $\bigcup_{n\in\mathbb{N}} \|n \in \dot{x}\| \cap D_n(W)$. □

1.3.7 Third-order arithmetic

We now turn to representing sets of reals.

Proposition 1.3.27. *If W, W' witness arithmetic, then the following are equivalent:*

(i) W' is subordinate to W and mapping $X' \in \mathcal{X}(W')$ to $X \in \mathcal{X}(W)$ with $X \supseteq X'$ defines an isomorphism of Boolean-valued semirings.

(ii) All $D_n(W') \setminus D_n(W)$ are nowhere dense.

Proof. Assume (i). By induction on n, the $X \in \mathcal{X}(W)$ with $X \supseteq D_n(W')$ equals $D_n(W)$ up to a nowhere dense set. Thus $D_n(W') \setminus D_n(W)$ is nowhere dense.

To deduce (i) from (ii), working locally we may assume $X' = D_n(W')$. Set $X = D_n(W)$. This is unique up to nowhere dense sets by Lemma 1.3.16, whence we obtain a well-defined isomorphism. □

In this case, we call W' **arithmetically subordinate** to W. Being arithmetically subordinate is definable in monadic logic.

Proposition 1.3.28. *If W' is arithmetically subordinate to W, then for each $X' \in \mathcal{X}(W')$ there is $X \in \mathcal{X}(W)$ with $X \supseteq X'$, and for each $X \in \mathcal{X}(W)$ there is $X' \in \mathcal{X}(W')$ with $X' \subseteq X$. They are unique up to nowhere dense sets.*

Proof. Locally, X' or X is some D_n, and thus $X' = D_n(W') \subseteq D_n(W) = X$ up to nowhere dense sets. □

Definition 1.3.29. [She88, Definition 2.10, Definition 2.11]

(a) $(W', W_{\in\mathbb{R}}, W_R)$ W-**represents a set of reals** is arithmetically subordinate to W.

(b) $(W', W_{\in\mathbb{R}}, W_R)$ W-**represents** a name $\dot{A}$ of a set of reals if additionally for all names $\dot{x}$ of reals, $\left\|\dot{x} \in \dot{A}\right\|$ is the join $S(\dot{x})$ of all regular open U such that there is an $X \in \mathcal{X}(W_{\in\mathbb{R}}, W_R)$ such that $U \leq \|R(D_0(W'), X)\|$ and all W'-representations Y of a natural $\dot{n}$ satisfy

$$U \wedge \|\dot{n} \in \dot{x}\| = U \wedge \|R(Y+1, X)\|.$$

Since W' is subordinate to W, we can switch between W-representations and W'-representations Y of naturals definably and $S(\dot{x})$ can thus be defined from a W-representation of $\dot{x}$.

Proposition 1.3.30. *[She88, Claim 2.13] W^* W-represents a set of reals if and only if there is some name $\dot{A}$ of a set of reals such that W^* W-represents $\dot{A}$. This $\dot{A}$ is unique up to equality in $V^{(B)}$.*

Proof. Sufficiency and uniqueness are clear. Define $\dot{A}$ by

$$\left\|\dot{y} \in \dot{A}\right\| = \bigvee_{\dot{x}} \|\dot{x} = \dot{y}\| \wedge S(\dot{x}).$$

To show that $\dot{A}$ does the job it suffices to show that $U \leq \|\dot{x} = \dot{y}\|$ implies $U \wedge S(\dot{x}) = U \wedge S(\dot{y})$. Substitute $U \wedge \|\dot{n} \in \dot{x}\| = U \wedge \|\dot{n} \in \dot{y}\|$ in the definition of S. □

Proposition 1.3.31. *[She88, Claim 2.10A] For each name $\dot{A}$ of a set of reals there is a W^* that W-represents $\dot{A}$.*

Proof. Let $X_n \subseteq D_n(W)$ be countable dense subsets. Fix $i_n \in \mathbb{N}^\mathbb{N}$ for $n \in \mathbb{N}$. By Lemma 1.1.5 there is an autohomeomorphism σ with $\sigma(X_n) = D_{i_n}$ for D_i of Theorem 1.3.7.

Since there are only $2^{\aleph_0}$ names of reals (they are determined by the functions $\mathbb{N} \to B, n \mapsto \|n \in \dot{x}\|$), there is a surjection $j \mapsto \dot{x}_j$ from $\mathbb{N}^\mathbb{N}$ onto these names. Define a map from $\mathbb{N}^\mathbb{N} \times \mathbb{N}^\mathbb{N}$ to the regular open sets by $R(i_0, j) = \sigma \left\| \dot{x}_j \in \dot{A} \right\|$, $R(i_{n+1}, j) = \sigma \|n \in \dot{x}_j\|$ and $R(i, j) = \bot$ else. Let $W'_\in, W'_+, W'_\cdot, W_{\in\mathbb{R}}, W_R$ be as in Theorem 1.3.7 such that $\mathcal{X}(W')$ witnesses arithmetic with $D_n(W') = D_{i_n}$ and $\mathcal{X}(W_{\in\mathbb{R}}, W_R)$ is another structure on the same D_i.

We claim that $W^* = (\sigma^{-1}W'_\in, \sigma^{-1}W'_+, \sigma^{-1}W'_\cdot, \sigma^{-1}W_{\in\mathbb{R}}, \sigma^{-1}W_R)$ does the job. It clearly witnesses arithmetic. It is arithmetically subordinate to W as

$$D_n(\sigma^{-1}W') = \sigma^{-1}D_n(W') = \sigma^{-1}D_{i_n} = X_n \subseteq D_n(W).$$

We first show that $\left\| \dot{x}_j \in \dot{A} \right\| \leq S(\dot{x}_j)$. Let $U = \left\| \dot{x}_j \in \dot{A} \right\|$ and $X = \sigma^{-1}D_j$. Then $\left\| R(\sigma^{-1}D_0(W'), X) \right\| = U$. Let Y be a $\sigma^{-1}W'$-representation of a natural, by localizing say $Y = \sigma^{-1}D_{i_n}$. Then $\|\dot{n} \in \dot{x}_j\| = \|R(Y + 1, X)\|$.

We finally show that $S(\dot{x}_j) \leq \left\| \dot{x}_j \in \dot{A} \right\|$. Thus let U be regular open and $X \in \mathcal{X}(W_{\in\mathbb{R}}, W_R)$ be such that $U \leq \left\| R(D_0(\sigma^{-1}W'), X) \right\|$ and for all $\sigma^{-1}W'$-representations Y of a natural $\dot{n}$ we have $U \wedge \|\dot{n} \in \dot{x}_j\| = U \wedge \|R(Y + 1, X)\|$. Localizing, assume $X = \sigma^{-1}D_{j'}$. Then

$$U \leq \left\| R(D_0(\sigma^{-1}W'), \sigma^{-1}D_{j'}) \right\| = \sigma^{-1} \|R(D_{i_0}, D_{j'})\| = \left\| \dot{x}_{j'} \in \dot{A} \right\|.$$

For all n, setting $Y = \sigma^{-1}D_{i_{n+1}}$, we have

$$U \wedge \|n \in \dot{x}_j\| = U \wedge \sigma^{-1} \left\| R(D_{i_{n+1}}, D_{j'}) \right\| = U \wedge \|n \in \dot{x}_{j'}\|.$$

Thus, $U \leq \|\dot{x}_j = \dot{x}_{j'}\|$. We conclude that $U \leq \left\| \dot{x}_j \in \dot{A} \right\|$. □

1.3.8 Conclusion

Theorem 1.3.32. *[She88, Main Lemma 2.17] Let W witness arithmetic. We can compute a map $\varphi \mapsto \psi$ from formulas of third-order arithmetic to monadic formulas such that for all tuples x of W-representations of naturals, reals and sets of reals, we have that*

$$\|\varphi(x)\| = U \text{ if and only if } \mathbb{N}^\mathbb{N} \models \psi(U, x, W).$$

Proof. We use recursion on φ. The Boolean cases are straightforward. We have to show that the atomic formulas, which can be taken to only contain $+, \cdot, \in$, for W-representations can be encoded, and, to establish the induction step for quantifiers, that being a W-representation can be encoded. This follows from the definitions and Proposition 1.3.19, as well as Proposition 1.3.9 for the first-order case. □

Corollary 1.3.33. *[She88, Conclusion 2.18] We can compute a map $\varphi \mapsto \psi$ from sentences of third-order arithmetic to monadic sentences such that for all x that are Cohen reals over V we have $\mathbb{N}^\mathbb{N} \models \psi$ if and only if $V[x] \models \varphi$.*

Proof. Since the Boolean algebra of regular open sets is homogeneous, we have $\|\varphi\| = \bot$ or $\|\varphi\| = \top$. In particular, whether $V[x] \models \varphi$ is independent of x. Let φ state that there is some W that witnesses arithmetic such that $\mathbb{N}^\mathbb{N} \models \psi(\top, W)$. □

Example 1.3.34. The monadic theory of topology of $\mathbb{N}^\mathbb{N}$ can express whether the continuum hypothesis holds in the ambient model of set theory. Indeed, since the continuum hypothesis is expressible in third-order arithmetic, the monadic theory expresses the continuum hypothesis in the generic extension by a Cohen real, but the continuum hypothesis is absolute for this extension.

Remark 1.3.35. [She88, Concluding remarks] It is possible to extend undecidability to total orders or topological spaces without a combinatorial property called p-**modesty** for $p \in \mathbb{N}$. In contrast, modest orders behave like countable orders [GS79].

Remark 1.3.36. [Boj+14, Theorem 3.1] Shelah's undecidability proof presented here also applies to some restricted monadic theories. For example, it applies to the class of projective sets (or levels of the projective hierarchy) if they contain a well-order of the reals. In this case, the set W of Lemma 1.3.4 can be chosen to be projective.

While the construction for third-order arithmetic fails for projective sets, the construction for second-order arithmetic in the forcing extension still works. Conversely, second-order arithmetic trivially interprets the projective monadic theory of order.

Chapter 2

The Borel monadic theory of order

Our aim in this chapter is to prove Theorem 0.1.22. We begin by outlining the proof strategy.

We describe the structure of the chapter. Section 2.1 provides prerequisites in topology, order theory and Ramsey theory. Section 2.2 presents point-set topological lemmas used in section 2.4 and uses them to show the definability of meagerness. In section 2.3, we present an adjusted version of results of [She75]. We restrict quantification and simplify by restricting to complete total orders, for example, using Cantor sets instead of convex equivalence relations. Finally, sections 2.4 and 2.5 combine the previous parts to prove the main result, explicitly describing a small amount of data sufficient to determine uniform types.

2.1 Introduction

2.1.1 Proof sketch

Before establishing undecidability of the unrestricted monadic theory, [She75] also introduced a novel technique for showing decidability of monadic theories of order. Here, we extend Shelah's methods using the Baire property in the Borel setting.

In the monadic language of order, for fixed $m, n \in \mathbb{N}$, there are up to equivalence only finitely many formulas in m free variables with n quantifiers. Thus, decidability amounts to computability of the subset of this finite set given by true formulas, uniformly in m, n. Using lexicographic sums and Ramsey theoretic methods, Shelah reduced this problem to computability for formulas only containing quantifiers over sufficiently uniform sets and quantifiers over Cantor sets. A set is sufficiently uniform if it satisfies the same formulas with n quantifiers on every open interval. In the unrestricted monadic theory, these quantifiers still suffice to encode arithmetic. This encoding requires a (uniform) parameter defined by transfinite recursion.

We show that the Borel monadic theory defines the class of meager sets. Using the definability and the Baire property, we describe uniform formulas in variables $X_1, \dots, X_n$ combinatorially. Such formulas merely specify the existence or absence of possibly nested Cantor sets meeting certain X_i while avoiding others.

In the following, we sketch both the technique of [She75] and its extension to prove Theorem 0.1.22.

For fixed $m \in \mathbb{N}$ and $k \in \mathbb{N}^n$ define a set $\mathsf{fTh}^-_k(m)$ of monadic formulas in m free variables by recursion on n. Let $\mathsf{fTh}^-_{()}(m)$ be the set of atomic formulas. For $n > 0$ let $\mathsf{fTh}^-_k(m)$ be the set of formulas $\exists X_1 \dots \exists X_{k_n} \varphi$ with φ ranging over the Boolean combinations of elements of $\mathsf{fTh}^-_{(k_1,\dots,k_{n-1})}(m + k_n)$. Since the language is relational, each $\mathsf{fTh}^-_k(m)$ is finite. For a tuple $X = (X_1, \dots, X_m)$ of Borel sets, set $\mathsf{Th}_k(\mathbb{R}, X) = \left\{\varphi \in \mathsf{fTh}^-_k(m) \,\middle|\, \mathbb{R} \models \varphi(X)\right\}$. Our aim is to compute $\mathsf{Th}_k(\mathbb{R})$ (setting $m = 0$) as k varies. (Actually, we use a slightly different definition of Th_k, considering first-order quantifiers separately to eliminate more quantifiers.)

Generalizing to arbitrary total orders, consider a lexicographic sum $\sum_{i \in I} X_i$ of total orders. For finite index sets I, $\mathsf{Th}_k(\sum_i X_i)$ is determined by and computable from the $\mathsf{Th}_k(X_i)$. This generalizes to arbitrary lexicographic sums if the input incorporates the monadic theory of I in an appropriate sense (Theorem 2.3.6). For example, if $\mathbb{R}$ is the disjoint union of half-open intervals $(a_i, a_{i+1}]$ with a common type $\mathsf{Th}_k((a_i, a_{i+1}], X) = t$ for all i, then $\mathsf{Th}_k(\mathbb{R}, X)$ can be computed from t, and if $C \subseteq \mathbb{R}$ is a Cantor set whose complementary

intervals share a common type t, then $\mathsf{Th}_k(\mathbb{R}, X)$ can be computed from $\mathsf{Th}_k(C, X)$ and t.

Call a tuple X of Borel sets k-uniform if any pair of open intervals $(a, b), (c, d)$ satisfies $\mathsf{Th}_k((a, b), X) = \mathsf{Th}_k((c, d), X)$. For example, for k large enough a single countable set is k-uniform if and only if it is either empty or dense. Despite the restrictiveness of this property, by a Ramsey theoretic argument every tuple is k-uniform on some open interval. Thus, the set of points with a k-uniform neighborhood is open dense. In fact, its complement is empty or a Cantor set $C \subseteq \mathbb{R}$. In the latter case, by the aforementioned computation for lexicographic sums, $\mathsf{Th}_k(\mathbb{R}, X)$ is essentially determined by $\mathsf{Th}_k(C, X)$.

Employing this line of reasoning, all monadic sentences can be transformed into sentences containing only the following two kinds of restricted quantifiers (Theorem 2.3.24):

- $\exists Y \varphi(X_1, \ldots, X_m, Y)$ quantifying over all Y such that $(X_1, \ldots, X_m, Y)$ is still uniform,
- $\exists C \varphi(X_1, \ldots, X_m, C)$ quantifying over all Cantor sets C for which $(X_1, \ldots, X_m)$ is also uniform on C.

The first kind of quantifier can often be eliminated or reduced to the second kind. The second kind can be more problematic.

The techniques described so far also apply to the unrestricted monadic theory and originate in [She75]. However, in the unrestricted theory, certain sets encode Boolean-valued first-order structures as discussed in chapter 1. In contrast, in the Borel setting, Baire category arguments control Cantor sets.

A prototypical instance of the remaining question is: Given a k-uniform tuple $(X_1, \ldots, X_m)$, is there a Cantor set C such that $X_1, \ldots, X_\ell$ are dense in C, while $X_{\ell+1}, \ldots, X_m$ are disjoint from C? We do not get full quantifier elimination since the answer depends not only on the monadic quantifier-free type of $X_1, \ldots, X_m$. For example, consider $m = 3, \ell = 2$. If X_1, X_2 are disjoint meager F_σ sets and $X_3 = \mathbb{R} \setminus (X_1 \cup X_2)$, then no such Cantor set exists, but if X_1 is comeager instead, then such a set does exist.

Based on these ideas, we define meagerness in the Borel monadic theory (section 2.2.3). By the Baire property, a set is either meager or comeager on some open interval. Hence, a k-uniform set is either meager or comeager everywhere.

After Shelah's reduction, the decision procedure consists of two parts.

First, we define a coarse version cTh_n of Th_k restricted to quantifiers of the second kind, where n is the number of quantifier alternations. (The actual definition proceeds in an expanded language.) We compute the set of satisfiable coarse types of uniform tuples (Corollary 2.4.31). In other words, we decide sentences with an existential block of quantifiers of the first kind followed by a formula containing only quantifiers of the second kind.

Each meager subset of the reals is included in a countable union of Cantor sets. Consider a type $\mathsf{cTh}_n(\mathbb{R}, X_1, \ldots, X_n)$ and assume without loss of generality that all X_i are meager. By a Baire category argument, $\mathsf{cTh}_n(\mathbb{R}, X_1, \ldots, X_n)$ is determined by the types $\mathsf{cTh}_n(C_m)$ of the countably many Cantor sets C_m with $\bigcup_m C_m \supseteq \bigcup_i X_i$. In other words, these Cantor sets do not interact. We finally show that every coarse type $\mathsf{cTh}_n(\mathbb{R}, P)$ arises from a finite iteration of a construction termed "uniform sum" (Proposition 2.4.30).

Intuitively, the uniform sum of Cantor sets $C_1, \ldots, C_n$ (equipped with tuples of subsets) is the simplest way to construct uniform meager unions of copies of the C_i. For instance, consider $n = 1$: First embed C_1 into $\mathbb{R}$; then embed C_1 into each complementary interval of the previous embedding, obtaining a new Cantor set that is a countable union of copies of C_1; finally iterate embedding C_1 into the complementary intervals thus obtained.

The second part of the decision procedure computes Th_k from cTh_n for appropriate n. If sufficiently complicated Cantor subsets exist, this step is straightforward. Otherwise, the reals can be decomposed into Boolean combinations of few F_σ sets, which can be analyzed independently. This dichotomy follows from the determinacy of a "separation game" (section 2.5.2), a generalized Wadge game.

2.1.2 Notation

Let $\mathbb{R}$ denote the reals with the usual ordering. Let $\overline{\mathbb{C}}$ denote the Cantor set $\prod_{n \in \mathbb{N}} \{0, 1\}$ with the lexicographic ordering and $\mathbb{C} = \overline{\mathbb{C}} \setminus \{0^\infty, 1^\infty\}$ be obtained by removing endpoints. All total orders carry the order topology. Equivalently, $\overline{\mathbb{C}}$ carries the product topology of the discrete topologies and $\mathbb{C}$ carries the subspace topology of $\overline{\mathbb{C}}$.

Fact 2.1.1. If $(Y, \leq)$ is a total order and $X \subseteq Y$ is compact in the subspace topology, then the order topology and the subspace topology on X coincide.

In particular, for subspaces homeomorphic to $\overline{\mathbb{C}}$, and thus also for subspaces homeomorphic to $\mathbb{C}$, the order topology and the subspace topology coincide. A subset of a space is **a Cantor set** if it is homeomorphic to $\mathbb{C}$. Note the deviation from the usual meaning of being homeomorphic to $\overline{\mathbb{C}}$.

Definition 2.1.2. Let $(C, \leq)$ be a total order with $C \cong \mathbb{C}$. Consider the equivalence relation that identifies a, b if $a = b$ or (a, b) is a jump or (b, a) is a jump. Let $\pi_C \colon C \to R$ denote the quotient map for this equivalence relation.

In particular, $R \cong \mathbb{R}$ and π_C is increasing. We sometimes also write $\pi_C \colon C \to \mathbb{R}$ for a composite with such an isomorphism.

2.1.3 Ramsey theory

Definition 2.1.3. Let $(X, \leq)$ be a total order and C be a set.

(a) A **coloring** of X by the set of colors C is a map $f \colon \{(x, y) \in X \times X \mid x < y\} \to C$.

(b) A coloring f is **additive** if $(C, +)$ is a semigroup such that $f(x, y) + f(y, z) = f(x, z)$ for all x, y, z.

(c) A subset $A \subseteq X$ is **homogeneous** if f is constant on $(A \times A) \cap \operatorname{dom} f$.

Let us state a particular case of Ramsey's theorem using this notation.

Fact 2.1.4. Every coloring of an infinite set by a finite set of colors has an infinite homogeneous subset.

Additivity of a coloring may be rephrased as functoriality if the preorder X and the monoid $C \cup \{0\}$ are regarded as categories as usual. The following theorem is proven as [She75, Theorem 1.3]. Shelah's definition of additive coloring is superficially more general, but our version easily implies his version. We present an optimized proof using idempotents in semigroups.

Theorem 2.1.5. *Let f be an additive coloring of an infinite dense total order X by a finite semigroup $(C, +)$. Then there is a homogeneous subset of X that is somewhere dense.*

Proof. Define

$$F \colon X \times 2^X \to 2^C, (a, J) \mapsto \bigcap_{X \ni b > a} \{f(a, x) \mid x \in J, a < x < b\}.$$

Since C is finite, for all $a \in X, J \subseteq X$ there is $b > a$ with $F(a, J) = \{f(a, x) \mid x \in J, a < x < b\}$.

We define a decreasing sequence of somewhere dense $J_i \subseteq X$ for $i \in \mathbb{N}$ by recursion. Set $J_0 = X$. Given $J_i \subseteq X$ somewhere dense, let $J_D = \{a \in J_i \mid F(a, J_i) = D\}$ for $D \subseteq X$. Since $J_i = \bigcup_D J_D$ is somewhere dense, there is $D_i \subseteq C$ such that J_{D_i} is somewhere dense. Set $J_{i+1} = J_{D_i}$.

Choose $D \subseteq C$ such that $\{i \in \mathbb{N} \mid D_i = D\}$ has at least three elements $k_0 < k_1 < k_2$. Set $J^i = J_{k_i+1}$ for $i \in \{0, 1, 2\}$.

We show that D is a subsemigroup. Given $d_1, d_2 \in D$, pick $a \in J^1$. Let $b > a$ be arbitrary. Since $F(a, J_{k_1}) = D \ni d_1$, there is $x \in J_{k_1} \subseteq J^0$ with $a < x < b$ and $f(a, x) = d_1$. Since $F(x, J_{k_0}) = D \ni d_2$, there is $y \in J_{k_0}$ with $x < y < b$ and $f(x, y) = d_2$. Thus

$$d_1 + d_2 = f(a, x) + f(x, y) = f(a, y) \in \{f(a, z) \mid z \in J_{k_0}, a < z < b\}.$$

Since b is arbitrary, $d_1 + d_2 \in F(a, J_{k_0}) = D$.

Since J^2 is somewhere dense, there are $J^2 \ni a < b$ such that J^2 is dense in (a, b). Replacing b if necessary, we may assume that

$$D = F(a, J_{k_0}) = \{f(a, x) \mid x \in J_{k_0}, a < x < b\}.$$

Since finite unions of nowhere dense sets are nowhere dense, there is $d \in D$ such that $\left\{x \in (a, b) \cap J^2 \;\middle|\; f(a, x) = d\right\}$ is somewhere dense, say in $(a', b') \subseteq (a, b)$.

Since D is a finite semigroup, there is $p \in \mathbb{Z}_{>0}$ such that $p \cdot d$ is idempotent. We claim that

$$J = \left\{x \in (a', b') \cap J^1 \;\middle|\; f(a, x) = p \cdot d\right\}$$

is homogeneous of color $p \cdot d$ and dense in (a', b').

For homogeneity consider $x, z \in J$ with $x < z$. Since $F(x, J_{k_1}) = D \ni p \cdot d$, there is $y \in J_{k_1}$ such that $x < y < z$ and $f(x, y) = p \cdot d$. Now

$$f(x, z) = f(x, y) + f(y, z) = p \cdot d + f(y, z).$$

Thus

$$f(x, z) = p \cdot d + f(y, z) = p \cdot d + p \cdot d + f(y, z) = p \cdot d + f(x, z).$$

Since $x, z \in J$, we have that $f(a, x) = f(a, z) = p \cdot d$ and so

$$f(x, z) = p \cdot d + f(x, z) = f(a, x) + f(x, z) = f(a, z) = p \cdot d.$$

For showing density, let $a'', b'' \in X$ be such that $a' < a'' < b'' < b'$. Since $\left\{x \in J^2 \,\middle|\, f(a, x) = d\right\}$ is dense in (a', b'), by increasing a'' we may assume that $a'' \in J^2$ and $f(a, a'') = d$. Since $F(a'', J_{k_2}) = D \ni (2p-1) \cdot d$, there is $x \in J_{k_2} \subseteq J^1$ such that $a'' < x < b''$ and $f(a'', x) = (2p-1) \cdot d$. Hence

$$f(a, x) = f(a, a'') + f(a'', x) = d + (2p-1) \cdot d = p \cdot d$$

and so $x \in (a'', b'') \cap J$. □

In particular, there is a homogeneous subset of order type $\mathbb{Q}$.

Corollary 2.1.6. *Let f be an additive coloring of $\mathbb{C}$. Then $\mathbb{C}$ has a somewhere dense homogeneous subset.*

Proof. Apply Theorem 2.1.5 to the subset of Fact 0.1.10, which is dense. □

2.2 Constructing Cantor sets

Consider a uniform labeling of the reals as defined in section 2.1.1. Uniformly labeled Cantor subsets with the same comeager label as the reals exist in all reasonable circumstances. The purely topological argument is provided here as Lemma 2.2.2. Uniformly labeled Cantor subsets whose comeager label differs from the comeager label of the reals might not exist. Whenever they do not exist, this follows from Baire category arguments in section 2.4.

Lemma 2.2.1. *Let $C \subseteq \mathbb{R}$ be compact, let U be a neighborhood of C and let $D_1, D_2 \subseteq \mathbb{R}$ be dense. Then there are $a_1 < b_1 < a_2 < \cdots < a_n < b_n$ with $a_i \in D_1, b_i \in D_2$ such that $C \subseteq \bigcup_i (a_i, b_i) \subseteq U$.*

Proof. We may assume that U is open. Then U is a union of bounded open intervals. Since C is compact, we may assume that the union is finite and disjoint. Let $a_1 < b_1 \leq a_2 < \cdots \leq a_n < b_n$ be such that $U = \bigcup_i (a_i, b_i)$. As C is closed with $a_i, b_i \notin U \supseteq C$, there are $a_i', b_i' \in \mathbb{R}$ such that $a_i < a_i' < b_i' < b_i$ and $[a_i, a_i'), (b_i', b_i]$ are disjoint from C. Since D_1 and D_2 are dense, there are $a_i'' \in D_1 \cap (a_i, a_i')$ and $b_i'' \in D_2 \cap (b_i', b_i)$. Clearly $C \subseteq \bigcup_i (a_i'', b_i'') \subseteq U$. □

Given a meager F_σ set A and Cantor sets $B_1, B_2, \ldots$, we find a Cantor set C disjoint from A such that C includes many B_n's but $\bigcup_n B_n$ is meager in C. Moreover, we can control the jumps of C.

Lemma 2.2.2. *Let $A \subseteq \mathbb{R}$ be meager F_σ, let $B = \bigcup_{n \in \mathbb{N}} B_n$ be with each $B_n \subseteq \mathbb{R}$ closed nonempty, and let $D_1, \ldots, D_\ell \subseteq \mathbb{R}$ be countable dense, such that $A, B_1, B_2, \ldots, D_1, \ldots, D_\ell$ are pairwise disjoint and each nonempty open subset of $\mathbb{R}$ includes some B_n. Let $E \subseteq \{1, \ldots, \ell\} \times \{1, \ldots, \ell\}$ be nonempty. Then there is a Cantor set $C \subseteq \mathbb{R}$ such that*

(a) C is disjoint from A;

(b) B is meager in C;

(c) each nonempty relatively open subset of C includes some B_n;

(d) for each jump (a, b) in C, there is $(i, j) \in E$ with $a \in D_i$ and $b \in D_j$;

(e) if $(i,j) \in E$, then the set of $a \in C$ such that there is a jump (a,b) in C with $a \in D_i$ and $b \in D_j$ is dense in C;

(f) every element of $C \cap \bigcup_i D_i$ is a component of a jump in C.

Proof. Note that B is meager. Indeed, otherwise there is n_0 such that B_{n_0} has nonempty interior. Then no proper nonempty open subset of B_{n_0} includes a B_m by disjointness of the B_n. This contradicts that every nonempty open subset includes some B_n. Adding a countable dense subset disjoint from the meager $B \cup \bigcup_i D_i$, we may assume that A is dense. Similarly, we may assume $\#B_n \geq 2$ for all n. Namely, decompose $\mathbb{N}$ into pairs $\{i,j\}$ such that each nonempty open subset includes some $B_i \cup B_j$ and replace the B_n by the $B_i \cup B_j$.

Write $A = \bigcup_{n\in\mathbb{N}} A_n$ with A_n compact nowhere dense and $D_i = \{d_{in} \mid n \in \mathbb{N}\}$.

Claim A. *Let $e\colon \mathbb{N} \to E$. There are sequences $(S_n)_{n\in\mathbb{N}}, (T_n)_{n\in\mathbb{N}}$ of subsets of $\mathbb{R}$ such that for all $n \in \mathbb{N}$*

(i) S_n is closed, nowhere dense, nonempty, and disjoint from A;

(ii) $\mathbb{R} \setminus T_n$ is a finite union of open intervals, and T_n has no isolated points;

(iii) for each complementary interval (a,b) of T_n we have that $a \in D_i$ and $b \in D_j$ for $e(k) = (i,j)$ with k minimal such that (a,b) is a complementary interval of T_k;

(iv) $S_n \subseteq T_n$;

and if $n > 0$

(v) $S_{n-1} \subseteq S_n$ and $T_n \subseteq T_{n-1}$;

(vi) each complementary interval of T_{n-1} is a complementary interval of T_n;

(vii) if U is a connected component of T_{n-1}, then $U \cap S_n$ includes some B_i;

(viii) each connected component of T_{n-1} is disconnected in T_n;

(ix) T_n is disjoint from $A_{n-1} \cup (\{d_{i,n-1} \mid i \in \{1,\dots,\ell\}\} \setminus \partial T_{n-1})$.

Proof. We construct S_n, T_n by recursion on n. Start with an arbitrary $S_0 \subseteq \mathbb{R} \setminus A$ with $S_0 \cong \overline{\mathbb{C}}$ and $T_0 = \mathbb{R}$.

Given S_n, T_n satisfying (i)-(ix), let $U_1, \dots, U_m$ be the connected components of T_n. By (ii) each U_k includes an open interval. Hence, we can pick $B_{i_k} \subseteq U_k$. Set $S_{n+1} = S_n \cup \bigcup_{k=1}^m B_{i_k}$. Then (i) and (vii) hold.

Pick $u_j \in \mathring{U}_j \setminus S_{n+1}$ for $j \in \{1,\dots,m\}$. By (iii), $\partial T_n \subseteq \bigcup_k D_k$. Since A_n is disjoint from $\bigcup_k D_k$ and from S_{n+1}, we deduce that

$$(A_n \cap T_n) \cup (\{d_{in} \mid i\} \cap \mathring{T}_n) \cup \{u_j \mid j\} \subseteq \mathring{T}_n \setminus S_{n+1}.$$

Write $(i_n, j_n) = e(n)$. Lemma 2.2.1 yields finitely many $a_\ell \in D_{i_n} \setminus \partial T_n, b_\ell \in D_{j_n} \setminus \partial T_n$ such that $a_1 < b_1 < a_2 < \cdots < b_L$ and

$$(A_n \cap T_n) \cup (\{d_{in} \mid i\} \cap \mathring{T}_n) \cup \{u_j \mid j\} \subseteq \bigcup_\ell (a_\ell, b_\ell) \subseteq T_n \setminus S_{n+1}.$$

Set $T_{n+1} = T_n \setminus \bigcup_\ell (a_\ell, b_\ell)$.

Part (ii) follows as $a_\ell, b_\ell \in \mathring{T}_n$ are pairwise distinct, (iii) and (v) are clear, and (iv) follows from

$$S_{n+1} \subseteq S_n \cup \bigcup_k U_k = T_n \text{ and } S_{n+1} \cap \bigcup_\ell (a_\ell, b_\ell) \subseteq S_{n+1} \cap T_n \setminus S_{n+1} = \emptyset.$$

Part (vi) follows from $(a_\ell, b_\ell) \subseteq T_n$ and (viii) from $\{u_j \mid j\} \subseteq \bigcup_l (a_\ell, b_\ell)$. Since

$$A_n \cap T_{n+1} = A_n \cap T_n \setminus \bigcup_\ell (a_\ell, b_\ell) = \emptyset,$$

part (ix) follows as

$$\text{if } d_{jn} \notin \partial T_n, \text{ then } d_{jn} \notin \overline{T_n} \supseteq T_{n+1} \text{ or } d_{jn} \in \{d_{in} \mid i\} \cap \mathring{T}_n \subseteq \bigcup_\ell (a_\ell, b_\ell) \subseteq \mathbb{R} \setminus T_{n+1}. \qquad \square$$

Let $e\colon \mathbb{N} \to E$ be such that every fiber is infinite. Let S_n, T_n satisfy (i)-(ix). Let C be $\bigcap_n T_n$ with minimum and maximum removed if they exist.

Then (a) follows from (ix). For (c) let $a, b \in \mathbb{R}$ be such that $(a,b) \cap C \neq \emptyset$, say $x \in (a,b) \cap C$. Since A is dense, there are $a', b' \in A$ with $a < a' < x < b' < b$. By (a) we obtain n such that $a', b' \notin T_n$. Since $x \in C \subseteq T_n$, there is a connected component U of T_n with $x \in U \subseteq (a,b)$. By (vii) the set $(a,b) \cap C \supseteq U \cap S_{n+1}$ includes some B_i.

Since $\#B_n \geq 2$, by (c) the space $\overline{C}$ has no isolated points and (b) follows by the same argument as meagerness of B in $\mathbb{R}$ above. Thus, as all T_n are closed, $\overline{C} = \bigcap_n T_n$. Since $S_0 \subseteq \overline{C}$, the set C is nonempty. Since A is dense, $\overline{C}$ has empty interior by (a). Using Fact 0.1.5 in $\mathbb{R} \cup \{-\infty, \infty\}$, we conclude that C is a Cantor set.

By (ii) and (vi) an interval is a complementary interval of C if and only if it is a complementary interval of some T_n. Thus (d) follows from (iii) and (f) from (ix). It remains to show (e). Since C has empty interior, for each nonempty relatively open subset V of C for cofinitely many n there is a connected component U of T_n with $U \cap C \subseteq V$. Pick such an n with $e(n) = (i,j)$ and such an U. By (viii) and (iii), there is a complementary interval $(a,b) \subseteq U$ with $a \in D_i$ and $b \in D_j$. We conclude

$$V \supseteq U \cap C \supseteq [a,b] \cap C = \{a, b\}\,. \qquad \square$$

Corollary 2.2.3. *Let $A \subseteq \mathbb{R}$ be meager F_σ, and let $B = \bigcup_{n\in\mathbb{N}} B_n$ be with each $B_n \subseteq \mathbb{R}$ closed nonempty, such that $A, B_1, B_2, \ldots$ are pairwise disjoint and each nonempty open subset of $\mathbb{R}$ includes some B_n. Then there is a Cantor set $C \subseteq \mathbb{R}$ such that*

(a) C is disjoint from A.

(b) B is meager in C.

(c) Each nonempty relatively open subset of C includes some B_n.

Proof. Let $\ell = 1$ and D_1 be a countable dense subset disjoint from A and B. Apply Lemma 2.2.2. $\square$

2.2.1 Cantor subsets

We present lemmas on the point set topology of the Cantor set.

Lemma 2.2.4. *Let X be a Hausdorff space.*

(a) Any countable family of copies of $\overline{\mathbb{C}}$ in X has a countable disjoint refinement consisting of copies of $\overline{\mathbb{C}}$.

(b) Suppose that X is second countable. Let $(C_{ni})_{n\in\mathbb{N}, i\in I \amalg \{\star\}}$ be a countable family of copies of $\overline{\mathbb{C}}$ in X such that all C_{ni} have empty interior and for all $i \in I$ each nonempty open subset includes some C_{mi}. Then the refinement of (a) can be chosen such that for each nonempty open $U \subseteq X$ there is a $C_{ni} \subseteq U$ that is also contained in the refinement.

Proof. For (a), consider a family $(C_n)_{n\in\mathbb{N}}$. Since X is Hausdorff, each C_n is closed. Thus each $C_n \setminus (C_0 \cup \cdots \cup C_{n-1})$ is open in C_n, hence a countable disjoint union of clopen subsets of C_n. By Fact 0.1.6 the family of all these clopen subsets is the desired refinement.

For (b), since the C_{ni} are nowhere dense, for all $i \in I$ and finite $A \subseteq \{C_{mj} \mid m \in \mathbb{N}, j \in I\}$ and nonempty open U there is n with $C_{ni} \subseteq U \setminus \bigcup_{C\in A} C$. Thus, using second countability, we may rearrange $(C_{ni})_{ni}$ as $(D_k)_{k\in\mathbb{N}}$ such that for all $i \in I$ and nonempty open U there are $k, n \in \mathbb{N}$ with $C_{ni} = D_k \subseteq U \setminus (D_0 \cup \cdots \cup D_{k-1})$. If we carry out the construction of (a) with this enumeration, then the conclusion of (b) holds. $\square$

Lemma 2.2.5. *Let $\pi_{\mathbb{C}}\colon \mathbb{C} \to \mathbb{R}$ be as in Definition 2.1.2. Sending C to its image $\pi_{\mathbb{C}}(C)$ defines a bijection from meager Cantor subsets of $\mathbb{C}$ to Cantor subsets of $\mathbb{R}$. The inverse maps C to the Cantor-Bendixson derivative $\pi_{\mathbb{C}}^{-1}(C)'$ of the preimage.*

Proof. We instead show the equivalent statement that the extension $\pi\colon \overline{\mathbb{C}} \to [0,1]$ of $\pi_{\mathbb{C}}$ yields a bijection between meager copies of $\overline{\mathbb{C}}$ in $\overline{\mathbb{C}}$ and copies of $\overline{\mathbb{C}}$ in $[0,1]$. Note that π is a continuous closed map with finite fibers. Preimages of arbitrary sets with empty interior have empty interior, and images of closed sets with empty interior have empty interior.

We show that the map is well-defined. It suffices to show that $\pi(C) \cong \overline{\mathbb{C}}$ whenever $C \cong \overline{\mathbb{C}}$. By the above $\pi(C)$ is closed and nonempty and has empty interior. No x is isolated in $\pi(C)$ as otherwise $\pi^{-1}(\{x\}) \cap C$ is nonempty, finite and open in C.

We show that the inverse map is well-defined. It suffices to show that $\pi^{-1}(C)' \cong \overline{\mathbb{C}}$ whenever $C \cong \overline{\mathbb{C}}$. By the above $\pi^{-1}(C)$ and so $\pi^{-1}(C)'$ are closed and have empty interior. Since C has no isolated points, each isolated point in $\pi^{-1}(C)$ is part of a two-element fiber (of a component of a jump) and the other element of the fiber is not isolated. Hence $\pi^{-1}(C)'$ has no isolated points.

Moreover, this yields $\pi(\pi^{-1}(C)') = \pi(\pi^{-1}(C)) = C$.

Clearly $\pi^{-1}(\pi(C))' \supseteq C' = C$, so suppose that $x \in \pi^{-1}(\pi(C))'$. Then $\pi(x) = \pi(y)$ for some $y \in C$. If $x = y$, we are done, and else by symmetry assume $x < y$. Then, as x is not isolated, each open interval (z, x) meets $\pi^{-1}(\pi(C))$ in uncountably many points and, in particular, in some point that is no component of a jump. Thus (z, x) meets C and so $x \in \overline{C} = C$. □

2.2.2 Iso-uniform orders

Here we provide a first construction of uniform sums, which satisfies a stricter uniformity property involving isomorphism classes instead of finite partial types. The description for partial types is in section 2.4.

Definition 2.2.6. Let $X \subseteq \mathbb{R}$ be locally closed and let P be a labeling of X. We call P **iso-uniform** if all nonempty convex $I, J \subseteq X$ without endpoints satisfy $(I, P) \cong (J, P)$ as labeled orders.

If P is clear from the context, we say that X is **iso-uniformly labeled**. In particular, if $Y \subseteq X$ is locally closed, then we say that Y is iso-uniformly labeled if $P|_Y$ is iso-uniform. Similar conventions apply to notions as k-uniformity introduced later.

Definition 2.2.7. Let T be a finite set of labeled order types of Cantor sets and points. Let P be a labeling of $X \cong \mathbb{R}$ and c_0 be a label. We call (X, P) an **iso-uniform sum** of T with comeager label c_0 if

(a) for every nonempty open $U \subseteq X$ and $t \in T$, there is a $C \subseteq U$ such that (C, P) realizes t;

(b) there are countably many closed nowhere dense $C_i \subseteq X$ such that for each i there is $t \in T$ such that (C_i, P) can be embedded into a realization of t and $\mathbb{R} \setminus \bigcup_i C_i$ has label c_0.

Proposition 2.2.8. *Let T be a finite set of labeled order types of Cantor sets and points and c_0 be a label.*

(a) Iso-uniform sums of T are iso-uniformly labeled.

(b) There is an iso-uniform sum of T with comeager label c_0.

(c) All iso-uniform sums of T with comeager label c_0 are isomorphic.

Proof. If T is empty, all statements are clear, so assume T nonempty. We may also assume that no $t \in T$ has c_0 as single realized label as otherwise the iso-uniform sums of T are precisely the iso-uniform sums of $T \setminus \{t\}$.

For (c), let P, P' be iso-uniform sums of T with comeager label c_0, with $(C_i)_{i\in\mathbb{N}}, (C_i')_{i\in N}$ as in Definition 2.2.7 (b). We first define by recursion partial isomorphisms $f_i \colon A_i \cong A_i'$ with $A_i \supseteq C_0 \cup \cdots \cup C_{i-1}$ and $A_i' \supseteq C_0' \cup \cdots \cup C_{i-1}'$ closed nowhere dense.

Start with $A_0 = A_0' = \emptyset$. Given f_i, we obtain an order isomorphism g_i between complementary intervals of A_i and A_i' since these are given by jumps in A_i and A_i'. For each complementary interval J of A_i, by definition $(C_i \cap J, P)$ embeds into a realization of some $t \in T$. The interval $g_i(J)$ includes a realization D of t by Definition 2.2.7 (a). Extend f_i by mapping $C_i \cap J$ into D for each J. Since J is a complementary interval, this is a partial isomorphism. Repeating this with the inverse of this extension yields f_{i+1}.

Set $A = \bigcup_i A_i$ and $A' = \bigcup_i A_i'$. The f_i combine to $f \colon A \cong A'$. By assumption on T, every nonempty open subset includes some C_i and some C_i'. Thus, A and A' are dense in $\mathbb{R}$. By uniqueness of Dedekind completions, f can be extended to an order automorphism of $\mathbb{R}$ (explicitly, $x \mapsto \sup_{A \ni a \leq x} f(a)$). This extension is still an isomorphism of labelings as both $\mathbb{R} \setminus A$ and $\mathbb{R} \setminus A'$ have label c_0.

If (X, P) is an iso-uniform sum of T and $Y \subseteq X$ is nonempty convex, then (Y, P) is an iso-uniform of T with the same comeager label. Thus, (a) follows from (c). It remains to prove (b).

For $t \in T$, let $\bar{t}$ be the labeled order type that equals t if t describes a point and is obtained by adding endpoints of label c_0 to t else. Let $C = \sum_{t \in T} \bar{t}$ be the labeled order type of their lexicographic sum. Let Q be the order type of the jumps of C. For a closed embedding $C \to (0,1)$ of orders, Q equals the order type of the complementary intervals of the image.

We define sets $\mathcal{C}_n$ of pairwise disjoint realizations of C and a map f from finite sequences in Q to nonempty convex subsets of $(0,1)$ by simultaneous recursion. Start with $\mathcal{C}_0 = \emptyset$ and $f() = (0,1)$. Given $q = (q_1, \dots, q_n)$, since T is nonempty, we may take a realization D_q of C in $f(q)$ all whose complementary intervals have length at most $\frac{1}{2} \operatorname{diam} f(q)$. For $r \in Q$, let $f(q,r)$ be the r-th complementary interval of D_q. Set $\mathcal{C}_{n+1} = \{D_q \mid |q| \leq n\}$.

We now define a labeling P of $(0,1)$. If $x \in \bigcup_n \mathcal{C}_n$, label x as in C, and else label x with c_0. For showing that $((0,1), P)$ is an iso-uniform sum, decompose all summands of elements of $\bigcup_n \mathcal{C}_n$ into countably many points and copies of $\overline{\mathbb{C}}$, and let the C_i enumerate all these sets. □

2.2.3 Definability in the Borel monadic theory of order

We provide an alternative proof of definability of the class of meager sets that works in every sufficiently stable Boolean algebra whose members have the Baire property. This provides an easier example of our methods and motivates adding meagerness to the language in section 2.4.1.

Lemma 2.2.9. *Let X be a Polish space without isolated points and let $A \subseteq X$ be meager. There is a countable dense $B \subseteq X$ such that there is no Cantor set $C \subseteq A \cup B$ with B dense in C.*

Proof. Let $A' \supseteq A$ and A_i be closed nowhere dense with $A' = \bigcup_i A_i$. Then there is a countable dense $B \subseteq X \setminus A'$. Suppose that $C \subseteq A \cup B$ is a Cantor set. Then some A_i is nonmeager in C and thus has nonempty interior in C. Thus, B is not dense in C as $A_i \cap B \subseteq A' \cap B = \emptyset$. □

Lemma 2.2.10. *Let $A \subseteq \mathbb{R}$ be comeager. For all dense $B \subseteq \mathbb{R}$ there is a Cantor set $C \subseteq A \cup B$ with B dense in C.*

Proof. We may assume that A is G_δ and $B = \{b_n \mid n\}$ is countable. Then Corollary 2.2.3 with $\mathbb{R} \setminus (A \cup B)$ as A and $\{b_n\}$ as B_n yields C. □

Proposition 2.2.11. *Let $A \subseteq \mathbb{R}$ have the Baire property and $\mathcal{B} \subseteq 2^{\mathbb{R}}$ contain all countable sets. Then the following are equivalent:*

(i) A is meager.

(ii) For all nonempty open $U \subseteq \mathbb{R}$ there is a dense subset $B \in \mathcal{B}$ of U such that there is no Cantor set $C \subseteq A \cup B$ with B dense in C.

Proof. If A is meager, use Lemma 2.2.9 with U for X and $A \cap U$ for A. Else, by the Baire alternative choose an open interval I with A comeager in I and apply Lemma 2.2.10 to $I \cong \mathbb{R}$ with $A \cap I$ for A. □

2.3 Shelah's decidability technique

This section repeats the results of [She75] concerning decidability for dense total orders. They are rephrased in our language, straightforwardly extended to restricted quantifiers, and specialized to complete orders for the sake of simplicity. For the first part, other references are [Gur85; Gur79a]. We start with notation.

Let A, B be sets together with fixed injections to $\mathbb{N}$. We call a map $A \to B$ **computable** if there is a computable map $\mathbb{N} \to \mathbb{N}$ making the diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ \mathbb{N} & \overset{\exists}{\dashrightarrow} & \mathbb{N} \end{array}$$

commute. We will use this notation for example if $A \subseteq \mathbb{N}^{<\omega}$ or if A is a (not necessarily computable) set of formulas in a finite language, without explicitly mentioning the injection to $\mathbb{N}$.

A family of functions $(f_i)_i$ is **computable uniformly in** i if the map $(i, x) \mapsto f_i(x)$ is computable. Analogously, a family of sets is **computable uniformly in** i if their indicator functions are so, and a sequence of theories is **decidable uniformly in** i if the set of valid formulas is computable uniformly in i.

Definition 2.3.1. Let X be a class and $f\colon X \to A, g\colon X \to B$ be maps, where A, B are sets together with fixed injections to $\mathbb{N}$. (Below, X is a class of structures, or families of structures, A, B are sets of formulas, and f, g map a structure to its partial theory.)

Let $f(x) \mapsto g(x)$ denote the relation with graph $\{(f(x), g(x)) \mid x \in X\}$. We say that $f(x) \mapsto g(x)$ is **well-defined** if this relation is a partial function. Similarly, the family $f_i(x) \mapsto g_i(x)$ is **well-defined** if for all i the relation $f_i(x) \mapsto g_i(x)$ is a partial function. We usually specify X by saying that the map is **well-defined in** x where the variable x is known to range over X.

When proving that such functions are well-defined and computable, we usually describe a computable extension $A \to B$ of $f(x) \mapsto g(x)$ informally by describing how to compute $g(x)$ only depending on $f(x)$. In our settings, the image of f will not generally be a computable subset of A.

2.3.1 Finite partial types

We first describe Shelah's constructions in a first-order setting before specializing to the monadic setting.

We call a first-order theory T in a finite language **essentially relational** if there is a computable map f from atomic formulas to quantifier-free formulas in the same free variables such that $T \models \varphi \Leftrightarrow f(\varphi)$ for all φ and for each fixed finite set V of variables the image of formulas with free variables in V under f is finite. For instance, theories in relational languages are essentially relational via $f = \mathrm{id}$. The theory of Boolean algebras in the language $\{\leq, \vee, \wedge, \neg, \bot, \top\}$ is essentially relational via f replacing all terms by their disjunctive normal forms.

Definition 2.3.2. Let T be a first-order theory and let f witness that T is essentially relational. Let $A \models T$, $a \in A^m$ and $k \in \mathbb{N}^n$.

(a) We define $\mathsf{fTh}^-_k(m)$ by recursion on n:

- For $n = 0$, $\mathsf{fTh}^-_{()}(m)$ is the image under f of the atomic formulas with m free variables.
- If $n > 0$, then $\mathsf{fTh}^-_k(m)$ is the power set of $\mathsf{fTh}^-_{k^-}(m + k_n)$.

(b) We define $\mathsf{Th}^1_k(A, a) \subseteq \mathsf{fTh}^-_k(m)$, the k-**theory** of (A, a), by recursion on n:

- For $n = 0$, set $\mathsf{Th}^1_{()}(A, a) = \left\{\varphi \in \mathsf{fTh}^-_{()}(m) \,\middle|\, (A, a) \models \varphi\right\}$.
- For $n > 0$, set $\mathsf{Th}^1_k(A, a) = \left\{\mathsf{Th}^1_{k^-}(A, ab) \mid b \in A^{k_n}\right\}$.

We abbreviate $\mathsf{Th}^1_k(A) = \mathsf{Th}^1_k(A, ())$.

(c) Let $\mathsf{fTh}^1_k(m) = \mathsf{fTh}^-_{(k,0)}(m)$ be the set of **formally possible** k-**theories**.

This definition deviates from the preliminary definition of Th in section 2.1.1 for technical convenience. We first observe how the two definitions are equivalent. To this end, define by induction on $k \in \mathbb{N}^n$ formulas of **quantifier type** k. A formula has quantifier type $()$ if it is atomic. For $n > 0$, a formula has quantifier type k if it is of the form $\exists x_1 \ldots \exists x_{k_n} \varphi$ with φ a Boolean combination of formulas of quantifier type k^-. By induction, $\mathsf{Th}^1_k(A, a) = \mathsf{Th}^1_k(B, b)$ if and only if (A, a) and (B, b) satisfy the same formulas of quantifier type k.

Thus, if we call a partial type in T k-complete whenever it only consists of formulas of quantifier type k and is maximal among such types, then elements of $\mathsf{Th}^1_k(A)$ correspond to k^--complete types in k_n free variables. Since the Boolean algebras generated by formulas with fixed quantifier type are finite, all filters are principal. Types, that is, filters of formulas, are thus determined by single formulas. Hence, we will use the terms "type" and "theory" interchangeably for these Th^1_k.

Furthermore, every $\mathsf{Th}^1_k(A) \in \mathsf{fTh}^1_k(0)$ is hereditarily finite and $(k, m) \mapsto \mathsf{fTh}^1_k(m)$ is computable. Computability of $k \mapsto \mathsf{Th}^1_k(A)$ is equivalent to decidability of the first-order theory of A.

Definition 2.3.3. Let X be a class, let $x \in X$, let F be a map from X to a class of $\mathscr{L}_F$-structures and G be a map from X to a class of $\mathscr{L}_G$-structures. We say that the maps are $\mathsf{Th}^1_{O(k)}(F(x)) \mapsto \mathsf{Th}^1_k(G(x))$ are **well-defined in x and computable** if there is a computable $\ell\colon \mathbb{N}^{<\omega} \to \mathbb{N}^{<\omega}$ such that $|\ell(k)| \leq |k|$ for all k and the maps $\mathsf{Th}^1_{\ell(k)}(F(x)) \mapsto \mathsf{Th}^1_k(G(x))$ are well-defined in x and computable uniformly in k.

Example 2.3.4. The well-known completeness, decidability and quantifier elimination of the theory of dense total orders without endpoints can be rephrased in this setting, simultaneously computing types. Let A be a dense total order without endpoints and $a \in A^n$.

We first describe the quantifier-free types. Namely, there is a computable bijection Φ from the set of possible $\mathsf{Th}^1_{()}(A, a)$ onto the set of surjections $\{1, \dots, n\} \to \{1, \dots, m\}$ for some m. Here, $\Phi(\mathsf{Th}^1_{()}(A, a))$ is the unique f such that $a_i \mapsto f(i)$ is a well-defined order isomorphism $\{a_1, \dots, a_n\} \cong \{1, \dots, m\}$.

Quantifier elimination states that $\mathsf{Th}^1_{()}(A, a) \mapsto \mathsf{Th}^1_k(A, a)$ is well-defined in A, a. By recursion, it suffices to consider $\mathsf{Th}^1_{()}(A, a) \mapsto \mathsf{Th}^1_{(k_1)}(A, a)$. We describe the image $\Phi(\mathsf{Th}^1_{(k_1)}(A, a)) = \left\{ \Phi(\mathsf{Th}^1_{()}(A, ab)) \,\middle|\, b \in A^{k_1} \right\}$ in terms of $\Phi(\mathsf{Th}^1_{()}(A, a))$. Namely, $f\colon \{1, \dots, n + k_1\} \to \{1, \dots, m\}$ equals $\Phi(\mathsf{Th}^1_{()}(A, ab))$ for some b if and only if $f|_{\{1,\dots,n\}} = \Phi(\mathsf{Th}^1_{()}(A, a))$. Necessity is clear and sufficiency follows since A is dense without endpoints.

This also describes types: Each type is determined by $\mathsf{Th}^1_{()}(A, a)$ and thus corresponds to a surjection $\{1, \dots, n\} \to \{1, \dots, m\}$. In particular, since there is a unique type of the empty tuple, the theory is complete. Finally, decidability follows from computability of $\mathsf{Th}^1_{()}(A, a) \mapsto \mathsf{Th}^1_k(A, a)$.

This formalism also applies to restricted monadic theories in the language of order (or any finite relational language) by our previous reduction to first-order theories in section 0.1.2. In other words, we work in expansions of the language $(\leq, \subseteq, \cup, \cap, \setminus, X, \mathsf{isAtom})$ of Boolean algebras with a predicate for atoms and an order $\leq$ on the atoms. Instead of Th^1_k as for first-order theories, we denote partial monadic theories by Th_k.

If Y is a substructure of a restricted monadic structure X and $P = (P_1, \dots, P_n)$ is a tuple of subsets of X, then we abuse notation by writing

$$\mathsf{Th}_k(Y, P) = \mathsf{Th}_k(Y, (Y \cap P_1, \dots, Y \cap P_n)).$$

In this case, if (X, P) and k are clear from the context, we often use "the type of Y" to denote $\mathsf{Th}_k(Y, P)$.

Although equivalent, it is often more convenient to work with labelings instead of subsets.

Definition 2.3.5. Let $(X, \leq)$ be a total order, regarded as restricted monadic structure with quantification over $\mathcal{B}$. A **$\mathcal{B}$-labeling** of X is a map $P\colon X \to I$ to a finite set I such that all fibers lie in $\mathcal{B}$. We also write P_i for the fiber $P^{-1}(\{i\})$ and $\mathsf{Th}_k(X, P)$ for $\mathsf{Th}_k(X, (P_i)_{i \in I})$.

Alternatively, a labeling could be encoded by adding a new sort and the map itself, leading to the same expressive power but not eliminating some quantifiers. Given $\mathsf{Th}_{()}(X, (P_i)_{i \in I})$ for a tuple $(P_i)_i$ of subsets, it is computable whether the P_i assemble into a labeling $X \to I$. Namely, they do if and only if they are pairwise disjoint with $\bigcup_i P_i = X$. For now, we assume that the only expansions of the language are by tuples of subsets or labelings.

2.3.2 Monadic sums

Next, we show how to compute the monadic theory of a lexicographic sum given the theories of its summands and the index set in an appropriate sense. Let $(I, \leq)$ be a total order and for $i \in I$ let $(X_i, \leq)$ be total orders. We regard them as restricted monadic structures with quantification over $\mathcal{B}_i \leq 2^{X_i}$ and I as monadic structure with unrestricted quantification.

Let $\sum_{i \in I} X_i$ be the total order with underlying set the disjoint union $\coprod_i X_i$ and the lexicographic ordering. We interpret monadic constant symbols P by the union of their interpretations P_i in the summands. In other words, we define the sum of orders with a distinguished subset by

$$\sum_i (X_i, P_i) = \left(\sum_i X_i, \bigcup_i P_i \right).$$

Finally, $\sum_i X_i$ is regarded as a monadic structure with quantification restricted to elements of

$$\mathcal{B}_{\sum_i X_i} = \left\{ \bigcup_i A_i \mid A_i \in \mathcal{B}_i \right\}.$$

If I is countable, all $X_i \subseteq \mathbb{R}$ are G_δ and $\mathcal{C}$ is a sufficiently stable family of Boolean algebras, then, since $\{\bigcup_i A_i \mid A_i \in \mathcal{C}_{X_i}\} = \mathcal{C}_{\sum_i X_i}$, our definitions coincide.

Theorem 2.3.6. *[She75, Theorem 2.4] Let I be a total order and (X_i, P_i) for $i \in I$ be total orders with tuples of m subsets. Consider the labeling*

$$T^{kP} \colon I \to \mathsf{fTh}_k(m), i \mapsto \mathsf{Th}_k(X_i, P_i).$$

Then the maps $\mathsf{Th}_{O(k)}(I, T^{kP}) \mapsto \mathsf{Th}_k(\sum_i (X_i, P_i))$ are well-defined in I, X_i, P_i and computable.

In particular, as the theory of the summands determines the theory of the sum uniquely, the lexicographic sum induces well-defined operations $\sum_{i \in I} \colon (\mathsf{fTh}_k(m))^I \to \mathsf{fTh}_k(m)$ with

$$\sum_{i \in I} \mathsf{Th}_k(X_i, P_i) = \mathsf{Th}_k\left(\sum_{i \in I} (X_i, P_i) \right).$$

Proof. Write $(X, P) = \sum_i (X_i, P_i)$. Following our conventions in Definition 2.3.1, we only need to effectively describe $\mathsf{Th}_k(X, P)$ in terms of $\mathsf{Th}_\ell(I, T^{kP})$ for some ℓ with $|\ell| = k$ by recursion on $|k|$. Recall that $(T^{kP})_t$ denotes the fiber $(T^{kP})^{-1}(\{t\})$. Thus, $i \in (T^{kP})_t$ if and only if $\mathsf{Th}_k(X_i, P_i) = t$.

For $|k| = 0$, let s and s' be terms in the monadic language of order. We distinguish cases for atomic formulas.

First, $X \models s \subseteq s'$ if and only if all $t \in \mathsf{fTh}_k(m)$ with $(s \not\subseteq s') \in t$ satisfy $I \models (T^{kP})_t = \emptyset$. Indeed, after unraveling definitions this states that $X \models s \subseteq s'$ if and only if $\{i \mid X_i \models s \not\subseteq s'\} = \emptyset$. The argument for $s = s'$ is analogous.

Next, $X \models \mathsf{isAtom}(s)$ if and only if there is a unique $t \in \mathsf{fTh}_k(m)$ such that

- $\mathsf{isAtom}(s) \in t$;
- $I \models \mathsf{isAtom}((T^{kP})_t)$;
- all $t' \in \mathsf{fTh}_k(m)$ with $t' \neq t$ satisfy $(s = \emptyset) \in t'$ or $I \models (T^{kP})_{t'} = \emptyset$.

We finish the base case by describing when $X \models s \leq s'$. We require that $X \models \mathsf{isAtom}(s)$ and $X \models \mathsf{isAtom}(s')$, say witnessed by $t, t' \in \mathsf{fTh}_k(m)$ with $\mathsf{isAtom}(s) \in t$ and $\mathsf{isAtom}(s') \in t'$ and $I \models \mathsf{isAtom}((T^{kP})_t)$ and $I \models \mathsf{isAtom}((T^{kP})_{t'})$. Then, we demand that

- $I \models (T^{kP})_t < (T^{kP})_{t'}$; or
- both $t = t'$ and $(s \leq s') \in t$.

Finally, suppose that $|k| > 0$. Given $\mathsf{Th}_{O(k)}(I, T^{kP})$, we need to compute the set of possible $\mathsf{Th}_{k^-}(X, PP')$ as P' varies over tuples of subsets of X. Since $\mathsf{Th}_{k^-}(X, PP') = \sum_i \mathsf{Th}_{k^-}(X_i, PP')$, by recursion it suffices to compute the set of possible $\mathsf{Th}_{O(k^-)}(I, T^{k^-, PP'})$. We thus describe for a labeling f of I in terms of $\mathsf{Th}_{O(k^-)}(I, T^{kP}, f)$ whether $f \in \left\{ T^{k^-, PP'} \,\middle|\, P' \right\}$. Such a P' corresponds to tuples P'_i of subsets of X_i, whose theories comprise $T^{kP}(i) = \mathsf{Th}_k(X_i, P_i)$. Thus,

$$f \in \left\{ T^{k^-, PP'} \,\middle|\, P' \right\} \text{ if and only if } f(i) \in T^{kP}(i) \text{ for all } i \text{ if and only if } f_t \subseteq \bigcup_{t' \ni t} (T^{kP})_{t'} \text{ for all } t.$$

This can be computed even from $\mathsf{Th}_{()}(I, T^{kP}, f)$. □

Corollary 2.3.7. *Let $(X,P),(Y,Q)$ be labeled total orders. The maps*

$$(\mathsf{Th}_k(X,P),\mathsf{Th}_k(Y,Q)) \mapsto \mathsf{Th}_k(X+Y,P+Q)$$

are well-defined in X,Y,P,Q and computable uniformly in k.

Proof. Compose $(\mathsf{Th}_k(X,P),\mathsf{Th}_k(Y,Q)) \mapsto \mathsf{Th}_{O(k)}(\{1,2\},T^{kP})$ with the map of Theorem 2.3.6. Since $T^{kP}(1)=\mathsf{Th}_k(X,P)$ and $T^{kP}(2)=\mathsf{Th}_k(Y,P)$ are given and $\{1,2\}$ is finite, the type $\mathsf{Th}_{O(k)}(\{1,2\},T^{kP})$ can be computed. □

To illustrate these methods, we reprove decidability of S1S. We only work with unrestricted monadic theories.

Lemma 2.3.8. *(a) The monadic theory of $\{1,\dots,n\}$ is decidable uniformly in n.*

(b) The common monadic theory of the class of finite total orders is decidable.

Proof. The theories of $\emptyset,\{1\}$ are decidable by finiteness. The theory of $\{1,\dots,n\}$ can be computed uniformly by iterating the algorithm of Corollary 2.3.7.

For (b), let k be a tuple of subsets. By the pigeonhole principle, $\mathsf{Th}_k(\{1,\dots,p\})=\mathsf{Th}_k(\{1,\dots,p+q\})$ for some $p,q\in\mathbb{N}$ with $q>0$. It suffices to show that each finite total order I has the same Th_k as some $\{1,\dots,n\}$ with $n<p+q$. Write $\#I=p+mq+r$ with $r<q$. Then $n=p+r$ works since we can prove by induction on m that

$$\mathsf{Th}_k(I)=(\mathsf{Th}_k(\{1,\dots,p\})+m\mathsf{Th}_k(\{1,\dots,q\}))+\mathsf{Th}_k(\{1,\dots,r\})=\mathsf{Th}_k(\{1,\dots,p\})+\mathsf{Th}_k(\{1,\dots,r\}).\ \square$$

Theorem 2.3.9 (S1S)**.** *[She75, Theorem 3.4] The monadic theory of $(\mathbb{N},\leq)$ is decidable.*

Proof. We compute $\mathsf{Th}_k(\mathbb{N})$ by recursion on $|k|$, and $|k|=0$ is trivial. Suppose that $|k|=n>0$.

It suffices to show that $\mathsf{Th}_k(\mathbb{N})=\left\{\mathsf{Th}_{k^-}(\mathbb{N},P)\,\middle|\,P\in(2^{\mathbb{N}})^{k_n}\right\}$ is the set of $s+\sum_{i\in\mathbb{N}}t$ for s,t ranging over all k^--types of nonempty finite total orders with a tuple of k_n subsets. Indeed, the set of all possible s,t can be computed by Lemma 2.3.8, and the map $(s,t)\mapsto s+\sum_i t$ can be computed by Theorem 2.3.6. Indeed, the computation of $t\mapsto\sum_i t$ requires $\mathsf{Th}_{O(k^-)}(\mathbb{N},T^{k^-,P})$ for the constant map $P\colon i\mapsto t$. It is computable from $\mathsf{Th}_{O(k^-)}(\mathbb{N})$ and thus by recursion.

Clearly, all $s+\sum_i t$ are possible types of tuples of subsets of $\mathbb{N}$. Conversely, let $P\in(2^{\mathbb{N}})^{k_n}$ be arbitrary. Color pairs of $i<j$ by $t_{ij}=\mathsf{Th}_{k^-}(\{i,\dots,j-1\},P)$. By Ramsey's theorem, there are t and an infinite set of $n_0<n_1<\dots$ such that $t_{n_i,n_{i+1}}=t$ for all i. Then $\mathsf{Th}_{k^-}(\mathbb{N},P)=t_{0,n_0}+\sum_i t$. □

Corollary 2.3.10. *The monadic theories of the order dual $\mathbb{N}^{\mathrm{op}}$ and of $\mathbb{Z}$ are decidable.*

Corollary 2.3.11. *Let X be a total order. The maps $\mathsf{Th}_k(X)\mapsto\mathsf{Th}_k(\sum_{i\in\mathbb{N}}X)$ and $\mathsf{Th}_k(X)\mapsto\mathsf{Th}_k(\sum_{i\in\mathbb{Z}}X)$ are well-defined in X and computable.*

2.3.3 Special quantifiers

Using monadic quantifiers, quantifiers over elements or labelings can be expressed. We now define analogues of Th_k for such quantifier types.

First-order quantifiers

Definition 2.3.12. Let Φ be an arbitrary map whose domain are structures X with $\mathcal{B}\subseteq 2^X$ and $P\in\mathcal{B}^k$. Let $m\in\mathbb{N}$. Set $\mathsf{asub}_m(\Phi)(X,P)=\{\Phi(X,PP')\mid P'\in\mathcal{B}^m\}$.

Informally, asub describes the standard restricted monadic quantifier. More precisely, $\mathsf{Th}_k=\mathsf{asub}_{k_n}(\mathsf{Th}_{k^-})$, hence inductively $\mathsf{Th}_k=\mathsf{asub}_{k_n}(\mathsf{asub}_{k_{n-1}}(\dots(\mathsf{asub}_{k_1}(\mathsf{Th}_{()})\dots)))$.

Next, we want to combine first-order theories with monadic theories. If $\mathcal{B}$ contains all singletons $\{x\}$, define $\mathsf{elem}(\Phi)(X,P)=\{\Phi(X,P\{x\})\mid x\in X\}$. For $m\in\mathbb{N},k\in\mathbb{N}^n$ define the mixed theory

$$\mathsf{Th}_{m;k}=\mathsf{asub}_{k_n}(\mathsf{asub}_{k_{n-1}}(\dots(\mathsf{asub}_{k_1}(\underbrace{\mathsf{elem}(\dots(\mathsf{elem}}_{m\text{ times}}(\mathsf{Th}_{()})\dots)))))).$$

Our previous Th_k is $\mathsf{Th}_{0;k}$. We obtain an analogue of Theorem 2.3.6.

Theorem 2.3.13. *Let I be a total order and (X_i, P_i) for $i \in I$ be total orders with tuples of subsets. Let $T^{m;k,P}\colon i \mapsto \mathsf{Th}_{m;k}(X_i, P_i)$. The maps $\mathsf{Th}_{m;O(k)}(I, T^{m;k,P}) \mapsto \mathsf{Th}_{m;k}(\sum_i (X_i, P_i))$ are well-defined in I, X_i, P_i and computable.*

Proof. We again work by recursion on (m, k) and the base case $(0, ())$ is unchanged. For $|k| > 0$, the recursion step from $(m, O(k^-))$ to (m, k) is exactly the same as the previous recursion step from $O(k^-)$ to k. Thus it remains to handle the step from $(m, ())$ to $(m+1, ())$.

Write $(X, P) = \sum_i (X_i, P_i)$. Given $\mathsf{Th}_{m+1;()}(I, T^{m;k,P})$, we need to compute the set of possible $\mathsf{Th}_{m;()}(X, P\{x\})$ as x varies over X. Again, since $\mathsf{Th}_{m;()}(X, P\{x\}) = \sum_i \mathsf{Th}_{m;()}(X_i, P\{x\})$, by recursion it suffices to compute the set of possible $\mathsf{Th}_{m;()}(I, T^{m;(),P\{x\}})$. We thus describe for a labeling f of I in terms of $\mathsf{Th}_{m+1;()}(I, T^{m;k,P}, f)$ whether $f \in \left\{T^{m;(),P\{x\}} \,\middle|\, x\right\}$. There is $x \in X$ with $f = T^{m;(),P\{x\}}$ if and only if there is $i \in I$ such that $f(i) \in T^{m+1;(),P}(i)$ (equivalently, there is $x \in X_i$ with $f(i) \in T^{m;(),P\{x\}}(i)$) and for $j \neq i$ we have that $f(j) = T^{m;(),P\emptyset}(j)$. This can be described using elem for the index set. □

In the following, it is convenient to work in a finite partial Morleyization of the language of order. This simplifies the treatment of low quantifier ranks, for example in Remark 2.3.23. More precisely, for $m \in \mathbb{N}$ let Φ_m be the set of monadic formulas such that

- the formulas contain only first-order quantifiers,
- the formulas are in prenex normal form,
- existential and universal quantifiers alternate,
- there are exactly m quantifiers.

For instance, $\mathsf{Th}_{m;()}(X, P) = \mathsf{Th}_{m;()}(Y, Q)$ if and only if (X, P) and (Y, Q) satisfy the same formulas in Φ_m. A formula is equivalent to an element of $\bigcup_m \Phi_m$ if and only if it only contains first-order quantifiers. From now on, we work in the monadic language obtained by adding predicates for all elements of some fixed Φ_m with m defined below. Then Th_k in this new language is $\mathsf{Th}_{m;k}$ in the old language. Thus, Theorem 2.3.6 and its corollaries continue to hold.

Let us specify m. It is desirable to express any first-order definable order-theoretic or topological property by a formula without monadic quantifiers (for example, being closed, being dense, having isolated points, being bounded on either side, being an open or closed interval, or having an immediate successor or predecessor). We will use this only for one property at a time, hence only for finitely many properties. Fix m large enough such that Φ_m expresses all these finitely many properties used in the remaining text.

Labelings

In our setting, labelings are more convenient to work with than subsets, but quantifying over them has the same expressive power. Thus, we adapt Th_k to work with $\mathcal{B}$-labelings.

For a structure X with a $\mathcal{B}$-labeling $P\colon X \to I$ and $f\colon J \to I$, we define the possible types of refinements,

$$\mathsf{refin}_f(\Phi)(X, P) = \{\Phi(X, PP') \mid P' \text{ a } \mathcal{B}\text{-labeling with } f \circ P' = P\}.$$

For a sequence k of surjections $I_n \xrightarrow{k_n} I_{n-1} \xrightarrow{k_{n-1}} \dots \xrightarrow{k_2} I_1 \xrightarrow{k_1} I_0$ of finite sets, let

$$\mathsf{pTh}_k = \mathsf{refin}_{k_n}(\mathsf{refin}_{k_{n-1}}(\dots(\mathsf{refin}_{k_1}(\mathsf{Th}_{()})\dots))).$$

For such a sequence k of surjections we use the same notation as for tuples of natural numbers. For example, we write $|k|$ for its length and k^- for the sequence without k_n.

Definition 2.3.14. We define a preorder on the set of sequences of surjections by $k \leq k'$ if $|k| \leq |k'|$ and there is a commutative diagram

$$\begin{array}{ccccccccc}
 & & & & I_{|k|} & \xrightarrow{k_{|k|}} & I_{|k|-1} & \xrightarrow{k_{|k|-1}} \cdots \xrightarrow{k_1} & I_0 \\
 & & & & \vdots\exists & & \vdots\exists & & \vdots\exists \\
I'_{|k'|} & \xrightarrow{k'_{|k'|}} \cdots \xrightarrow{k'_{|k|+1}} & I'_{|k|} & \xrightarrow{k'_{|k|}} & I'_{|k|-1} & \xrightarrow{k'_{|k|-1}} & \cdots \xrightarrow{k'_1} & I'_0 &
\end{array}$$

with the vertical maps injective.

This ordering is analogous to the componentwise ordering on tuples of natural numbers. In particular, if $k \leq k'$, then $\mathsf{pTh}_k(X,P)$ can be computed from $\mathsf{pTh}_{k'}(X,P)$ uniformly in k,k'. Again, we write $O(k)$ for a large enough uniformly computable sequence of the same length as k.

Proposition 2.3.15. *The maps $\mathsf{Th}_{O(k)}(X,P) \mapsto \mathsf{pTh}_k(X,P)$ and $\mathsf{pTh}_{O(k)}(X,P) \mapsto \mathsf{Th}_k(X,P)$ are well-defined in X,P and computable.*

The proof generalizes the observations that a subset of X can be coded by its characteristic function, which is a labeling, while a labeling can be coded by its fibers, which are subsets.

Proof. By recursion it suffices to show for $P\colon X \to I$ that:

1. Given $f\colon J \to I$ we can compute $m \in \mathbb{N}$ such that from $\mathsf{asub}_m(\Phi)(X,P)$ we can compute $\mathsf{refin}_f(\Phi)(X,P)$ uniformly in f.
2. Given $m \in \mathbb{N}$ we can compute $f\colon J \to I$ such that from $\mathsf{refin}_f(\Phi)(X,P)$ we can compute $\mathsf{asub}_m(\Phi)(X,P)$ uniformly in I,m.

For (a), check from $\Phi(X,PP')$ if $P'_1,\dots,P'_m$ form the fibers of a refinement of P by a quantifier-free monadic formula. For (b), let $J = I \times 2^{\{1,\dots,m\}}$ and $f\colon J \to I$ be the projection. Then the type after adding subsets $P'_1,\dots,P'_m$ can be detected from the labeling $X \to J, x \mapsto (P(x), \{k \mid x \in P'_k\})$. □

For the singleton order $\{\star\}$, for all k,k' the data of $\mathsf{Th}_{()}(\{\star\},P), \mathsf{Th}_k(\{\star\},P), \mathsf{pTh}_{k'}(\{\star\},P)$ are readily computable from each other (they only state the label $P(\star)$). We do not distinguish them in the following, but just write $\mathsf{Th}(\{\star\},P)$ for any of them. Similarly, we denote any $\mathsf{Th}_k(\emptyset,P)$ or $\mathsf{pTh}_{k'}(\emptyset,P)$ by $\mathsf{Th}(\emptyset)$.

Labeling Cantor sets

From now on, we fix a sufficiently stable Boolean algebra $\mathcal{B}$.

Definition 2.3.16. Let I,J be finite sets, $X \cong \mathbb{C}$ and $Y \cong \mathbb{R}$.

(a) A labeling $P\colon X \to I \amalg J$ is **jump-normalized** if it maps left endpoints of jumps into J and all other points into I.

(b) Let $C \subseteq Y$ be a Cantor set and $P\colon Y \to I$ be a $\mathcal{B}$-labeling. Let $J = \mathsf{fTh}_{O(k)}(\#I)$ and $\mathsf{exten}_Y(P)\colon C \to I \amalg J$ be the jump-normalized labeling that satisfies $\mathsf{exten}_Y(P)(x) = \mathsf{pTh}_k([x,y),P)$ for all jumps (x,y) and coincides with P on all other points.

The property of being jump-normalized is quantifier-free definable as being a jump is quantifier-free definable. Our next aim is to reconstruct $\mathsf{pTh}_k(Y,P)$ from the refinement $\mathsf{exten}_Y(P)$ of $\mathsf{pTh}_k(C,P)$.

Proposition 2.3.17. *Let $Y \cong \mathbb{R}$ and $P\colon Y \to I$ be a $\mathcal{B}$-labeling and $C \subseteq Y$ be a Cantor set. The maps*

$$\mathsf{recon}\colon \mathsf{pTh}_{O(k)}(C, \mathsf{exten}_Y(P)) \mapsto \mathsf{pTh}_k(Y,P)$$

are well-defined in P and computable.

Since (Y,P) is the lexicographic sum $\sum_{x\in C}(I_x,P)$ where $I_x = [x,y)$ for all jumps (x,y) and $I_z = \{z\}$ for all other points z, for Proposition 2.3.17 it suffices to show the following lemma.

Lemma 2.3.18. *Let $X,Y \subseteq \mathbb{R}$ be locally closed and I_x for $x \in X$ be total orders. Suppose that $\{I_x \mid x \in X\} / \cong$ is finite. Let P be a $\mathcal{B}$-labeling of Y with $(Y,P) \cong \sum_{x\in X}(I_x,P)$. Set $T^{kP}\colon x \mapsto \mathsf{pTh}_k(I_x,P)$. Then the maps $\mathsf{pTh}_{O(k)}(X,T^{kP}) \mapsto \mathsf{pTh}_k(Y,P)$ are well-defined in X, I_x, P and computable.*

Proof. The only difference to Theorem 2.3.6 is that the index set X does not admit unrestricted quantification but only quantification over $\mathcal{B}$. We show that the same algorithm remains correct.

The base case is unchanged. For the induction step, we use the same notation as in the previous proof. Now P' varies over tuples in $\mathcal{B}$ and f varies over $\mathcal{B}$-labelings. Thus, for correctness of the previous algorithm it suffices to show that, for a refinement P' of P such that all $P'|_{I_x}$ are $\mathcal{B}$-labelings,

(i) if P' is a $\mathcal{B}$-labeling, so is $T^{k^-,P'}$;

(ii) if $T^{k^-,P'}$ is a $\mathcal{B}$-labeling, then there is a $\mathcal{B}$-refinement Q' of P with $T^{k^-,P'} = T^{k^-,Q'}$.

Each I_x is convex in Y, thus locally closed in $\mathbb{R}$. By convexity $\#I_x \leq 1$ for all but countably many x. Write $C = \{x \,|\, \#I_x = 1\}$ and $D = \sum_{x\in C} I_x \subseteq Y$.

For (i), it suffices to show that the restriction of $T^{k^-,P'}$ to $\{x \,|\, \#I_x \leq 1\} \in \mathcal{B}$ is a $\mathcal{B}$-labeling. Since Y is locally closed, so is $\{x \,|\, I_x \neq \emptyset\}$. (Indeed, working locally, suppose that Y is closed. Since Y is Dedekind complete, $\{x \,|\, I_x \neq \emptyset\}$ is Dedekind complete, thus locally closed.) Since $(C, T^{k^-,P'}) \cong (D, P')$ (identifying labels $\mathsf{Th}(\{x\}, P')$ with $P'(x)$), also $T^{k^-,P'}|_C$ is a $\mathcal{B}$-labeling.

For (ii), for each type s of some I_x and isomorphism class t of I_x as unlabeled order with $\#I_x > 1$, choose a realization Q'_{st} of s and t. Define Q' by $Q'|_D = P'|_D$ and $Q'|_{I_x} = Q'_{\mathsf{pTh}_{k^-}(I_x,P'),I_x}$. Then $T^{k^-,P'} = T^{k^-,Q'}$ by construction and it remains to show Q' a $\mathcal{B}$-labeling.

Since $(D, Q') = (D, P') \cong (C, T^{k^-,P'})$, the restriction $Q'|_D$ is a $\mathcal{B}$-labeling. To show that $Q'|_{Y\setminus D}$ is a $\mathcal{B}$-labeling, let $\mathcal{A}$ be the generator of $\mathcal{B}$ in Definition 0.1.21. First, since there are finitely many Q'_{st}, there is a finite bound on the number of elements of $\mathcal{A}$ used in some Q'_{st}. Second, by the defining property of $\mathcal{A}$ for the partition into D and the I_x, taking an element of $\mathcal{A}_{I_x}$ for each x yields an element of $\mathcal{A}_Y$. We conclude that each fiber of $Q'|_{Y\setminus D}$ is a finite Boolean combination of elements of $\mathcal{A}$. □

Definition 2.3.19. (a) Let $X \subseteq \mathbb{R}$ be locally closed and $P\colon X \to I$ be a $\mathcal{B}$-labeling. We define

$$\mathsf{cant}(\Phi)(X, P) = \{\Phi(C, \mathsf{exten}_X(P)) \mid C \subseteq X \text{ a Cantor set}\}.$$

If $t \in \mathsf{cant}(\Phi)(X, P)$, we often say that t occurs in (X, P).

(b) Let $C \cong \mathbb{C}$ and $P\colon C \to I$ be a $\mathcal{B}$-labeling. We obtain a labeling $\mathbb{R}_C(P)\colon R \to I \amalg I \times I$ with $R \cong \mathbb{R}$ as in Definition 2.1.2 by leaving all points of C that are no component of a jump unchanged and replacing all jumps (a, b) by a point of label $(P(a), P(b))$.

By Lemma 2.3.18, $\mathsf{pTh}_{O(k)}(C, P)$ and $\mathsf{pTh}_{O(k)}(R, \mathbb{R}_C(P))$ are computable from each other. Indeed, there are decompositions $C \cong \sum_{r\in R} I_r$ with $\#I_r \in \{1, 2\}$ and $R \cong \sum_{c\in C} J_c$ with $\#J_c \in \{0, 1\}$.

2.3.4 Uniform labelings

Our next aim is to show that over the reals, we can reduce to quantifying over Cantor sets and quantifying over sets satisfying a certain uniformity condition (Theorem 2.3.24).

Proposition 2.3.20. *[She75, Lemma 5.3] Let $X \subseteq \mathbb{R}$ be locally closed, P be a $\mathcal{B}$-labeling of X and k be a sequence of surjections. The following conditions are equivalent*

(i) all nonempty relatively convex $I, J \subseteq X$ without endpoints satisfy $\mathsf{pTh}_k(I, P) = \mathsf{pTh}_k(J, P)$;

(ii) there are a dense $D \subseteq X$ and types p, q such that all $a, b \in D$ satisfy $\mathsf{pTh}_k((a, b), P) = p$ and $\mathsf{Th}(\{a\}, P) = q$.

Proof. Assuming (i), let D be a nonempty fiber of P. Since every nonempty relatively convex subset has the same type as X and this type guarantees an element of D in any realization, the set D is dense.

Suppose that (ii) holds and let I, J be nonempty convex without endpoints. Since D is dense, there are strictly increasing sequences $(a_n)_{n\in\mathbb{Z}}, (b_n)_{n\in\mathbb{Z}}$ in D with $a_{-n} \to \inf I, a_n \to \sup I, b_{-n} \to \inf J, b_n \to \sup J$ as $n \to \infty$. Thus

$$\mathsf{pTh}_k(I, P) = \sum_{n\in\mathbb{Z}} \mathsf{Th}(\{a_n\}, P) + \mathsf{pTh}_k((a_n, a_{n+1}), P) = \sum_{n\in\mathbb{Z}} (q + p)$$
$$= \sum_{n\in\mathbb{Z}} \mathsf{Th}(\{b_n\}, P) + \mathsf{pTh}_k((b_n, b_{n+1}), P) = \mathsf{pTh}_k(J, P).$$

□

We call P k-**uniform** if these equivalent conditions hold.

A $\mathcal{B}$-labeling P of $\mathbb{C}$ is k-uniform if and only if $\mathbb{R}_{\mathbb{C}}(P)$ is k'-uniform for k' the image of k under the endofunctor $I \mapsto I \amalg I \times I$ of finite sets. Moreover, iso-uniform $\mathcal{B}$-labelings are k-uniform for all k.

Lemma 2.3.21. *Let P be a $\mathcal{B}$-labeling of X with $X \cong \mathbb{R}$ or $X \cong \mathbb{C}$ and let k be a sequence of surjections. Then there is an interval $I \subseteq X$ that is k-uniformly labeled by P.*

Proof. By Theorem 2.1.5 or Corollary 2.1.6 there are an interval I and a dense $D \subseteq I$ that is homogeneous for the additive coloring $(a, b) \mapsto (\mathsf{pTh}_k((a, b), P), \mathsf{Th}(\{a\}, P))$. □

Remark 2.3.22. If P is k-uniform and $C \subseteq \mathbb{R}$ is a Cantor set, then given $\mathsf{pTh}_{O(k)}(C, P)$, or equivalently $\mathsf{pTh}_{O(k)}(\mathbb{R}, \mathbb{R}_C(P))$, together with $\mathsf{pTh}_k(\mathbb{R}, P)$, we can compute $\mathsf{pTh}_k(C, \mathsf{exten}_{\mathbb{R}}(P))$. Indeed, every complementary interval has type $\mathsf{pTh}_k(\mathbb{R}, P)$.

Let Φ be an element of some $\mathsf{fTh}_k(\#P)$ for P a $\mathcal{B}$-labeling of X with $X \cong \mathbb{R}$ or $X \cong \mathbb{C}$. We define $\mathsf{urefin}^k_f(\Phi)(X, P)$ as the set of $\Phi(X, PP')$ for P' ranging over all jump-normalized k-uniform $\mathcal{B}$-refinements of P along f.

We define a uniform analogue of cant, called $\mathsf{ucant}^k(\Phi)(X, P)$, as the set of $\Phi(C, P) \in \mathsf{cant}(\Phi)(X, P)$ for k-uniformly labeled C. Note that $\mathsf{urefin}^k_f(\mathsf{ucant}^k(\Phi)) = \mathsf{urefin}^k_f(\mathsf{cant}(\Phi))$ as coarsening a k-uniform $\mathcal{B}$-labeling yields a k-uniform $\mathcal{B}$-labeling. We call this composite also $\mathsf{urefincant}^k_f(\Phi)$.

Now suppose that P is k-uniform. We define $\mathsf{uTh}_k(X, P)$ by recursion on $|k|$. Set $\mathsf{uTh}_{()}(X, P) = \mathsf{Th}_{()}(X, P)$. If $|k| > 0$, then $\mathsf{uTh}_k(X, P) = (\mathsf{uTh}^1_k(X, P), \mathsf{uTh}^2_k(X, P))$ with $\mathsf{uTh}^1_k(X, P) = \mathsf{urefin}^{k^-}_{k_n}(\mathsf{uTh}_{k^-})(X, P)$ and $\mathsf{uTh}^2_k(X, P) = \mathsf{urefincant}^{k^-}_{k_n}(\mathsf{uTh}_{k^-})(X, P)$.

Remark 2.3.23. For uTh^2_k, [She75, Definition 5.2] used convex equivalence relations instead of Cantor sets. This also allows handling incomplete total orders, but is equivalent for the reals. We use Cantor sets for the more direct applicability of Baire category methods. He worked with Th instead of pTh (although he also used an analogue of pTh elsewhere), and the difference is irrelevant. Due to Morleyizing the language, we obtain that $\mathsf{pTh}_{O(k)}(\mathbb{R}, P) \mapsto \mathsf{uTh}_k(\mathbb{R}, P)$ is well-defined and computable, while he needed to increase $|k|$ for this [She75, Lemma 5.1].

Theorem 2.3.24. *[She75, Theorem 5.4] The maps $\mathsf{pTh}_{O(k)}(\mathbb{R}, P) \mapsto \mathsf{uTh}_k(\mathbb{R}, P)$ and $\mathsf{uTh}_{O(k)}(\mathbb{R}, P) \to \mathsf{pTh}_k(\mathbb{R}, P)$ are well-defined in k-uniform P and computable.*

Proof. Computing $\mathsf{refin}(\mathsf{cant})$ from refin is easy using Fact 0.1.5. Thus, for computing $\mathsf{pTh}_{O(k)}(\mathbb{R}, P) \mapsto \mathsf{uTh}_k(\mathbb{R}, P)$ it suffices to decide, given $\mathsf{pTh}_{O(k)}(\mathbb{R}, P)$, whether P is k^--uniform. This is easy by quantifying over intervals.

For the converse computation, let $P\colon \mathbb{R} \to I$. By recursion it suffices to compute pTh_k from $\mathsf{urefin}^{k^-}_f(\mathsf{pTh}_{k^-})$ and $\mathsf{urefincant}^{k^-}_f(\mathsf{pTh}_{k^-})$ for a large enough (in the sense of Definition 2.3.14) map f onto I. Informally, we show that each element of $\mathsf{pTh}_k(\mathbb{R}, P)$ is obtained from uniform elements (in (i) below) by repeated lexicographic addition ((ii) and (iii) below), and by an operation (iv) that combines the given types of uniformly labeled Cantor sets and previously obtained types of complementary intervals. Formally, define E as the smallest subset of $\mathsf{fTh}_{k^-}(\#P)$ such that the following conditions hold:

(i) $\mathsf{urefin}^{k^-}_{k_n}(\mathsf{pTh}_{k^-})(\mathbb{R}, P) \subseteq E$.

(ii) Let $t_1, t_2 \in E$ be arbitrary and $t = \mathsf{Th}(\{x\}, Q)$ for some $x \in \mathbb{R}$ and refinement Q of P. Then $t_1 + t + t_2 \in E$.

(iii) Let $t_1 \in E$ be arbitrary and $t = \mathsf{Th}(\{x\}, Q)$ for some $x \in \mathbb{R}$ and refinement Q of P. Then $\sum_{i \in \mathbb{N}}(t_1 + t) \in E$ and $\sum_{i \in \mathbb{N}^{\mathrm{op}}}(t + t_1) \in E$.

(iv) Define $J = \mathsf{fTh}_{O(k^-)}(\#I)$ and let $g\colon I \amalg J \to I$ be the identity on I and such that $g(\mathsf{pTh}_{O(k^-)}([x, y), P)) = P(x)$. Let $h\colon J \to J$ be such that $h(\mathsf{pTh}_{O(k^-)}([x, y), P)) = \mathsf{pTh}_{O(k^-)}((x, y), P)$. Consider an arbitrary $\mathsf{pTh}_{k^-}(\mathbb{C}, Q) \in \mathsf{urefincant}^{k^-}_g(\mathsf{pTh}_{k^-})(\mathbb{R}, P)$ with $h(Q(\mathbb{C}) \cap J) \subseteq E$. Then $\mathsf{recon}(\mathsf{pTh}_{k^-}(\mathbb{C}, Q)) \in E$.

Since E is clearly computable, it suffices to show that $E = \mathsf{pTh}_k(\mathbb{R}, P)$.

For showing $E \subseteq \mathsf{pTh}_k(\mathbb{R}, P)$, we use induction on the construction. The base case (i) follows from

$$\mathsf{urefin}_{k_n}^{k^-}(\mathsf{pTh}_{k^-})(\mathbb{R}, P) \subseteq \mathsf{refin}_{k_n}(\mathsf{pTh}_{k^-})(\mathbb{R}, P) = \mathsf{pTh}_k(\mathbb{R}, P).$$

For (ii), suppose that $t_1, t_2 \in \mathsf{pTh}_k(\mathbb{R}, P)$ and $t = \mathsf{Th}(\{x\}, Q)$ for some refinement Q of P. The k-uniformity of P yields $t_1 \in \mathsf{pTh}_k(\mathbb{R}, P) = \mathsf{pTh}_k((-\infty, x), P)$ and $t_2 \in \mathsf{pTh}_k(\mathbb{R}, P) = \mathsf{pTh}_k((x, \infty), P)$, say Q_1, Q_2 are $\mathcal{B}$-refinements of P with $t_1 = \mathsf{pTh}_{k^-}((-\infty, x), Q_1)$ and $t_2 = \mathsf{pTh}_{k^-}((x, \infty), Q_2)$. Then Q_1, Q_2, Q combine to Q' with

$$t_1 + t + t_2 = \mathsf{pTh}_{k^-}((-\infty, x), Q_1) + \mathsf{Th}(\{x\}, Q) + \mathsf{pTh}_{k^-}((x, \infty), Q_2) = \mathsf{pTh}_{k^-}(\mathbb{R}, Q') \in \mathsf{pTh}_k(\mathbb{R}, P).$$

Similarly, splitting $\mathbb{R}$ into countably many intervals separated by points we obtain (iii), and splitting $\mathbb{R}$ into intervals separated by a Cantor set we obtain (iv).

For the converse inclusion, let $\mathsf{pTh}_{k^-}(\mathbb{R}, Q) \in \mathsf{pTh}_k(\mathbb{R}, P)$ be arbitrary. We need the following result.

Claim A. *Let Q be a $\mathcal{B}$-labeling of $\mathbb{R}$ and $I \subseteq \mathbb{R}$ be convex without endpoints such that for all $x \in I$ there are $a, b \in I$ with $a < x < b$ such that all a', b' with $a < a' < b' < b$ satisfy $\mathsf{pTh}_{k^-}((a', b'), Q) \in E$. Then $\mathsf{pTh}_{k^-}(I, Q) \in E$.*

Proof. By assumption we can cover I with open intervals J such that for all subintervals $K \subseteq J$ without endpoints we have that $\mathsf{pTh}_{k^-}(K, Q) \in E$. By compactness we may assume that every closed subinterval of I meets finitely many J. Writing I as an increasing union of such closed intervals, we obtain a strictly increasing sequence $(a_n)_{n\in\mathbb{Z}}$ with $a_{-n} \to \inf I, a_n \to \sup I$ as $n \to \infty$ and $\mathsf{pTh}_{k^-}((a_n, a_{n+1}), Q) \in E$ for all $n \in \mathbb{Z}$. Using (ii), we obtain $\mathsf{pTh}_{k^-}((a_m, a_n), Q) \in E$ for all integers $m < n$. Apply Ramsey's theorem to the colorings $(m, n) \mapsto (\mathsf{pTh}_{k^-}((a_m, a_n), Q), \mathsf{Th}(\{a_m\}, Q), \mathsf{Th}(\{a_n\}, Q))$ of $\mathbb{N}$ and $-\mathbb{N}$ separately. Replacing $(a_n)_{n\in\mathbb{Z}}$ by the subsequences thus obtained we may assume that there are s, t, s_1, t_1 with $\mathsf{Th}(\{a_n\}, Q) = s$ for $n < 0$, $\mathsf{Th}(\{a_n\}, Q) = t$ for $n > 0$, $\mathsf{pTh}_{k^-}((a_{n-1}, a_n), Q) = s_1$ for $n < 0$, and $\mathsf{pTh}_{k^-}((a_n, a_{n+1}), Q) = t_1$ for $n > 0$. By (ii) and (iii)

$$\mathsf{pTh}_{k^-}(I, Q) = \sum_{i\in\mathbb{N}^{\mathrm{op}}} (s + s_1) + \mathsf{Th}(\{a_{-1}\}, Q) + \mathsf{pTh}_{k^-}((a_{-1}, a_1), Q) + \mathsf{Th}(\{a_1\}, Q) + \sum_{i\in\mathbb{N}} (t_1 + t) \in E. \quad \square$$

Let $C \subseteq \mathbb{R}$ be the set of x such that for all a, b with $a < x < b$ there are a', b' with $a < a' < b' < b$ and $\mathsf{pTh}_{k^-}((a', b'), Q) \notin E$. Clearly C is closed.

We show that C has no isolated points. Suppose for the sake of contradiction that $C \cap (a, b) = \{x\}$. By Claim A all subintervals I of $(a, x), (x, b)$ satisfy $\mathsf{pTh}_{k^-}(I, Q) \in E$. Since $x \in C$, there are a', b' with $a < a' < b' < b$ and $\mathsf{pTh}_{k^-}((a', b'), Q) \notin E$. Then $a' < x < b'$ as otherwise we may set $I = (a', b')$, and similarly $\mathsf{pTh}_{k^-}((a', x)), \mathsf{pTh}_{k^-}((x, b')) \in E$. This contradicts (ii).

We show that C has empty interior. Let I be any interval. By Lemma 2.3.21 there is an open subinterval $J \subseteq I$ such that Q is k-uniform on J. Thus for $a', b' \in J$ with $a' < b'$ by (i)

$$\mathsf{pTh}_{k^-}((a', b'), Q) \in \mathsf{urefin}_{k_n}^{k^-}(\mathsf{pTh}_{k^-})(\mathbb{R}, P) \subseteq E.$$

In other words, $J \cap C = \emptyset$. Hence $I \nsubseteq C$.

Applying Facts 0.1.5 and 0.1.8 in $\mathbb{R} \cup \{-\infty, \infty\}$ we conclude that C is order isomorphic to one of $\emptyset, \mathbb{C}, \{\star\} + \mathbb{C}, \mathbb{C} + \{\star\}, \overline{\mathbb{C}}$. If C is empty, then Claim A with $I = \mathbb{R}$ shows $\mathsf{pTh}_{k^-}(\mathbb{R}, Q) \in E$.

Else C includes a subset $D \cong \mathbb{C}$ that is convex in C. Shrinking D, by Lemma 2.3.21 we may assume that Q is k-uniform on D. Let $I = (\inf D, \sup D)$ be the convex hull of D in $\mathbb{R}$ and $J \subseteq I$ be any subinterval. Then by definition of C and Claim A, the type of each complementary interval of D in J lies in E, whence (iv) yields $\mathsf{pTh}_{k^-}(J, Q) \in E$. This shows $D \subseteq I \cap C = \emptyset$, which is absurd. $\square$

For later use, we record a consequence of the proof.

Lemma 2.3.25. *Let P be a $\mathcal{B}$-labeling of $\mathbb{R}$ and k be a sequence of surjections. Then there is a realization P' of $\mathsf{Th}_k(\mathbb{R}, P)$ such that $\mathbb{R}$ can be decomposed as a countable disjoint union of open intervals, Cantor sets and points on which P' is k-uniform.*

Proof. Let E be as in the proof of Theorem 2.3.24. Since $\mathsf{Th}_k(\mathbb{R}, P) \in E$, we obtain a realization P' such that $(\mathbb{R}, P')$ is constructed by a finite number of applications of the operations of (i) to (iv). Each of these operations preserves the property to admit such a decomposition. □

Example 2.3.26. [She75, Theorem 6.5] Consider the Boolean algebra generated by the countable sets as $\mathcal{B}$. This requires minor modifications since $\mathcal{B}$ is not sufficiently stable and the proof of Lemma 2.3.18 fails. However, the reader may check that, using the notation of the proof, the subclass of the $T^{k^-,PP'}$ such that there is a $\mathcal{B}$-labeling Q' with $T^{k^-,PP'} = T^{k^-,PQ'}$ is computable. Thus Lemma 2.3.18 persists.

We prove decidability of the restricted monadic theory. Informally, we obtain full quantifier elimination for uniform sets. More precisely, by Theorem 2.3.24 it suffices to show the following. Let $X_1, \dots, X_n \in \mathcal{B}$ be the fibers of a valid ()-uniform $\mathcal{B}$-labeling. In other words, $\mathbb{R}$ is the disjoint union of the X_i and each X_i is empty or dense. Then the labeling is k-uniform for all k and the maps $\mathsf{uTh}_{()}(\mathbb{R}, X_1, \dots, X_n) \mapsto \mathsf{uTh}_k(\mathbb{R}, X_1, \dots, X_n)$ are well-defined in the X_i and computable uniformly in k. By recursion, it suffices to compute $\mathsf{uTh}_{(k_1)}(\mathbb{R}, X_1, \dots, X_n)$.

For $\mathsf{uTh}^1_{(k_1)}$ with $k_1 \colon \{1, \dots, m\} \to \{1, \dots, n\}$, as in Example 2.3.4 we show that $\mathsf{uTh}^1_{(k_1)}(\mathbb{R}, X_1, \dots, X_n)$ is the set of all $\mathsf{uTh}_{()}(\mathbb{R}, Y_1, \dots, Y_m)$ with $Y_i \subseteq X_{k_1(i)}$ for all i. Indeed, $\mathsf{uTh}_{()}(\mathbb{R}, X_1, \dots, X_n)$ only contains the information which of the X_i is the unique cocountable one and which ones are empty. Thus the statement follows from the following two facts.

- If $X \subseteq \mathbb{R}$ is dense, then there are two disjoint dense subsets of X.
- If $X \subseteq \mathbb{R}$ is cocountable, then there are two disjoint dense subsets of X one of which is countable.

Thus, for computing $\mathsf{uTh}^2_{(k_1)}$ it suffices to compute ucant. Similarly to the previous case, we show that a type t of a uniformly labeled Cantor set occurs in $(\mathbb{R}, X_1, \dots, X_n)$ whenever there are X'_i with $\mathsf{uTh}_{()}(\mathbb{R}, X'_1, \dots, X'_n) = \mathsf{uTh}_{()}(\mathbb{R}, X_1, \dots, X_n)$ and such that t occurs in $(\mathbb{R}, X'_1, \dots, X'_n)$. A partial version is the following statement.

- If $A_1, \dots, A_m, B_1, \dots, B_n \subseteq \mathbb{R}$ are countable, dense and pairwise disjoint, then there is a Cantor set C such that all A_j are disjoint from C and all B_i are dense in C.

The general statement also considers endpoints of jumps and follows easily from Lemma 2.2.2.

[She75, Theorem 6.2] used a variant of this argument to compute the common unrestricted monadic theory of the rationals and all Specker orders (also called Aronszajn lines).

2.4 Nesting Cantor sets

From now on, we assume that all members of the sufficiently stable Boolean algebra $\mathcal{B}$ have the Baire property. This is a strictly weaker assumption than determinacy.

We define a notion of coarse types, which contain a limited amount of data that nonetheless determines the type of a uniform $\mathcal{B}$-labeling. Essentially, we may forget all data about refinements, urefin, and only need to remember the types of Cantor subsets, ucant.

In this section, we define coarse types and compute which ones are satisfiable. In the coming section 2.5 we show how to compute uniform refinements of a coarse type.

2.4.1 Expanding the language

To eliminate more quantifiers, we now expand the Morleyized monadic language of section 2.3.3 by a predicate for meager sets, thus changing the meaning of types like pTh_k. The results of the previous sections, in particular Theorem 2.3.24, persist in the expanded language, as argued here.

The same proofs apply if we obtain analogues of Theorem 2.3.6 for countable index sets and Lemma 2.3.18. Since the induction steps are handled as before, it suffices to consider the base case $|k| = 0$. We distinguish three cases:

(i) For Theorem 2.3.6, the index set is countable.

(ii) For the first application of Lemma 2.3.18 in Proposition 2.3.17, the union of all summands that are singletons is a Cantor set in $\mathbb{R}$, thus meager.

(iii) For the remaining applications of Lemma 2.3.18 in computing $\mathsf{pTh}_{O(k)}(\mathbb{C}, P)$ and $\mathsf{pTh}_{O(k)}(\mathbb{R}, \mathbb{R}_{\mathbb{C}}(P))$ from each other, all summands are finite.

In case (i) or (ii), a subset of the sum is meager if and only if it is meager in all infinite summands. In case (iii), a subset A is meager if and only if the set of summands I_x with $A \cap I_x \neq \emptyset$ is meager.

2.4.2 Coarse types

Definition 2.4.1. Let I be a finite set of labels and $n \in \mathbb{N}$.

(a) We define $\mathsf{fcTh}_n(I)$ by recursion on n. Set $\mathsf{fcTh}_0(I) = I \times 2^I$, and for $(i, J) \in \mathsf{fcTh}_0(I)$ call i the **comeager label**, elements of J **realized labels**, and elements of $I \setminus J$ **omitted labels**. Set $\mathsf{fcTh}_{n+1}(I) = \mathsf{fcTh}_0(I) \times 2^{\mathsf{fcTh}_n(I \amalg I \times I)}$, with the first component called **underlying 0-type** and elements of the second component called **Cantor n-subtypes**.

(b) We define whether a $\mathcal{B}$-labeling P of $R \cong \mathbb{R}$ or $C \cong \mathbb{C}$ by I **realizes** $t \in \mathsf{fcTh}_n(I)$ by recursion on n, and whether it is **coarsely n-uniform**.

- A $\mathcal{B}$-labeling of R realizes a coarse 0-type (i, J) if
  - the fiber of the comeager label i is indeed comeager;
  - the fibers of the realized labels $j \in J$ are dense; and
  - the fibers of the omitted labels $j' \notin J$ are empty.
- Given the notion of realizing coarse n-types for $\mathcal{B}$-labelings of $\mathbb{R}$, we say that
  - a $\mathcal{B}$-labeling Q of C with m labels realizes a coarse n-type $t \in \mathsf{fcTh}_n(I \amalg I \times I)$ if $\mathbb{R}_C(Q)$ realizes t; and
  - a $\mathcal{B}$-labeling of $R \cong \mathbb{R}$ or $C \cong \mathbb{C}$ is coarsely n-uniform if it realizes some $\mathsf{fcTh}_n(I)$.
- A $\mathcal{B}$-labeling P of R realizes a coarse $(n+1)$-type (t_0, T) if
  - P realizes the underlying 0-type t_0;
  - for each Cantor subtype $t \in T$, every nonempty open subset of R has a Cantor subset that realizes t; and
  - each coarsely n-uniformly labeled Cantor subset of (R, P) realizes some Cantor subtype $t \in T$.

(c) For a coarsely n-uniform $\mathcal{B}$-labeling P of $R \cong \mathbb{R}$ or $C \cong \mathbb{C}$, write $\mathsf{cTh}_n(R, P)$ (respectively $\mathsf{cTh}_n(C, P)$) for the unique element of $\mathsf{fcTh}_n(I)$ (respectively $\mathsf{fcTh}_n(I \amalg I \times I)$) realized by P, and call it the **coarse type** of P.

As for usual types, we use "realizes" and "satisfies" synonymously for coarse types. We sometimes use the terminology for coarse types for arbitrary labelings, calling labels **realized** if their fiber is nonempty and **omitted** else, and referring to types of coarsely n-uniformly labeled Cantor sets of a labeled order X as the **Cantor n-subtypes** of X.

Example 2.4.2. We give an equivalent description of coarse uniformity and coarse n-types that is more directly analogous to uTh_k.

Given a $\mathcal{B}$-labeling $P\colon \mathbb{R} \to I$ and $n \in \mathbb{N}$, we define a preliminary n-theory $T_n(P)$. For $n = 0$, let $T_0(P)$ be the pair of the set of labels with empty fiber and the set of labels with meager fiber. These data are encoded in $\mathsf{uTh}_{()}(\mathbb{R}, P) = \mathsf{Th}_{()}(\mathbb{R}, P)$, and for ()-uniform P they determine $\mathsf{Th}_{()}(\mathbb{R}, P)$ completely. Let $T_{n+1}(P)$ be the set of Cantor n-subtypes of $(\mathbb{R}, P)$. In other words, $T_{n+1}(P)$ is the set of $\mathsf{cTh}_n(\mathbb{R}, \mathbb{R}_C(P))$ as C varies over n-uniformly labeled Cantor subsets.

Then P is coarsely n-uniform if and only if $T_n(P) = T_n(P|_J)$ and $T_0(P) = T_0(P|_J)$ for all nonempty open intervals J. For $n = 0$, although there is no formal equality, the data encoded by $\mathsf{cTh}_0(\mathbb{R}, P)$ and $T_0(P)$ coincide for coarsely 0-uniform P since by the Baire property there is a unique comeager label. For $n > 0$, we have that $\mathsf{cTh}_n(\mathbb{R}, P) = (\mathsf{cTh}_0(\mathbb{R}, P), T_n(P))$ for coarsely n-uniform P.

We will investigate which coarse types are satisfiable (by a $\mathcal{B}$-labeling of $\mathbb{R}$), leading to Proposition 2.4.30.

Here we can already answer the question of satisfiability for 0-types: By a straightforward argument, $(i, J) \in \mathsf{fcTh}_0(I)$ is satisfiable if and only if $i \in J$. For a coarse type with labels in $I \amalg I \times I$ to be satisfiable by a $\mathcal{B}$-labeling of $\mathbb{C}$, we need an additional condition: The labels in $I \times I$ of the jumps must contain at least one realized label, but must not contain the comeager label. For 0-types, it is straightforward that these conditions suffice. Also for general n, we call coarse n-types satisfiable by a $\mathcal{B}$-labeling of $\mathbb{C}$ **Cantor-satisfiable**. Cantor-satisfiable types are also satisfiable by a $\mathcal{B}$-labeling of $\mathbb{R}$.

Lemma 2.4.3. *$O(k)$-uniform $\mathcal{B}$-labelings are coarsely $|k|$-uniform.*

Proof. Since coarse $|k|$-uniformity means that every label or recursively Cantor subtype that occurs in some open subset occurs in every open subset, this follows from the fact that usual types $\mathsf{Th}_{O(k)}(\mathbb{R}, P)$ determine coarse types $\mathsf{cTh}_{|k|}(\mathbb{R}, P)$. □

Lemma 2.4.4. *(a) Open intervals in coarsely n-uniformly labeled sets are coarsely n-uniformly labeled of the same type.*

(b) For all n, every labeled set has a coarsely n-uniformly labeled subinterval.

Proof. (a) is clear and (b) follows from Lemmas 2.3.21 and 2.4.3. □

For the remaining document, we abbreviate "coarsely uniform" as "uniform" and will not use the previous notion of uniformity. Similarly, we will often abbreviate "coarse type" as "type" and reserve the unqualified use of "type" for coarse types.

Corollary 2.4.5. *For $n \geq k$, the map $\mathsf{cTh}_n(\mathbb{R}, P) \mapsto \mathsf{cTh}_k(\mathbb{R}, P)$ is well-defined in P and computable uniformly in n, k. We denote this map also by $t \mapsto t|k$. If the Cantor subtypes of t are $t_1, \ldots, t_m$, then the Cantor subtypes of $t|k$ are $t_1|(k-1), \ldots, t_m|(k-1)$.*

Proof. By recursion, it suffices to consider $\mathsf{cTh}_{n+1}(\mathbb{R}, P) \mapsto \mathsf{cTh}_n(\mathbb{R}, P)$. The case $n = 0$ holds by definition of cTh_1. For the induction step it suffices to show that if the Cantor subtypes of an $(n+1)$-type t are $t_1, \ldots, t_m$, then the Cantor subtypes of $t|n$ are $t_1|(n-1), \ldots, t_m|(n-1)$. Clearly each $t_i|(n-1)$ is realized. The converse follows as by Lemma 2.4.4 each Cantor subtype of $t|n$ also has an n-uniform realization, which realizes some t_i. □

2.4.3 Uniform sums

We first observe that iso-uniform sums make sense for $\mathcal{B}$-labeled orders. More precisely, if $\mathcal{B}$ is a sufficiently stable Boolean algebra and T is a finite set of $\mathcal{B}$-labeled order types, then the iso-uniform sum of T is a $\mathcal{B}$-labeled order type by Definition 0.1.21 (d). The notion of uniform sum defined below describes the coarse type of an iso-uniform sum (this will follow from Proposition 2.4.16), but is more flexible.

Let $n \in \mathbb{N}$ and $T \subseteq \mathsf{fcTh}_n(I \amalg I \times I)$.

Definition 2.4.6. A labeled total order C is **compatible with** T if every Cantor n-subtype of C is realized by a Cantor subset of some realization of some $t \in T$.

In particular, realizations of elements of T are compatible with T, and singleton orders are compatible with all T. Moreover, subsets of sets compatible with T are compatible with T.

Proposition 2.4.7. *Let (X, P) be a labeled total order and $(C_i)_{i \in \mathbb{N}}$ be a sequence of closed subsets of X compatible with T. Then $\bigcup_i C_i$ is compatible with T.*

Proof. Let $D \subseteq \bigcup_i C_i$ be an n-uniformly labeled Cantor subset. Then some C_i is nonmeager in D, thus by closedness has nonempty interior. By uniformity C_i includes a realization of $\mathsf{cTh}_n(D, P)$. Since C_i is compatible with T, we conclude that $\mathsf{cTh}_n(D, P)$ can be embedded in some realization of some $t \in T$. □

Definition 2.4.8. Let P be a $\mathcal{B}$-labeling of $X \cong \mathbb{R}$ or $X \cong \mathbb{C}$. We call (X, P) an n-**uniform sum** of T if

- every nonempty open subset of X includes realizations of all $t \in T$ by meager Cantor sets, and

- there is a meager F_σ set $M \subseteq X$ compatible with T such that $X \setminus M$ has a single realized label.

The definition for $X \cong \mathbb{C}$ is equivalent to $(\mathbb{R}, \mathbb{R}_{\mathbb{C}}(P))$ being an n-uniform sum of T, using for (a) the bijection of meager Cantor subsets of Lemma 2.2.5. Thus, when proving statements about uniform sums (X, P) and coarse types, it usually suffices to consider the case $X = \mathbb{R}$.

Given realizations of all summands, an iso-uniform sum of these realizations is an n-uniform sum. Coarsening an (iso- or n-)uniform sum along a map of labels yields a uniform sum of the set of coarsened types.

By Lemma 2.4.4, whenever C is compatible with a set $\{t_1, \dots, t_m\}$ of n-types, then C is also compatible with $\{t_1|k, \dots, t_m|k\}$ for $k \leq n$. In particular, any n-uniform sum of $\{t_1, \dots, t_m\}$ is a k-uniform sum of $\{t_1|k, \dots, t_m|k\}$.

Lemma 2.4.9. *Let P be a $\mathcal{B}$-labeling of $\mathbb{R}$. Then $(\mathbb{R}, P)$ is an n-uniform sum of T if and only if there are countably many Cantor subsets C_j with pairwise disjoint closures such that every C_j is compatible with T, for all $t \in T$ every nonempty open subset includes a C_j realizing t, and $\mathbb{R} \setminus \bigcup_j C_j$ has a cocountable label.*

Proof. For sufficiency, let M be the union of all $\overline{C_j}$'s and countable labels of $\mathbb{R} \setminus \bigcup_j C_j$. Then M is compatible with T by Proposition 2.4.7.

For necessity, choose for all nonempty bounded basic open U_k and $t \in T$ a $D_{kt} \subseteq U_k$ realizing t. Let $(E_i)_{i \in \mathbb{N}}$ be a sequence of compact, nowhere dense sets that are compatible with T such that $\mathbb{R} \setminus \bigcup_i E_i$ has a cocountable label. Removing isolated points, we may assume $E_i \cong \overline{\mathbb{C}}$. Lemma 2.2.4 (b) for $I = T$ and $\left\{\overline{D_{kt}} \,\middle|\, k, t\right\} \cup \{E_i \,|\, i\}$ yields a sequence of pairwise disjoint copies of $\overline{\mathbb{C}}$ compatible with T such that for all $t \in T$ every nonempty open subset includes such a copy of the form $\overline{D_{kt}}$. Removing endpoints from these copies yields the desired sequence $(C_j)_j$ of Cantor sets. □

Lemma 2.4.10. *If P is $(n+1)$-uniform, then $(\mathbb{R}, P)$ is an n-uniform sum of the set T of Cantor n-subtypes of $(\mathbb{R}, P)$.*

Proof. Every subset is compatible with T. Thus, we may choose M to be any meager F_σ set that includes the complement of the comeager label. By $(n+1)$-uniformity, every element of T occurs in every nonempty open subset. □

Corollary 2.4.11. *Every satisfiable n-type s is realized by an n-uniform sum of some set T of satisfiable n-types. Moreover, for $n > 0$ we may choose T such that $\{t|(n-1) \,|\, t \in T\}$ is the set of Cantor subtypes of s.*

Proof. By Lemma 2.4.4, s has an $(n+1)$-uniform realization P. By Lemma 2.4.10, $(\mathbb{R}, P)$ is an n-uniform sum of Cantor its n-subtypes. Their $(n-1)$-types are the Cantor subtypes of s. □

2.4.4 Computing uniform sums

The aim of this subsection is to prove computability of the coarse type of a uniform sum from the coarse types of its summands (Proposition 2.4.16). Making the computation compatible with our choice to regard labelings $P\colon \mathbb{C} \to I$ as labelings $\mathbb{R}_{\mathbb{C}}(P)\colon \mathbb{R} \to I \amalg I \times I$ requires a fine grained analysis of the endpoints of jumps in nested Cantor sets.

Let $P\colon \mathbb{C} \to I$ be an $(n+1)$-uniform $\mathcal{B}$-labeling and $D \subseteq \mathbb{C}$ be an n-uniformly labeled meager Cantor subset. For π as in Definition 2.1.2, consider the following commutative diagram:

$$\begin{array}{ccccc} \mathbb{C} & \supseteq & D & \xrightarrow{\pi_D} & \mathbb{R} \\ \downarrow{\scriptstyle \pi_{\mathbb{C}}} & & \downarrow{\scriptstyle \pi_{\mathbb{C}}} & & \| \\ \mathbb{R} & \supseteq & \pi_{\mathbb{C}}(D) & \xrightarrow{\pi_{\pi_{\mathbb{C}}(D)}} & \mathbb{R} \end{array}$$

Then D induces two labelings of the reals. One of them, let us call it $Q_1(P, D)$, with labels in $I \amalg I \times I$, is $\mathbb{R}_D(P|_D)$ for the labeling of D with m labels obtained as a subset of $\mathbb{C}$. The other one, let us call it $Q_2(P, D)$, with labels in $(I \amalg I \times I) \amalg (I \amalg I \times I) \times (I \amalg I \times I)$, is $\mathbb{R}_{\pi_{\mathbb{C}}(D)}(\mathbb{R}_{\mathbb{C}}(P)|_{\pi_{\mathbb{C}}(D)})$ for the labeling of $\pi_{\mathbb{C}}(D)$ with labels in $I \amalg I \times I$ obtained as a restriction of $\mathbb{R}_{\mathbb{C}}(P)$ in the lower left corner.

Lemma 2.4.12. *The maps $f\colon \mathsf{cTh}_n(\mathbb{R}, Q_2(P,D)) \mapsto \mathsf{cTh}_n(\mathbb{R}, Q_1(P,D))$ are well-defined in P, D and computable uniformly in n. In fact, $Q_1(P,D)$ is obtained by postcomposing $Q_2(P,D)$ with a map*

$$(I \amalg I \times I) \amalg (I \amalg I \times I) \times (I \amalg I \times I) \to I \amalg I \times I.$$

Proof. We abbreviate $Q_i = Q_i(P,D)$. Recall that $\mathbb{R}_E(S)$ maps x to the ordered 1- or 2-tuple of labels $S(y)$ of elements y of the fiber $\pi_E^{-1}(\{x\})$.

In particular, $Q_1\colon \mathbb{R} \to I \amalg I \times I$ maps x to the ordered 1- or 2-tuple $(P(y))_{y \in \pi_{\mathbb{C}}^{-1}(\{x\})}$ of labels $P(y)$ of elements y of the fiber of x under $\pi_{\mathbb{C}}\colon \mathbb{C} \to \mathbb{R}$. For instance, if $\pi_{\mathbb{C}}^{-1}(\{x\}) = \{y_1, y_2\}$ with $y_1 < y_2$, then $Q_1(x) = (P(y_1), P(y_2))$.

Similarly, $Q_2\colon \mathbb{R} \to (I \amalg I \times I) \amalg (I \amalg I \times I) \times (I \amalg I \times I)$ maps x to the ordered tuple of ordered tuples $\left((P(z))_{z \in \pi_{\mathbb{C}}^{-1}(\{y\})}\right)_{y \in \pi_{\pi_{\mathbb{C}}(D)}^{-1}(\{x\})}$. First, suppose that $\pi_{\mathbb{C}}^{-1}(\{y\}) \subseteq D$, or equivalently $\pi_{\mathbb{C}}^{-1}(\{y\}) = \pi_{\mathbb{C}}|_D^{-1}(\{y\})$. If we equivalently regard Q_2 as a map

$$\mathbb{R} \to (I \amalg I \times I) \amalg (I \amalg I \times I) \times (I \amalg I \times I) \cong I \amalg I \times I \amalg I \times I \amalg I \times I \times I \amalg I \times I \times I \amalg I \times I \times I \times I,$$

then it maps such an x to an ordered tuple $(P(z))_{z \in \pi_D^{-1}(\{x\})}$ of labels of elements of the fiber of x under $\pi_{\pi_{\mathbb{C}}(D)} \circ \pi_{\mathbb{C}}|_D = \pi_D$. For instance, if $\pi_D^{-1}(\{x\}) = \{x_1, x_2\}$ with $x_1 < x_2$, then $Q_2(x) = (P(x_1), P(x_2))$. This already describes the required map of finite sets on the first three summands $I \amalg I \times I \amalg I \times I$. Namely, it is the identity on the first summand I and identifies the second summand $I \times I$ with the third summand $I \times I$.

If $\pi_{\mathbb{C}}^{-1}(\{y\}) \not\subseteq D$, then the fiber of $\pi_{\pi_{\mathbb{C}}(D)} \circ \pi_{\mathbb{C}}$ contains unnecessary labels. However, it is still possible to determine which levels arise from the fiber of π_D. Namely, since D is a Cantor subset of $\mathbb{C}$, the left endpoints of jumps in D are no right endpoints of jumps in $\mathbb{C}$ and vice versa. In other words, the fiber of $\pi_D = \pi_{\pi_{\mathbb{C}}(D)} \circ \pi_{\mathbb{C}}|_D$ consists of the first and the last element of the fiber of $\pi_{\pi_{\mathbb{C}}(D)} \circ \pi_{\mathbb{C}}$. Thus an element of a summand $I \times I \times I$ or $I \times I \times I \times I$ is mapped to the pair in $I \times I$ of its first and its last entry. □

Lemma 2.4.13. *The maps*

$$\mathsf{ucsub}\colon \mathsf{cTh}_{n+1}(\mathbb{C}, P) \mapsto \{\mathsf{cTh}_n(D, P) \mid D \subseteq \mathbb{C} \text{ an } n\text{-uniformly labeled Cantor subset}\}$$

are well-defined in P and computable uniformly in n. In fact, $\mathsf{ucsub}(t)$ is the union of the image under f of Lemma 2.4.12 of the Cantor n-subtypes of t with $\{t|n\}$.

Proof. If D is nonmeager, then by uniformity $\mathsf{cTh}_n(D,P) = \mathsf{cTh}_n(\mathbb{C},P)$. For meager D, the type $\mathsf{cTh}_n(D,P)$ is $\mathsf{cTh}_n(\mathbb{R}, Q_1(P,D))$ in the above discussion. Since $D \mapsto \pi_{\mathbb{C}}(D)$ in Lemma 2.2.5 is a well-defined surjection, the Cantor subtypes of $\mathsf{cTh}_{n+1}(\mathbb{C},P)$ are precisely the $\mathsf{cTh}_n(\mathbb{R}, Q_2(P,D))$ as D varies. □

Definition 2.4.14. A 0-type t_0 is **consistent with** a set T of n-types if all realized labels of elements of T are realized labels of t_0.

Lemma 2.4.15. *Let $(\mathbb{R}, P)$ be an $(n+1)$-uniform sum of T and let S denote the set of Cantor n-subtypes of $\mathsf{cTh}_{n+1}(\mathbb{R}, P)$. Then $t \in S$ if and only if at least one of the following conditions holds:*

(a) $t \in \bigcup_{s \in T} \mathsf{ucsub}(s)$.

(b) *t is the n-type of an n-uniform sum of some $T' \subseteq \bigcup_{s \in T} \mathsf{ucsub}(s)$. Moreover, $t|0$ is Cantor-satisfiable and consistent with T', has the same comeager label as P, and all components of realized labels of $t|0$ are realized by $(\mathbb{R}, P)$.*

Proof. Let $(C_j)_j$ be as in Lemma 2.4.9. We first show that every $t \in S$ satisfies (a) or (b).

First, suppose that some C_j is nonmeager in a realization of t. Thus C_j has nonempty interior and, by uniformity, C_j includes a realization of t. Since C_j is compatible with T, (a) follows.

Otherwise, let D be a realization of t such that all C_j are meager in D. We show (b). Since $\bigcup_j C_j$ is meager in D, D has the same comeager label as $\mathbb{R}$ and it remains to show that (D,P) is a uniform sum of some T'. We set T' as the set of $t' \in \bigcup_{s \in T} \mathsf{ucsub}(s)$ that are realized in D. By Lemma 2.4.4 we may assume D to be $(n+1)$-uniformly labeled, whence each element of T' is realized in every nonempty open subset. It remains to show that D includes countably many closed nowhere dense E_j compatible with T' such that

$D \setminus \bigcup_j E_j$ has a cocountable label. Set $E_j = \overline{C_j \cap D}$. Then $D \setminus \bigcup_j E_j \subseteq \mathbb{R} \setminus \bigcup_j C_j$ has a cocountable label. For showing that E_j is compatible with T', let F be an n-uniformly labeled Cantor subset of E_j. Then F is a subset of C_j, thus by compatibility of C_j a subset of a realization of some $s \in T$. Since $\mathsf{cTh}_n(F, P)$ is realized in $\overline{D} \supseteq E_j$ and thus also in D, we conclude that $\mathsf{cTh}_n(F, P) \in T'$.

Since all t satisfying (a) clearly lie in S, it remains to show that all t satisfying (b) lie in S. For all $t' \in T'$ or realized labels c of $t|0$ or components c' of realized labels of jumps, and nonempty basic open $U_m \subseteq \mathbb{R}$, choose realizations $E_{t'm} \subseteq U_m$ of t' and points $x_{cm} \in U_m, y_{c'm} \in U_m$ of labels c, c' such that all $E_{t'm}, \{x_{cm}\}, \{y_{c'm}\}$ are included in pairwise distinct $C_{j_{t'm}}, C_{j_{cm}}, C_{j_{c'm}}$. Set $B = \bigcup_m B_m$ with $B_m = \bigcup_{t' \in T'} \overline{E_{t'm}} \cup \bigcup_c \{x_{cm}\}$ and $D_{c'} = \{y_{c'm} \mid m\}$. Let A be the union of $\bigcup_j C_j \setminus (B \cup \bigcup_{c'} D_{c'})$ and all countable labels of $\mathbb{R} \setminus \bigcup_j C_j$. Then A is F_σ as $\bigcup_j C_j \setminus (B \cup \bigcup_{c'} D_{c'})$ is the union of all $C_{j_{t'm}} \setminus \overline{E_{t'm}}, C_{j_{cm}} \setminus \{x_{cm}\}, C_{j_{c'm}} \setminus \{y_{c'm}\}$ and all C_j including no $E_{t'm}, \{x_{cm}\}$ or $\{y_{c'm}\}$. Let C be the Cantor set obtained from Lemma 2.2.2 with these $A, B, D_{c'}$ and E given by the realized labels of jumps. Then C has the given labels of jumps by construction; since C contains an x_{cm} in each nonempty open subset, C realizes all realized labels of non-jumps; since $\bigcup_j C_j \subseteq A \cup B$ is meager in C, it has the same comeager label; and since C is disjoint from A and meets $\bigcup_{c'} D_{c'}$ only in jumps, C omits all omitted labels of non-jumps. Thus, C realizes $t|0$. Furthermore, C includes an $E_{t'm}$ in each nonempty open subset, and since C is disjoint from A, the restriction $C \setminus \bigcup_{t'm} E_{t'm}$ has a cocountable label, while each $C \cap E_{t'm}$, being a subset of $E_{t'm}$, is compatible with T'. Thus, (C, P) is a uniform sum of T'. □

Proposition 2.4.16. *Let $(\mathbb{R}, P)$ be an n-uniform sum of T. Then the maps $(\mathsf{cTh}_0(\mathbb{R}, P), T) \mapsto \mathsf{cTh}_n(\mathbb{R}, P)$ are well-defined in T and n-uniform sums $(\mathbb{R}, P)$ and computable uniformly in n.*

Proof. We proceed by recursion on n, and $n = 0$ is vacuous. If $(\mathbb{R}, P)$ is an $(n+1)$-uniform sum of T, then $\bigcup_{t \in T} \mathsf{ucsub}(t)$ is computable by Lemma 2.4.13. Using Lemma 2.4.15, we may recursively compute the set of Cantor subtypes. □

Well-definedness in Proposition 2.4.16 yields uniqueness in the following the definition.

Definition 2.4.17. For a 0-type t_0 and a set T of n-types, the n-uniform sum of T with underlying 0-type t_0 is the unique n-type that restricts to t_0 and is realized by an n-uniform sum of T.

However, the conditions of Definition 2.4.8 are not expressible in terms of cTh_n. Thus, sometimes $(\mathbb{R}, P)$ is no n-uniform sum of T, but $\mathsf{cTh}_n(\mathbb{R}, P)$ is an n-uniform sum of T.

Proposition 2.4.18. *Let T be a set of Cantor-satisfiable n-types and t_0 be a 0-type consistent with T. Then the n-uniform sum of T with 0-type t_0 is satisfiable.*

Proof. The iso-uniform sum of realizations of the summands and the isomorphism classes of points of all realized labels is an n-uniform sum with the given 0-type t_0. □

2.4.5 Computing coarse types

The aim of this subsection is to prove decidability for coarse types (Corollary 2.4.31). For this purpose, we prove that each coarse type is realized by an iterated iso-uniform sum of types that have a single realized label (Proposition 2.4.30).

Lemma 2.4.19. *If $(\mathbb{C}, P)$ is an n-uniform sum of T and $T' \subseteq T$, then $\mathbb{C}$ includes a meager Cantor set with the same 0-type that is an n-uniform sum of T'.*

Proof. Replacing it by a subinterval, by Lemma 2.4.4 we may assume P to be $(n+1)$-uniform. Corollary 2.4.11 yields that $(\mathbb{C}, P)$ is an $(n+1)$-uniform sum of some set T'' with $T \subseteq \{t|n \mid t \in T''\}$. Now Lemma 2.4.15 (b) yields an n-uniform sum of $T' \subseteq \bigcup_{s \in T''} \mathsf{ucsub}(s)$ with 0-type $\mathsf{cTh}_0(\mathbb{C}, P)$. □

Corollary 2.4.20. *The set $\mathsf{ucsub}(t)$ is the image of the Cantor subtypes of t under f of Lemma 2.4.12.*

Proof. By Lemma 2.4.13 it suffices to show that $t|n$ lies in this image, that is, is realized by a meager Cantor set. Apply Lemma 2.4.19 with $T' = T$. □

Definition 2.4.21. For coarse n-types s, t, write $s \leq t$ if s and t have the same comeager label, the realized labels of s form a subset of the realized labels of t and the Cantor subtypes of s form a subset of the Cantor subtypes of t. An $(n+k)$-type s is k-**minimal** if s is least with respect to this order among the satisfiable t above $s|n$, that is, with $t|n = s|n$. We never regard an n-type for $n < k$ as k-minimal.

For $n > 0$, in $s \leq t$ the condition that the realized labels form a subset follows from the condition that the Cantor subtypes form a subset. By Corollary 2.4.20, $s \leq t$ implies $\mathsf{ucsub}(s) \subseteq \mathsf{ucsub}(t)$.

Lemma 2.4.22. *The $(n+1)$-uniform sum of a set T of Cantor-satisfiable $(n+1)$-types with a consistent 0-type is the least satisfiable $(n+1)$-type with the given comeager labels and whose Cantor subtypes include $\bigcup_{t\in T} \mathsf{ucsub}(t)$.*

Proof. The sum is satisfiable by Proposition 2.4.18.

Let s be a uniform sum of T and t be another such satisfiable $(n+1)$-type. Corollary 2.4.11 yields that t is an $(n+1)$-uniform sum of a set containing realizations of all elements of $\bigcup_{t'\in T} \mathsf{ucsub}(t')$. Then $s \leq t$ follows as the description in Lemma 2.4.15 shows that the set of Cantor subtypes of a uniform sum of T is $\subseteq$-isotone in $\bigcup_{t'\in T} \mathsf{ucsub}(t')$. □

Lemma 2.4.23. *The $(n+1)$-uniform sum of a set T of 1-minimal Cantor-satisfiable $(n+1)$-types with a consistent 0-type is the least satisfiable $(n+1)$-type with the given comeager label and whose Cantor subtypes include $\{t|n \mid t \in T\}$.*

Proof. This follows from Lemma 2.4.22 since for an 1-minimal $(n+1)$-type t, an $(n+1)$-type s includes $\mathsf{ucsub}(t)$ if and only if s contains $t|n$. Indeed, if s contains $t|n$, then each realization of s includes an $(n+1)$-uniform realization t' of $t|n$, and so includes $\mathsf{ucsub}(t') \supseteq \mathsf{ucsub}(t)$. □

Definition 2.4.24. (a) An n-type **construction** of **rank** at most $m \in \mathbb{N}$ is defined recursively as a pair of a satisfiable 0-type t and a set $T = \{t_1, \ldots, t_k\}$ of n-type constructions of rank less than m (for $m = 0$, this permits only $T = \emptyset$). We denote the construction by $\sum^t T$ or $\sum_i^t t_i$.

(b) The value of an n-type construction $\sum^t T$ is the n-type s with $s|0 = t$ that is a uniform sum of the values of the elements of T. For this definition we recursively require that the values of the elements of T are Cantor-satisfiable and 1-minimal and that t is consistent with T. If $\sum^t T$ has value t', we also say that $\sum^t T$ is a construction of t'.

(c) A type t is **constructible** of rank at most m if t has a construction of rank at most m. The type t is constructible if t is constructible of some rank. The rank of t, denoted by $\mathrm{rk}\, t$, is the minimal rank of a construction of t.

For example, each satisfiable 0-type t is constructible. Indeed, $\sum^t \emptyset$ is a construction of t.

Definition 2.4.25. The **canonical realization** of a construction $\sum^t T$ is defined recursively as the iso-uniform sum of the canonical realizations of all $s \in T$ and the isomorphism classes of points of all realized labels of t with the comeager label of t. A labeling P of $\mathbb{C}$ is a canonical realization of a construction if $\mathbb{R}_{\mathbb{C}}(P)$ is a canonical realization of this construction.

Different constructions of a type may lead to non-isomorphic canonical realizations.

Proposition 2.4.26. *Constructible types are satisfiable.*

Proof. This follows by induction on a fixed construction using Proposition 2.4.18. Explicitly, the canonical realization is a realization. □

Proposition 2.4.27. *For a label c of a constructible n-type t, the following conditions are equivalent:*

(i) c is meager and if $n > 0$, then c is meager in all Cantor subtypes of t.

(ii) c is meager and if $n > 0$, then, recursively on n, c satisfies (ii) in all Cantor subtypes of t.

(iii) c is countable in some realization of t.

(iv) If $n = 0$, then c is countable in the canonical realization of $\sum^t \emptyset$. If $n > 0$, then c is countable in the canonical realizations of all constructions.

Proof. If (i) fails, then for each realization, c is comeager in some Cantor set, thus uncountable, and so (iii) fails. Since clearly (iv) $\implies$ (iii), it suffices to show (i) $\implies$ (ii) $\implies$ (iv).

We show $\neg$(ii) $\implies$ $\neg$(i) by induction on n. The case $n = 0$ holds by definition. If c is not pseudo-countable in some Cantor subtype s, then by the induction hypothesis s contains a Cantor subtype s' with c comeager in s'. Then there is an $(n-1)$-uniform realization of s', which witnesses that c is comeager in some Cantor subtype of t.

For (ii) $\implies$ (iv), we use induction on the given construction. Since c is countable in all summands by (ii) (using $n > 0$) and the induction hypothesis, c is countable in all C_i in the definition of iso-uniform sums. It is countable in $\mathbb{R} \setminus \bigcup_i C_i$ as c is not the comeager label. □

A label is **pseudo-countable** if the conditions of Proposition 2.4.27 hold. Assuming determinacy for $\mathcal{B}$, a pseudo-countable label is countable in all realizations by the perfect set property. This fails for $\mathcal{B} = \sigma(\mathbf{\Sigma}^1_1)$ in the constructible universe.

Proposition 2.4.28. *A constructible type is Cantor-satisfiable if and only if there is a realized label of jumps and all labels of jumps are pseudo-countable.*

Proof. Necessity follows as all labels of jumps are countable in all realizations. For sufficiency consider a realization of t in $\mathbb{R}$ with all labels of jumps countable. Replacing each point of such a label by two points of the corresponding labels yields a Cantor set realizing t. □

In particular, Cantor-satisfiability of a constructible n-type t for $n > 0$ only depends on $t|1$.

Proposition 2.4.29. *(a) All $(n+1)$-uniform sums of 2-minimal Cantor-satisfiable constructible $(n+1)$-types are 1-minimal.*

(b) For every constructible n-type t, there is a unique 1-minimal $(n+1)$-type t' above t. Moreover, t' is constructible, $t \mapsto t'$ is computable, and t' is Cantor-satisfiable if and only if t is Cantor-satisfiable.

(c) If a $\mathcal{B}$-labeling of $\mathbb{R}$ has a Cantor subset with constructible n-type t, then it has a Cantor subset realizing t' of (b).

Proof. We use simultaneous induction on n.

(c) for $n-1$ implies (a) for n and (a) holds for $n = 0$. Indeed, let $t_1, \dots, t_m$ be 2-minimal $(n+1)$-types (note that $m = 0$ if $n = 0$), and let s be a satisfiable $(n+1)$-type above the uniform sum of $t_1|n, \dots, t_m|n$. We need to show that s includes the uniform sum of $t_1, \dots, t_m$. By Lemma 2.4.23 it suffices to show that s contains $t_1|n, \dots, t_m|n$. This follows from (c) as $s|n$, being the uniform sum of $t_1|n, \dots, t_m|n$, contains $t_1|(n-1), \dots, t_m|(n-1)$.

(a) for n implies (b) for n. Indeed, let t be constructible, say an n-uniform sum of 1-minimal Cantor-satisfiable constructible types $s_1, \dots, s_m$ of smaller rank. By induction on the rank, there are 2-minimal Cantor-satisfiable constructible $s'_1, \dots, s'_m$ above $s_1, \dots, s_m$. Their uniform sum t' is 1-minimal by (a), is constructible and lies above t. Uniqueness holds as $\leq$ is a partial order on coarse types. By induction on the rank, s'_i has the same pseudo-countable labels as s_i. Since a label in a uniform sum is pseudo-countable if and only if it is pseudo-countable in all summands and meager, t' has the same pseudo-countable labels as t. The claim about Cantor-satisfiability follows from Proposition 2.4.28.

(b) for n implies (c) for n. We use induction on $\operatorname{rk} t$. Let t be an n-uniform sum of 1-minimal types $s_1, \dots, s_m$ of smaller rank and P be a realization. Then P contains $s_1|(n-1), \dots, s_m|(n-1)$. By the induction hypothesis on n, P contains $s_1, \dots, s_m$ (note that this also works for $n = 0$ as then $m = 0$). By the induction hypothesis on the rank, P contains the 2-minimal $(n+1)$-types $s'_1, \dots, s'_m$ above $s_1, \dots, s_m$. Replacing it by a subinterval, we may assume that P is $(n+2)$-uniform. Lemma 2.4.10 yields that P is an $(n+1)$-uniform sum of $s'_1, \dots, s'_m$ and some other types and thus by Lemma 2.4.19 includes an $(n+1)$-uniform sum t' of $s'_1, \dots, s'_m$. This t' is the 1-minimal $(n+1)$-type above t as defined in the proof of (b). □

The proof shows that the 1-minimal type t' above t has the syntactically same construction as t (just regarded as $(n+1)$-type construction, not as n-type construction). In particular, they have equal canonical realizations.

Proposition 2.4.30. *A coarse type is satisfiable if and only if it is constructible. In fact, each satisfiable type is the uniform sum of the* 1*-minimal types above its Cantor subtypes.*

Proof. Sufficiency follows from Proposition 2.4.26. For necessity, let t be a satisfiable $(n+1)$-type. Then by Lemma 2.4.23 the uniform sum s of the 1-minimal $(n+1)$-types above the Cantor n-subtypes of t is the least satisfiable type with $s \geq t$. Thus $t = s$, which is constructible as the Cantor subtypes of t are constructible by induction. □

Corollary 2.4.31. *The set of satisfiable coarse* n*-types is computable.*

2.5 Uniform refinements

From now on, we assume that all members of $\mathcal{B}_{\overline{\mathbb{C}}}$ are determined. The aim of this section is to show the following result.

Proposition 2.5.1. *(a) The maps* $\mathsf{cTh}_{m+1}(\mathbb{R}, P) \mapsto \mathsf{urefincant}^{k^-}_{k_n}(\mathsf{cTh}_m)(\mathbb{R}, P)$ *are well-defined in* P *and computable uniformly in* k *and* m *when* k^- *is large enough in terms of* m*.*

(b) There is p*, computable from* k, m*, such that the maps* $\mathsf{cTh}_p(\mathbb{R}, P) \mapsto \mathsf{urefin}^{k^-}_{k_n}(\mathsf{cTh}_m)(\mathbb{R}, P)$ *are well-defined in* P *and computable uniformly in* k *and* m *when* k^- *is large enough in terms of* m*.*

We first deduce Theorem 0.1.22 from Proposition 2.5.1. For (a), by Theorem 2.3.24 it suffices to compute uTh, that is, compute recursively $\mathsf{urefincant}$ and urefin. Since increasing k^- does not change their value by Lemma 2.3.21, we may apply Proposition 2.5.1, using computability of cTh by Corollary 2.4.31.

It remains to show (b). Since our construction does not depend on the Boolean algebra, all sufficiently stable Boolean algebras share the same theory. In fact, if P is a labeling for the smaller Boolean algebra, then all refinements constructed in the definition of E in the proof of Theorem 2.3.24 and iso-uniform sums required to canonically realize the cTh_p, as well as the refinements constructed in section 2.5.4, can be chosen in the smaller Boolean algebra, and the labeling restricted to a Cantor subset lies in the smaller Boolean algebra.

Remark 2.5.2. Since S1S is reducible to the Borel monadic theory of order, any decision procedure requires iterated exponential time [Mey75]. The reduction of Th to uTh in Theorem 2.3.24 and the computation of cTh evidently only require this running time. However, the map of Proposition 2.5.1 (b) sending (k, m) to p requires us to compute higher levels of cTh, thus increasing the running time of our procedure (which remains primitive recursive).

The author does not expect this increase to be necessary. It even seems plausible that deciding the fragment of the theory expressed by uTh is possible in less than iterated exponential time and only the procedure of Theorem 2.3.24 is computationally expensive.

2.5.1 Refined Cantor subsets

We first discuss a general approach to both parts of Proposition 2.5.1. To obtain $t \in \mathsf{urefincant}^f_k(\mathsf{cTh}_n)(\mathbb{R}, P)$, respectively $t \in \mathsf{urefin}^f_k(\mathsf{cTh}_n)(\mathbb{R}, P)$, there are a few necessary conditions. First, t must be satisfiable for urefin and Cantor-satisfiable for $\mathsf{urefincant}$. Second, observe that the map of coarsening types along f, namely $\mathsf{cTh}_n(\mathbb{R}, Q) \mapsto \mathsf{cTh}_n(\mathbb{R}, f \circ Q)$, is well-defined and computable by an easy recursion on the definition of cTh_n. Then, coarsening the type t along f must yield $\mathsf{cTh}_{n-1}(\mathbb{R}, P)$ for urefin and a Cantor $(n-1)$-subtype for $\mathsf{urefincant}$.

One might try to show that these conditions suffice. While this succeeds for Proposition 2.5.1 (a), as we argue now, it fails for Proposition 2.5.1 (b), and we will later phrase the obstructions as winning strategies of Separator in separation games.

Lemma 2.5.3. *Let* S *be a construction of a coarse type* t*. Every Cantor set realizing* t *has a Cantor subset that is a canonical realization of* S*.*

Proof. We use induction on S. Write $S = \sum^{t|0} \mathcal{S}$. Let $(\mathbb{C}, P)$ be a realization of S.

The complement of the comeager label is included in a meager F_σ set M. Increasing M, by the induction hypothesis we may assume that for every nonempty open U, the restriction $M \cap U$ includes canonical realizations of all $S' \in \mathcal{S}$. By Lemma 2.2.4 (b), write $M = \bigcup_j C_j$ with pairwise disjoint closed C_j such that for each nonempty open U and $S' \in \mathcal{S}$, some $C_j \cap U$ includes a canonical realization of S'.

We now proceed as in Lemma 2.4.15. For all $S' \in \mathcal{S}$ or realized labels c of $t|0$ or components c' of realized labels of jumps, and nonempty basic open $U_m \subseteq \mathbb{R}$, choose canonical realizations $E_{S'm} \subseteq U_m$ of S' and points $x_{cm} \in U_m, y_{c'm} \in U_m$ of labels c, c' such that all $\overline{E_{S'm}}, \{x_{cm}\}, \{y_{c'm}\}$ are included in pairwise distinct $C_{j_{S'm}}, C_{j_{cm}}, C_{j_{c'm}}$. Set $B = \bigcup_m B_m$ with $B_m = \bigcup_{S' \in \mathcal{S}} \overline{E_{S'm}} \cup \bigcup_c \{x_{cm}\}$ and $D_{c'} = \{y_{c'm} \mid m\}$. Let A be the union of $\bigcup_j C_j \setminus (B \cup \bigcup_{c'} D_{c'})$ and all countable labels of $\mathbb{R} \setminus \bigcup_j C_j$. Then A is F_σ as $\bigcup_j C_j \setminus (B \cup \bigcup_{c'} D_{c'})$ is the union of all $C_{j_{S'm}} \setminus \overline{E_{S'm}}, C_{j_{cm}} \setminus \{x_{cm}\}, C_{j_{c'm}} \setminus \{y_{c'm}\}$ and all C_j including no $E_{S'm}, \{x_{cm}\}$ or $\{y_{c'm}\}$. Let C be the Cantor set obtained from Lemma 2.2.2 with these $A, B, D_{c'}$ and E given by the realized labels of jumps. Then C includes some $E_{S'm}, \{x_{cm}\}, \{y_{c'm}\}$ in each nonempty open subset. Let $(D_i)_i$ be an enumeration of the $E_{S'm}$ and the remaining points of labels distinct from c_0. Then the D_i witness that (C, P) is an iso-uniform sum of the canonical realizations of all $S' \in \mathcal{S}$ and the isomorphism classes of points of the realized labels, hence a canonical realization of S. □

Lemma 2.5.4. *Let $(\mathbb{R}, Q)$ be an n-uniform $\mathcal{B}$-refinement of $(\mathbb{R}, P)$. There is a construction S of $\mathsf{cTh}_n(\mathbb{R}, P)$ such that the canonical realization of S has an n-uniform $\mathcal{B}$-refinement realizing $\mathsf{cTh}_n(\mathbb{R}, Q)$.*

Proof. Choose a construction S' of $\mathsf{cTh}_n(\mathbb{R}, Q)$. Coarsening S' yields a construction S of $\mathsf{cTh}_n(\mathbb{R}, P)$. Then the canonical realization of S' is a refinement of the canonical realization of S and realizes $\mathsf{cTh}_n(\mathbb{R}, Q)$. □

Proposition 2.5.5. *Let $(\mathbb{C}, Q)$ be an n-uniform $\mathcal{B}$-refinement of $(\mathbb{C}, P)$ such that $\mathsf{cTh}_n(\mathbb{C}, P)$ is a Cantor subtype of an $(n+1)$-type t. Then every realization of t has a Cantor subset that has a $\mathcal{B}$-refinement realizing $\mathsf{cTh}_n(\mathbb{C}, Q)$.*

Proof. By Lemma 2.5.3 it suffices to consider the canonical realization $(\mathbb{C}, P')$ of the construction of t with summands all constructions of minimal types above Cantor subtypes. In particular, $(\mathbb{C}, P')$ contains a canonical realization of each construction of $\mathsf{cTh}_n(\mathbb{C}, P)$. Thus Lemma 2.5.4 finishes the proof. □

Proposition 2.5.5 yields Proposition 2.5.1 (a) as indicated above. Namely, the possible uniform $\mathcal{B}$-refinements of Cantor subsets of $(\mathbb{R}, P)$ are the satisfiable coarse types whose coarsening is a Cantor subtype of $(\mathbb{R}, P)$.

2.5.2 Separation games

We prove Proposition 2.5.1 (b) using games, by induction on the number of labels. In the game we describe below, Separator aims to show the existence of a partition into sets of low complexity in the difference hierarchy [Kec95, 22.E] of F_σ sets using fewer labels. Pathfinder aims to show that, locally on a Cantor set, all labels have high complexity in the difference hierarchy. We formalize this using the building blocks $\left\{y \in k^{\mathbb{N}} \,\middle|\, \limsup y = i\right\}$ for Wadge complete sets $\left\{y \in k^{\mathbb{N}} \,\middle|\, \limsup y \equiv i \mod 2\right\}$ for the first ω levels of the difference hierarchy.

Definition 2.5.6. Consider a labeling $P\colon X \to \{0, \dots, m-1\}$ of $X \subseteq \mathbb{R}$ and $k \in \mathbb{N}_{>0}$. We define the **separation game** of (P, k). There are two players, Pathfinder and Separator.

Alternatingly, Pathfinder plays either a digit in $\{0, \dots, 9\}$ or the radix point, and Separator plays an element of $\{0, \dots, k-1\}$. A play determines $x \in \mathbb{R}$ whose decimal representation is coded by Pathfinder and $y \in k^{\mathbb{N}}$ played by Separator. Pathfinder wins the play if $x \in X$ and $P(x) \equiv \limsup y \mod m$.

If $X \in \mathcal{B}_{\mathbb{R}}$ and P is a $\mathcal{B}$-labeling, then the separation game is determined.

Remark 2.5.7. The separation game for $m = 2$ is essentially a Wadge game for the sets $P^{-1}(\{0\})$ and $\left\{y \in k^{\mathbb{N}} \,\middle|\, \limsup y \equiv 0 \mod 2\right\}$. There are two differences:

- Instead of $x \in \mathbb{N}^{\mathbb{N}}$, Pathfinder produces $x \in \mathbb{R}$. This difference is insignificant: For our argument, after minor changes we could restrict play to $\mathbb{R} \setminus C \cong \mathbb{N}^{\mathbb{N}}$ for a countable subset C of the comeager label.

- Separator is not allowed to play waiting moves. In other words, our game is a Lipschitz game, not a Wadge game. However, playing $0 \in \{0, \dots, k-1\}$ is at least as good as a waiting move, whence the Lipschitz and the Wadge game are equivalent, or more formally, won by the same player.

Taking $m > 2$ corresponds to generalizing Wadge degrees to partitions ([Sel17]; for a more general account, see [KM19]). This observation could also be used to obtain an alternative proof of the following Lemma 2.5.8.

Lemma 2.5.8. *The following conditions are equivalent for a labeling P and $k \in \mathbb{N}$:*

(i) Separator has a winning strategy in the separation game of (P, k).

(ii) There is a partition $X = \bigcup_{i=0}^{k-1} X_i$ such that each $\bigcup_{i=0}^{j} X_i$ is F_σ in X and no label in $P(X_i)$ is congruent to i modulo m.

Proof. Suppose that Separator wins. Declare $x \in X_i$ for the minimal i such that there is a play that is compatible with the strategy and consists of a representation of x and some y with $\limsup y = i$. Since the strategy is winning, no label in $P(X_i)$ is congruent to i modulo m. Finally, $\bigcup_{i=0}^{j} X_i$ is the set of $x \in X$ such that for one of the at most two representations of x, if Pathfinder plays this representation, then the y played by Separator satisfies the F_σ condition $\limsup y \leq j$. Thus $\bigcup_{i=0}^{j} X_i$ is F_σ.

Conversely, consider a partition $X = \bigcup_{i=0}^{k-1} X_i$ as in (ii). For each $j \in \{0, \dots, k-1\}$, write $\bigcup_{i=0}^{j-1} X_i = \bigcup_{n \in \mathbb{N}} C_{jn} \cap X$ with C_{jn} closed. Given a position a, let $p_j(a) \in \mathbb{N} \cup \{\infty\}$ be the minimal n such that a has an extension in C_{jn}. Then Separator plays the maximal j such that p_j was increased by the immediately preceding move of Pathfinder. Let $x \in \mathbb{R}$ and $y \in k^{\mathbb{N}}$ form a play compatible with this strategy. Set $j = \limsup y$. Since p_j is increased infinitely often, $x \notin \bigcup_n C_{jn} \supseteq \bigcup_{i=0}^{j-1} X_i$. Since p_{j+1} is increased only finitely often, x has arbitrarily long prefixes with extensions in $C_{j+1,n}$, where $n = p_{j+1}(a)$ for almost all prefixes a of x. Since $C_{j+1,n}$ is closed, $x \in C_{j+1,n} \subseteq (\mathbb{R} \setminus X) \cup \bigcup_{i=0}^{j} X_i$. Hence, $x \notin X$ or $x \in X_j$, whence Separator wins. □

In particular, X_j is the intersection of the F_σ set $\bigcup_{i=0}^{j} X_i$ and the G_δ set $\bigcup_{i=j}^{k-1} X_i$. Using Proposition 0.1.24, for each k, the winner of the separation game is expressible in restricted monadic logic by a formula $\varphi_k(P)$.

Example 2.5.9. There are sufficiently stable Boolean algebras whose members have the Baire property and such that the separation game is not determined. For instance, let X be an uncountable set without a perfect subset. Then a separation game for $P\colon \mathbb{R} \to \{0, 1\}$ with $P_1 = X$ is undetermined. Indeed, a winning strategy of Separator for some k would imply that X is a Boolean combination of F_σ sets, while a winning strategy of Pathfinder for $k \geq 2$ would yield a perfect subset of X. This is the same argument as for classical Wadge games [Wad83, Theorem II.C.2].

2.5.3 Separable types

We now phrase consequences of winning strategies in terms of coarse types, which will be employed in Corollary 2.5.14.

Definition 2.5.10. (a) We define by recursion on k if an element of $\mathsf{fcTh}_n(\{0, \dots, m-1\})$ is **separable** of **index** k. A type t is separable of index at most k whenever $k-1$ is not congruent to the comeager label modulo m and t is a uniform sum with summands that are separable of index at most k (as base case, this permits only empty sets of summands). For 1-types and $k = 1$, we additionally demand that label 0 is omitted.

(b) We define by recursion on k if an element of $\mathsf{fcTh}_n(\{0, \dots, m-1\})$ is **anti-separable** of **index** $k \leq n$. A type is anti-separable of index at least k whenever $k-1$ is congruent to the comeager label modulo m and some Cantor $(n-1)$-subtype is anti-separable of index $k-1$. For $k = 1$, we demand that label 0 is realized instead of the condition on $k-1$.

Let $P\colon \mathbb{R} \to I$ be a $\mathcal{B}$-labeling.

If an n-type t is separable of some index, then so is $t|\ell$ for all $\ell \leq n$. Conversely, if $t|k$ is separable of index k, then so is t. An n-type t is anti-separable of index k if and only if $t|k$ is so. Thus, we call $(\mathbb{R}, P)$ (anti-)separable of index k if $\mathsf{cTh}_k(\mathbb{R}, P)$ is (anti-)separable of index k.

Relating separable types with winning strategies is straightforward:

Lemma 2.5.11. *If Separator wins the separation game of (P, k), then $(\mathbb{R}, P)$ is separable of index at most k.*

Proof. Lemma 2.5.8 yields a partition into an F_σ set $X_{<k-1}$ and a G_δ set X_{k-1} such that no label in $P(X_{k-1})$ is congruent to $k-1$ and Separator wins the separation game of $(P|_{X<k-1}, k-1)$. By induction on k, each Cantor subset of $X_{<k-1}$ is separable of index less than k. Since the label congruent to $k-1$ does not occur, each Cantor subset of X_{k-1} is separable of index at most k, and the comeager label is not congruent to $k-1$. □

To relate anti-separable types with winning strategies, we iterate the following lemma.

Lemma 2.5.12. *Let $i \in \{0, \dots, m-1\}$ be a label such that Pathfinder wins the separation game of $(P|_X, m \cdot k + i + 1)$ and $n \in \mathbb{N}$. Then there are an n-uniformly labeled $C \subseteq X$ and a relatively comeager G_δ subset Y of C such that C is an interval or Cantor set, Y has label i, and Pathfinder wins the separation game of $(P|_{C\setminus Y}, m \cdot k + i)$.*

Proof. For $m = 1$, Pathfinder wins on every nonempty X and the lemma is easy. Thus assume $m > 1$. The winning strategy of Pathfinder corresponds to a continuous map $\sigma\colon (m \cdot k + i + 1)^{\mathbb{N}} \to X$.

Let C be a subinterval of the image of σ that is n-uniformly labeled by the labeling that expands P by the subset $\sigma(\{y \,|\, \limsup y < m \cdot k + i\})$. Replacing a finite initial segment of σ, we may arrange that $C = \mathrm{im}(\sigma)$. Set

$$Y = C \setminus \sigma(\{y \,|\, \limsup y < m \cdot k + i\}).$$

Since the image of the F_σ subset $\{y \,|\, \limsup y < m \cdot k + i\}$ of a compact space is F_σ, its complement Y is G_δ. Since σ is winning, $Y \subseteq \sigma(\{y \,|\, \limsup y = m \cdot k + i\})$ has label i. Since $\sigma((m \cdot k + i)^{\mathbb{N}}) \subseteq C \setminus Y$, the restriction of σ is a winning strategy for the separation game of $(P|_{C\setminus Y}, m \cdot k + i)$.

We show that C is a closed interval or Cantor set. First, $C = \mathrm{im}(\sigma)$ is a nonempty compact subset of $\mathbb{R}$. We show that there is no isolated point x. Otherwise $\sigma^{-1}(\{x\})$ is nonempty open, thus meets the dense sets $\{y \,|\, \limsup y = \ell\}$ for $\ell \in \{0, 1\}$, which contradicts that σ is winning and $m > 1$.

It remains to show that Y is comeager. Since Y is G_δ, it suffices to show that Y is dense. By 0-uniformity, it suffices to show that Y is nonempty. Suppose otherwise. On the one hand, σ also witnesses that Pathfinder wins the separation game of $(P|_{\sigma((m\cdot k+i)^{\mathbb{N}})}, m \cdot k + i)$. On the other hand, Lemma 2.5.8 with

$$X_0 = \emptyset \text{ and } X_\ell = \sigma(\{y \,|\, \limsup y = \ell - 1\}) \setminus \sigma(\{y \,|\, \limsup y < \ell - 1\}) \text{ for } \ell > 0$$

shows that Separator wins this game. This is absurd. □

Lemma 2.5.13. *If Pathfinder wins the separation game of (P, k), then $(\mathbb{R}, P)$ is anti-separable of index at most k.*

Proof. First, replace $\mathbb{R}$ by the set C of Lemma 2.5.12. Since Y is comeager of the label congruent to $k-1$, it suffices to obtain a subset that is anti-separable of index $k-1$. Pathfinder wins the separation game of $(P|_{C\setminus Y}, k-1)$, thus also for some closed subset of $C \setminus Y$. We finish by induction on k. □

Corollary 2.5.14. *(a) Separator wins the separation game of (P, k) if and only if $(\mathbb{R}, P)$ is separable of index at most k.*

(b) Pathfinder wins the separation game of (P, k) if and only if $(\mathbb{R}, P)$ is anti-separable of index at least k.

Proof. Lemmas 2.5.11 and 2.5.13 establish necessity. Since by induction no type is both separable and anti-separable of the same index, sufficiency follows from determinacy. □

In particular, the winner of the separation game can be computed from a coarse type. This enables the case distinction in the proof of Proposition 2.5.1 (b) below.

2.5.4 Refinements

We now show Proposition 2.5.1 (b) for refinements of a $\mathcal{B}$-labeling $P\colon \mathbb{R} \to J$ along a fixed $f\colon I \to J$ by induction on the number of labels $\#J$.

Case 1: Separator wins the separation game of (P, k) for some k

Let $t \in \mathsf{fcTh}_n(\{0, \dots, m-1\})$. Let $k \in \mathbb{N}$ and let $i \in \{0, \dots, m-1\}$ be the label with $i \equiv k-1 \mod m$.

Definition 2.5.15. Let t be separable of index k.

(a) The **lower summands** of t are the Cantor $(n-1)$-subtypes of t that are separable of index less than k.

(b) The **upper part** of t is the type in $\mathsf{fcTh}_n(\{0, \dots, m-1\} \setminus \{i\})$ that is obtained from t by removing all Cantor subtypes with realized label i.

Lemma 2.5.16. *Let t be separable of index k. Each Cantor subtype of t without realized label i is a lower summand of t.*

Proof. Since no Cantor subset has comeager label i, this follows by recursion on a construction. □

Lemma 2.5.17. *Let t be separable of index k. If $(\mathbb{R}, P)$ realizes t, then there is a comeager G_δ set X such that the following conditions hold:*

(a) The labeling that equals P on X and is constantly equal to the comeager label of P outside X realizes the upper part of t; and

(b) each $(n-1)$-uniform Cantor subset of $\mathbb{R} \setminus X$ realizes a lower summand of t and each nonempty open subset of $\mathbb{R}$ includes Cantor subsets of $\mathbb{R} \setminus X$ realizing all lower summands of t.

Proof. Lemma 2.5.8 yields a G_δ set X such that label i does not occur in X and Separator wins the separation game of $(P|_{\mathbb{R}\setminus X}, k-1)$. Hence all types of Cantor subsets of $\mathbb{R} \setminus X$ are lower summands. Since some Cantor subtype of index k is not realized outside X, but each nonempty open subset includes a realization, the G_δ set X is dense, thus comeager.

Removing some Cantor sets from X, we may arrange that each lower summand is realized in every nonempty open subset by a Cantor subset of $\mathbb{R} \setminus X$, that is, (b) holds.

Let s be a Cantor subtype without realized label i. By Lemma 2.5.16, s is realized in every nonempty open subset by a Cantor subset of $\mathbb{R} \setminus X$. Write $\mathbb{R} \setminus X$ as a countable disjoint union of closed sets C_j. For all s and basic open U_m, choose realizations $E_{sm} \subseteq U_m$ of s that are subsets of pairwise distinct C_j. Replacing X by $X \cup \bigcup_{sm} E_{sm}$, which remains G_δ as each $C_j \cap \bigcup_{sm} E_{sm}$ is locally closed, we can arrange that all s occur in X in every nonempty open subset. This guarantees (a). □

Thus, X and the Cantor subsets of $\mathbb{R} \setminus X$ can be refined independently.

Corollary 2.5.18. *Let t be a separable n-type of index k. If $(\mathbb{R}, P)$ realizes t, then the possible refinements of $\mathsf{cTh}_n(\mathbb{R}, P)$ along f are obtained in the following way. Take any possible refinement r_0 of the upper part and for any lower summand s take a nonempty set R_s of possible refinements. Then a possible refinement is obtained by adding to a uniform sum decomposition of r_0 all elements of $\bigcup_s R_s$ as summands.*

Proof. Let X be as in Lemma 2.5.17. Write $\mathbb{R} \setminus X$ as a countable disjoint union of Cantor sets and points C_j. Since we will define the refinement below independently on each C_j, we may replace P on each C_j by a P' with $\mathsf{Th}_\ell(C_j, P) = \mathsf{Th}_\ell(C_j, P')$ for ℓ large enough to express the possible refinements. Thus, decomposing them further, by Lemmas 2.3.25 and 2.4.3 we may assume that all C_j are n-uniformly labeled.

Each Cantor set C_j realizes some lower summand of t. We modify the C_j such that every nonempty open subset includes C_j's realizing all lower summands s of t. For all s and basic open U_m, choose Cantor sets $E_{sm} \subseteq U_m$ realizing s that are subsets of pairwise distinct C_j. Decompose each C_j that includes an E_{sm} into this E_{sm} and countably many relatively open intervals and points. By uniformity of the C_j, all resulting Cantor sets realize lower summands.

Finally, refine X as indicated by r_0 and the C_j as indicated by the R_s such that each element of R_s is realized in every nonempty open subset. Since X and the C_j witness a uniform sum decomposition before refining, they do so afterwards. □

This enables a computation of the set of possible refinements by recursion. We first describe base cases. The refinements of types in a single label, that is, in $\mathsf{cTh}_n(\{\star\})$, are described by Corollary 2.4.31. Next, types that are separable of index 1 can be reduced to types in fewer labels.

The set of possible refinements of the upper part can be computed by recursion on the number of labels. The sets of possible refinements of lower summands can be computed by recursion on the index k of the separable type t. Corollary 2.5.18 finishes the computation of the set of possible refinements.

Remark 2.5.19. Instead of the partition of Lemma 2.5.17, we could have chosen other ways of distributing Cantor subsets between X and $\mathbb{R} \setminus X$. Our induction requires to place Cantor subsets containing label i outside X and Cantor subsets that are not separable of index less than k inside X. In our construction, Cantor subtypes that do not contain label i, and thus in particular are separable of index less than k by Lemma 2.5.16, are realized both outside and inside X. We could instead demand that they are realized only inside X unless they embed into other separable Cantor subsets. It is not clear to the author whether we could alternatively demand that they are realized only outside X.

Case 2: Pathfinder wins the separation game of (P, k) for large enough k

It suffices to show that in this situation every satisfiable refinement of $\mathsf{cTh}_n(\mathbb{R}, P)$ is realized by a refinement of $(\mathbb{R}, P)$, as discussed in the beginning of section 2.5.1.

The following strengthening of anti-separability can be obtained at the cost of increasing the quantifier rank.

Definition 2.5.20. A coarse n-type is **maximal** if it is maximal with respect to the partial order $\leq$ of Definition 2.4.21.

Equivalently, a 0-type is maximal if all labels are realized, and an $(n+1)$-type is maximal if all satisfiable types of Cantor sets are realized.

Lemma 2.5.21. *Let $m, n \in \mathbb{N}$ and $i \in \{0, \dots, m-1\}$. If a type $t \in \mathsf{fcTh}_{m\cdot(n+1)+i+1}(\{0, \dots, m-1\})$ is anti-separable of index at least $m \cdot (n+1) + i + 1$, then t has a Cantor subtype s of comeager label i such that $s|n$ is maximal.*

Proof. We proceed by induction on n. Passing to a Cantor subtype, we may assume that t is anti-separable of index exactly $m \cdot (n+1) + i + 1$. In this case, by Lemma 2.4.19 it suffices to show that $t|n$ is maximal.

By anti-separability, the comeager label is i. For $n = 0$, all labels are comeager in same Cantor subtype, thus realized, and maximality follows. Suppose $n > 0$. For all labels $j \in \{0, \dots, m-1\}$, by the induction hypothesis for $t|(m \cdot n + j + 1)$ there are Cantor subtypes t_j of comeager label j such that $t_j|(n-1)$ is maximal. □

We say that a type $t \in \mathsf{fcTh}_n(I)$ is **maximal in the labels** $I' \subseteq I$ if only labels in I' are realized in t, and the thus obtained type in $\mathsf{fcTh}_n(I')$ is maximal.

Lemma 2.5.22. *Let $(\mathbb{R}, Q)$ be an n-uniform $\mathcal{B}$-refinement of $(\mathbb{R}, P)$ along $f\colon I \to J$. Suppose that there is $I' \subseteq I$ with $f(I') = f(I)$ such that all $(n-1)$-types that are maximal in the labels I' are Cantor subtypes of $\mathsf{cTh}_n(\mathbb{R}, Q)$. Then every realization of $\mathsf{cTh}_n(\mathbb{R}, P)$ has an $(n-1)$-uniform $\mathcal{B}$-refinement realizing $\mathsf{cTh}_{n-1}(\mathbb{R}, Q)$.*

Proof. Let $P'\colon \mathbb{R} \to J$ realize $\mathsf{cTh}_n(\mathbb{R}, P)$. In the next paragraph, we proceed as in the second paragraph of Lemma 2.5.3.

The complement of the comeager label under P' is included in a meager F_σ set M. Increasing M, by Lemma 2.5.3 we may assume that for every nonempty open U, the restriction $M \cap U$ includes for each construction S of a Cantor subtype of $\mathsf{cTh}_n(\mathbb{R}, P)$ a canonical realization of S. By Lemma 2.2.4 (b), write $M = \bigcup_j C_j$ with pairwise disjoint closed C_j such that for each nonempty open U and S, some $C_j \cap U$ includes a canonical realization of S.

Using Lemma 2.5.4, refine these canonical realizations to obtain every Cantor subtype of $\mathsf{cTh}_{n-1}(\mathbb{R}, Q)$. Refine $\mathbb{R} \setminus \bigcup_j C_j$ by the comeager label of $(\mathbb{R}, Q)$. It remains to refine the remaining C_j in a compatible way. For this, let $g\colon f(I) \to I'$ be such that $f \circ g = \mathrm{id}$ and coarsen the remaining C_j along g. The resulting refined Cantor sets are compatible with the set of Cantor subtypes of $\mathsf{cTh}_n(\mathbb{R}, Q)$ since $\mathsf{cTh}_n(\mathbb{R}, Q)$ contains all maximal $(n-1)$-types in its realized labels. □

The assumptions of Lemma 2.5.22 are satisfied by the following lemma, whose assumptions follow from Lemma 2.5.21 for anti-separable types of sufficiently large index.

Lemma 2.5.23. *Suppose that a type t' is a refinement of t along a map $f\colon I \to J$. There is k, computable from f and n, such that whenever t has a maximal Cantor k-subtype, then there is $I' \subseteq I$ with $f(I') = J$ such that all n-types that are maximal in the labels I' are Cantor subtypes of t'.*

Proof. We use induction on $\#I$. If t' realizes a Cantor set omitting some $i \in I$ whose coarsening is a maximal k_1-type for k_1 large enough, then the induction hypothesis for $I \setminus \{i\}$ applies.

Otherwise, we show that we may choose $I' = I$ for large enough k. By induction on n, there is k_2 such that every Cantor subtype whose coarsening has a maximal k_2-type contains a maximal $(n-1)$-type. Set $k = k_1 + k_2$.

We show that an arbitrary maximal n-type, say with comeager label i, is realized. Write t' as a nested uniform sum of types with k_2-maximal coarsenings. Let t'_1 be the type that results from replacing each of these summands by its underlying 0-type without label i. Let t_1 be the coarsening of t'_1. Since t contains a k-maximal type, t_1 contains a k_1-maximal type. By choice of k_1, t'_1 contains label i. Thus some Cantor subtype of t' has comeager label i and a Cantor subtype t'_2 with k_2-maximal coarsening. Finally, t'_2 contains each $(n-1)$-maximal type, whence t' contains the given maximal n-type. □

Chapter 3

The Borel monadic theory of the binary tree

3.1 Introduction

While Shelah [She75, p. 393] deemed a proof of decidability of S2S using his methods "very interesting", the automata-theoretic approach towards the Borel monadic theory of the binary tree seems more promising. Here, we define prioritized Borel games, show that their finite memory determinacy would yield decidability, and provide partial results. The introduction of the thesis contains a more detailed discussion of the relations with monadic logic. Here, we give a direct motivation of the game model.

Finite memory determinacy has become a valuable tool in automata theory, starting with the proof of decidability of S2S from a variant of the Büchi-Landweber theorem [BL69; GH82]. We aim to extend S2S using automata operating on Borel labeled trees. This would require a corresponding generalization of the Büchi-Landweber theorem. Of course, finite memory determinacy requires some finiteness restrictions on the games played. The countably infinite information of Borel sets becomes accessible to the players by another possibly infinite piece of information, their position in the arena.

More specifically, in a game there is some "public" information, the position in the arena, and some "private" information, for example, a string of priorities that has been produced so far. After an infinite play, using a payoff set on all public and private information of the game, a winner is determined. Finite memory determinacy then is the existence of a winning strategy whose moves may depend on the entire public information, but only on a bounded finite fragment of the private information. There are two extreme cases:

The first extreme case is if there is no private information. This is the pure study of determinacy in set theory. If the payoff set is Borel, then determinacy holds in ZFC [Mar75], whence this will be no obstruction to our purposes. For notational simplicity, we restrict ourselves to Borel sets throughout, but larger classes of sets are shortly discussed in section 3.7.

The second extreme case is if the public information is finite. In this case, it suffices to consider payoff sets depending only on the private information. If not even determining the winner is possible with finite memory, then winning the game is hopeless. Thus, we should assume that, fixing the public part of a play, the payoff set is ω-regular in the private information. We conclude that the case of finite public information is solved by the Büchi-Landweber theorem.

Generalizations to infinite arenas have been studied, for example giving finite-memory determinacy if the payoff set is an ω-regular set in the private information. The necessary determinacy for S2S is [GH82, Theorem 1] - for graphs see also [Zie98]. For us, however, the payoff set also depends on the public information in an infinitary way. This motivates defining the payoff set as a Boolean combination of Borel conditions on the public information and ω-regular conditions on the private information, see Remark 3.2.5.

3.2 Games for Borel monadic logic

3.2.1 Prioritized Borel games

We define a common generalization of Gale-Stewart games and games on graphs with a built-in notion of finite memory. It is more general than we would need; the general case motivates later restrictions.

Definition 3.2.1. A **game** is played on a set Q of positions, with a set P of public positions and a private alphabet Σ. There are two players, Eve, often denoted by $\exists$, and Adam, often denoted by $\forall$. It consists of the following data:

- an initial position $q_0 \in Q$,
- a map $c\colon Q \to \{\exists, \forall\}$ that determines the current player in a position; we write $Q_\exists = c^{-1}(\{\exists\}), Q_\forall = c^{-1}(\{\exists\})$.
- a transition relation $\delta \subseteq Q \times \Sigma \times Q$; we usually write $p \xrightarrow{a} q$ for $(p, a, q) \in \delta$,
- a map $Q \to P, q \mapsto \langle q \rangle$ sending a position to its underlying public position,
- a payoff set $W \subseteq Q^{\mathbb{N}} \times \Sigma^{\mathbb{N}}$.

A **strategy** σ of a player p is a partial map $Q_p \times \Sigma^{<\omega} \to Q$ that respects the transition relation in the sense that $q \xrightarrow{a} \sigma(q, xa)$. A strategy σ requires only **finite memory** if there is a morphism $f\colon (Q \times \Sigma)^{<\omega} \to M$ to a finite monoid such that σ factors through $(Q_p \times \Sigma)^{<\omega} \to P^{<\omega} \times M, (q, x) \mapsto (p(q), f(q, x))$.

We regard two games as **equivalent** if the same player wins with the same amount of memory. While this is too crude for many purposes and our equivalences tend to yield stronger conclusions, we are only interested in finite memory determinacy.

Remark 3.2.2. This definition is slightly redundant to simplify some theorem statements. Every game is equivalent to a game with $\Sigma = \{\star\}$ by replacing Q by $Q \times \Sigma$. Alternatively, every game is equivalent to a game with $Q = P$ and $\langle q \rangle = q$ by replacing Σ by $Q \times \Sigma \times Q$.

It is straightforward to define winning plays and strategies. For definiteness, we declare that a player loses immediately if there is no possible move - thus the domain of a winning strategy σ contains every position compatible with σ. Finite memory strategies could also be defined using a finite set of memory states instead of a morphism to a finite monoid (this is analogous to the fact that a language is recognized by a finite automaton if and only if it is recognized by a finite monoid), and we will mostly consider such states in the following. Some details are omitted; the reader should think of the following examples.

Example 3.2.3. (a) A Gale-Stewart game on an alphabet A is a game in our sense by setting $Q = P = A^{<\omega}, \langle q \rangle = q, \Sigma = \{\star\}, q_0 = ()$ with $x \in Q_\exists$ if $|x|$ is even, $x \xrightarrow{\star} y$ if y is obtained by adding a single letter to x, and as payoff set the preimage under the usual payoff set under $(A^{<\omega})^{\mathbb{N}} \to A^{\mathbb{N}}, x \mapsto (n \mapsto x(n+1)(n))$.

(b) A game on a finite graph (V, E) is a game in our sense with $Q = V, P = \{\star\}$ and $\Sigma = E$. In our notation, the Büchi-Landweber theorem states that if the payoff set is ω-regular, then some player has a finite memory winning strategy.

In the following, we will exclusively consider the situation where the public part of the play moves downward in a tree, that is, $P \subseteq A^{<\omega}$ and $x \xrightarrow{a} y$ implies $p(x) \preceq p(y)$. Moreover, the payoff set will always only depend on the limiting branch in $A^{\leq\omega}$ and not the sequence of positions in $(A^{<\omega})^{\mathbb{N}}$, similarly to Example 3.2.3 (a). Such games are called **tree games** in the following.

Definition 3.2.4. A **prioritized Borel game** is a tree game such that the private alphabet Σ is a finite total order, whose elements are called priorities, and there is a Borel $W \subseteq P^{\mathbb{N}} \times \Sigma$ (using the product topology on the discrete topologies for Q and Σ) such that the payoff set is the preimage of W under $Q^{\mathbb{N}} \times \Sigma^{\mathbb{N}} \to P^{\mathbb{N}} \times \Sigma, (q, p) \mapsto (\langle q \rangle, \limsup p)$. We often write $W(p) = \left\{x \in P^{\mathbb{N}} \,\middle|\, (x, p) \in W\right\}$.

Remark 3.2.5. In the introduction, we suggested a more general definition, namely, that the payoff set is a Boolean combination of Borel subsets of $P^{\mathbb{N}}$ and ω-regular subsets of $Q^{\mathbb{N}} \times \Sigma^{\mathbb{N}}$. Similar to the reduction of the Büchi-Landweber theorem to parity languages, we can transform each such game into a prioritized Borel game without affecting finite memory determinacy.

Proof. By Remark 3.2.2, assume $\langle q \rangle = q$ for all q. Fix a deterministic parity automaton recognizing all given languages on the private alphabet. After taking a product of the game and this automaton, the Boolean combination for the payoff set is of the form required for prioritized Borel games. □

By Borel determinacy, every prioritized Borel game is determined.

3.2.2 State choices

While states $p, q \in Q$ cannot be meaningfully compared if their nodes $\langle p \rangle, \langle q \rangle$ differ in general tree games, in the games associated with automata they can be compared by their underlying state of the automaton. The following definition abstracts this comparison.

Definition 3.2.6. (a) A **state choice** of a game is a pair of a set S and a map $s\colon Q \to S$ such that the map $Q \to P \times S, q \mapsto (\langle q \rangle, s(q))$ is injective.

(b) A state choice s of a tree game on $P \subseteq A^{<\omega}$ is **uniform** if the current player and the transition relation of q only depend on $s(q)$ in the sense that there are $A \subseteq S$ with $Q_\exists = s^{-1}(A)$ and $B \subseteq S \times A^{<\omega} \times \Sigma \times S$ such that $p \xrightarrow{a} q$ if and only if there is x with $(s(p), x, a, s(q)) \in B$ and $\langle q \rangle = \langle p \rangle x$.

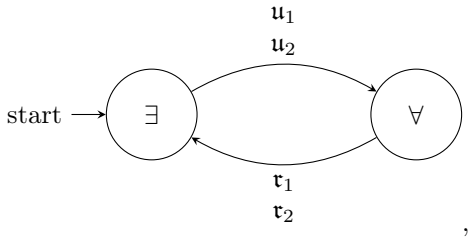
Example 3.2.7. (a) $S = Q$ with $s = \mathrm{id}$ always defines a uniform state choice.

(b) For a Gale-Stewart game, $S = \mathbb{Z}/2\mathbb{Z}$ with $s(q) = |q| + 2\mathbb{Z}$ defines a uniform state choice.

(c) In the associated game of a tree automaton (see section 3.3.1), there is a canonical uniform state choice with S the states of the automaton.

We will often represent prioritized Borel games with a uniform state choice by diagrams; then nodes represent states in S, the current player of a state is provided as a label, and the transition relation is indicated by arrows of the form $s \xrightarrow{x;a} t$ for $(s, x, a, t) \in B$. In our diagrams, the private alphabet Σ has elements $\mathfrak{r}_1 < \cdots < \mathfrak{r}_m$, while the public alphabet A has elements $\mathfrak{u}_1, \ldots, \mathfrak{u}_n$. We will use some abbreviations: Instead of two arrows $s \xrightarrow{x;a} t$ and $s \xrightarrow{y;b} t$ we often use a single arrow $s \xrightarrow[y;b]{x;a} t$, instead of $(); \mathfrak{r}_i$ we write $\mathfrak{r}_i$ and instead of $\mathfrak{u}_i; \mathfrak{r}_1$ we write $\mathfrak{u}_i$.

Example 3.2.8. In contrast to games on finite graphs, reducing the private part of the payoff set to a parity language does not enable memory-less determinacy. Consider the game



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where Eve wins a play in which $\mathfrak{u}_2$ occurs infinitely often if and only if $\mathfrak{r}_2$ occurs infinitely often. Then Eve wins by copying Adam's moves, but she requires memory.

3.2.3 Finiteness conditions

We define some finiteness conditions satisfied by associated games of Borel automata.

Definition 3.2.9. A tree game is **non-jumping** if $x \xrightarrow{a} y$ implies $|\langle y \rangle| \leq |\langle x \rangle| + 1$.

Definition 3.2.10. $n \in \mathbb{N}$ is a **bound** on a game if for all $x \in P$ there are at most n distinct $q \in Q$ with $\langle q \rangle = x$. A game is **bounded** if there is a bound.

Proposition 3.6.2 can be easily transformed into an example of an unbounded prioritized Borel game that requires infinite memory. This example still satisfies the following weaker property:

Definition 3.2.11. A game is **Kőnig** if for all $x \in P$ there are only finitely many $q \in Q$ with $\langle q \rangle = x$.

Clearly, bounded games are Kőnig. We isolated Kőnig games as they enable applications of Kőnig's lemma. We illustrate this in the following result, although the result persists for non-Kőnig games by Proposition 3.4.1.

Proposition 3.2.12. *A non-jumping Kőnig prioritized Borel game with $\Sigma = \{\star\}$ is memory-less determined.*

Proof. Let σ_0 be an arbitrary winning strategy of Eve. Let σ be the memory-less strategy that guesses a history h compatible with σ_0 in a given position of Eve and plays $\sigma_0(h)$. We show that σ is winning.

Let h_∞ be a play compatible with σ. For each $n \in \mathbb{N}$ let h_n be the history chosen by σ in position $h_\infty|_n$. Let T be the tree of h' such that there is an n with $h' \preceq h_n$. Since the game is non-jumping Kőnig, T is finitely branching. By Kőnig's lemma, T has an infinite branch. This branch yields a play h'_∞ consistent with σ_0, which thus is won by Eve. Since $\langle h'_\infty \rangle = \langle h_\infty \rangle$ and $\Sigma = \{\star\}$, also h_∞ is won by Eve. □

We can now state our main conjecture.

Conjecture 3.2.13. *Every bounded prioritized Borel game is finite memory determined. There is an upper bound on the memory that computably depends only on the cardinality of the private alphabet and the bound.*

Definition 3.2.14. A tree game with a uniform state choice and a finite public alphabet A is **regularly jumping** if for all states $s, t \in S$ and priorities $a \in \Sigma$ the language $\left\{ x \in A^{<\omega} \;\middle|\; s \xrightarrow{x;a} t \right\}$ is regular.

In particular, non-jumping prioritized Borel games are regularly jumping.

Definition 3.2.15. The **branching** of a tree game on $P \subseteq A^{<\omega}$ is the cardinality of A.

In particular, a tree game is **countably branching** if A is countable.

Example 3.2.16. Nullary branching tree games are equivalent to usual parity games on graphs. Namely, $\emptyset^{<\omega} = \{()\}$ is a singleton, whence there is no public information. The states and transition relation yield a game graph, and the payoff set is thus determined by a winning set $W \subseteq \Sigma$ of priorites. A nullary branching tree game is bounded if and only if the corresponding graph is finite, and always regularly jumping.

It is also straightforward to reduce any unary branching tree game to an equivalent nullary branching one. Thus, we will be mostly interested in at least binary branching.

3.3 Borel automata

Decidability of S2S is usually proved using automata operating on labelings $2^{<\omega} \to \Sigma$ of the infinite binary tree. We extend this to automata operating on Borel labelings $2^{\leq\omega} \to \Sigma$.

For context, we discuss [Rab69, Theorem 2.6]. We already mentioned that [Rab69, Theorem 2.9] states that the F_σ monadic theory of order of the reals is decidable. It is a corollary of [Rab69, Theorem 2.6], which gives an interpretation of the F_σ monadic theory of $(2^{\leq\omega}, \preceq, \leq)$ in S2S. This interpretation represents a subset of $2^{<\omega}$ by itself and an F_σ subset $X \subseteq 2^{\mathbb{N}}$ by a non-canonical set $A \subseteq 2^{<\omega}$ such that X is the set of branches that contain only finitely many elements of A. Thus, the core of the argument is the following observation.

Fact 3.3.1. For every F_σ subset $X \subseteq 2^{\mathbb{N}}$ there is $A \subseteq 2^{<\omega}$ such that X is the set of branches that contain only finitely many elements of A.

By Theorem 0.1.19, the Borel monadic type of a Borel subset of $2^{\mathbb{N}}$ can be approximated by Boolean combinations of F_σ sets. This suggests such an approximation for Borel subsets of $2^{\leq\omega}$. Since F_σ sets can be represented by sets of nodes by our discussion, in this case sets of nodes suffice to approximate Borel subsets of branches. This motivates our automata model with very limited interaction between nodes and branches.

3.3.1 Automata

We recall the well-known definitions of non-deterministic and alternating tree automata, mostly to fix notation. To obtain Borel automata, we will only change the winning conditions of the underlying games.

Definition 3.3.2. A **tree automaton** on a finite alphabet Σ consists of the following data:

(a) A finite set Q of states.

(b) An initial state $q_0 \in Q$.

(c) A transition relation $\delta \subseteq Q \times \Sigma \times Q \times Q$. We often write $p \xrightarrow{a} (q, r)$ for $(p, a, q, r) \in \delta$.

For an automaton and a tree $T\colon 2^{<\omega} \to \Sigma$, we define the **associated game**. Positions are pairs of nodes $v \in 2^{<\omega}$ and states $q \in Q$. The play proceeds in rounds between two players, Automaton and Pathfinder. In each round (v, p), first Automaton chooses q, r with $p \xrightarrow{T(v)} (q, r)$. Then Pathfinder chooses to move to either $(v0, q)$ or $(v1, r)$.

A **run** is a strategy for the associated game, and a run is **accepting** if the strategy is winning. While runs are often defined without mentioning the associated game explicitly, we chose this presentation to emphasize the analogy with alternating automata.

We will specify several possible winning conditions for these games. Each such winning condition yields a notion of language recognized by an automaton. Namely, an automaton **accepts** a tree if and only if Automaton has a winning strategy in the associated game. The set of trees accepted by an automaton is called the language **recognized** by this automaton. A language is called **regular** if it is accepted by some automaton.

We now define **alternating tree automata** by changing the data in Definition 3.3.2. They consist of the same data as tree automata together with a map $p\colon Q \to \{\text{Automaton}, \text{Pathfinder}\}$ sending a state to the current player, and with the transition relation instead a subset of $Q \times \Sigma \times \{(), 0, 1\} \times Q$.

The associated game of an alternating tree automaton has the same set $2^{<\omega} \times Q$ of positions. In position (v, p), the current player chooses i, q with $p \xrightarrow{T(v)} (i, q)$ and moves to (vi, q).

In particular, every tree automaton with states Q yields an alternating tree automaton with states $Q \amalg Q \times Q$. In position (v, p), first Automaton chooses q, r and moves to (v, q, r). In position (v, q, r), Pathfinder moves to $(v0, q)$ or $(v1, r)$. The converse procedure, dealternating automata, is the core of decidability proofs.

3.3.2 Winning conditions

For comparison, we first recall the winning conditions of usual parity tree automata. A parity tree automaton is a tree automaton such that its set Q of states is totally ordered with a set W of accepting states. If $q \in Q^{\mathbb{N}}$ is the sequence of states occurring during a play, then Automaton wins if $\limsup q \in W$. This definition also works for alternating tree automata and it is immediate that regarding a non-deterministic automaton as alternating preserves the winner of its games.

For a Borel automaton, we also demand that Q is totally ordered. However, we replace $W \subseteq Q$ by a subset $W \subseteq Q \times \Sigma$, and declare that Automaton wins a play on a tree $T\colon 2^{\leq\omega} \to \Sigma$ with states $q \in Q^{\mathbb{N}}$ and on branch $b \in 2^{\mathbb{N}}$ if $(\limsup q, T(b)) \in W$.

Again, this also works for alternating automata and regarding a non-deterministic automaton as alternating preserves the winner of its games. Using the general procedure mentioned before, this finishes our definition of acceptance for Borel automata and regular languages of trees $2^{\leq\omega} \to \Sigma$.

3.3.3 Operations on automata

We prove the standard closure properties of regular languages for $2^{\leq\omega}$, but the complementation procedure requires Conjecture 3.2.13.

Proposition 3.3.3. *Finite unions and intersections of regular languages are regular.*

Proof. The empty set and the set of all trees are recognized by automata with a single state and a trivial acceptance condition. For binary unions and intersections we use product automata. Namely, if A is recognized by an automaton with states P and B is recognized by an automaton with states Q, then $A \cup B$ and $A \cap B$ are recognized by automata with states $P \times Q$ with componentwise transition relation. This will only yield a Muller automaton, which can be transformed into a parity automaton as usually using last appearance records. □

While languages recognized by alternating automata are also stable under finite unions and intersections by an analogous argument, the following argument only works for non-deterministic automata.

Proposition 3.3.4. *Projections of regular languages are regular.*

Proof. If a language on an alphabet $\Sigma \times T$ is recognized by an automaton $\mathcal{A}$, then its projection on Σ is recognized by an automaton with the same states and transitions $p \xrightarrow{a} (q, r)$ whenever there is some b with $p \xrightarrow{(a,b)} (q, r)$ in $\mathcal{A}$. For Borel automata, we additionally replace the payoff set $W \subseteq Q \times \Sigma \times T$ by its projection in $Q \times \Sigma$. □

For alternating automata, in contrast, we obtain complementation.

Proposition 3.3.5. *Languages recognized by alternating automata are stable under complementation.*

Proof. Switching the players in the alternating automaton yields a language for the complement by Borel determinacy. □

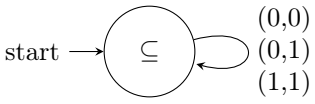
Proposition 3.3.6. *The relations $\subseteq, \preceq, \leq$ are regular.*

To make the statement of the proposition precise, we represent binary relations on subsets by the alphabet $\Sigma = \{0, 1\} \times \{0, 1\}$ and encode first-order relations as in Remark 0.1.1. Thus, a tree $T\colon \{0, 1\}^{\leq \omega} \to \{0, 1\} \times \{0, 1\}$

- is an element of $\subseteq$ if and only if $T(x) \neq (1, 0)$ for all x,
- is an element of $\preceq$ if and only if $T(x) = (1, a), T(y) = (b, 1)$ implies $x \preceq y$,
- is an element of $\leq$ if and only if $T(x) = (1, a), T(y) = (b, 1)$ implies $x \leq y$.

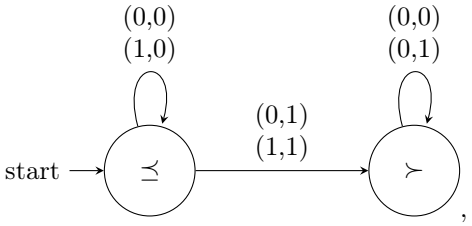
Proof. We draw automata recognizing these relations and specify the acceptance conditions. If several letters occur on a transition, this means that a copy of this transition occurs for each letter. While transitions of the form $p \xrightarrow{a} (q, q)$ are denoted by a simple arrow $p \to q$, transitions of the from $p \xrightarrow{a} (q, r)$ are split at some point, with the part $p \to q$ labeled 0 and the part $p \to r$ labeled 1.

For $\subseteq$, we use the automaton



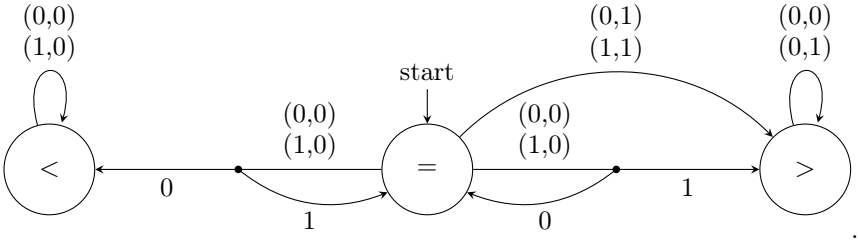
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and a play on a branch is accepting if the branch has label $(0, 0), (0, 1)$ or $(1, 1)$. This automaton works on each node individually. For the prefix relation, we use the automaton



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and a play that remains in state $\preceq$ is always accepting, while a play reaching state $\succ$ accepts only branches with labels $(0, 0), (0, 1)$. This automaton deterministically searches for a letter in the second set followed by a letter in the first set. For the lexicographic order, we use the automaton



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A play that remains in state $=$ is always accepting, while a play reaching state $<$ accepts only labels $(0,0),(1,0)$ and a play reaching state $>$ accepts only labels $(0,0),(0,1)$. This automaton proves $a \leq b$ for $a \in A, b \in B$ by guessing an element x such that $a \leq x \leq b$ for all $a \in A, b \in B$ and remaining in state $=$ until reaching x. □

Remark 3.3.7. If our only aim is to prove decidability of the Borel monadic theory of $(2^{\leq\omega}, \preceq, \leq)$, then the lexicographic order $\leq$ is irrelevant. More formally, removing it from the signature does not affect the Turing degree (although it is undefinable as the structure $(2^{\leq\omega}, \preceq)$ has automorphisms given by swapping two subtrees of a node), similar to the situation in Proposition 0.1.23. Namely, an isomorphic copy of the lexicographic order can be recovered from any subset containing exactly one child of every node, which can be guessed.

Proposition 3.3.8. *If Conjecture 3.2.13 holds, then a language is regular if and only if it is recognizable by an alternating automaton.*

Proof. Necessity does not depend on Conjecture 3.2.13 and has been shown. Conversely, suppose that L is recognized by an alternating automaton $\mathcal{A}$ over the alphabet Σ. Using Conjecture 3.2.13, let n be large enough such that with n memory states we can win all games that are at most as complex as the game associated to $\mathcal{A}$ for any tree.

Now, we repeat the proof for S2S, but a part of the argument must be done for each label of a branch independently. Intuitively, the non-deterministic automaton for L guesses a strategy τ using at most n memory states. Following τ is even possible for a deterministic automaton, using determinization of word automata.

More formally, let T be the set of functions mapping a state of the automaton to a possible move. We construct an automaton $\mathcal{B}$ on $\Sigma \times T$ whose language projects onto L. For $a \in \Sigma$, let K_a be the language of words $x \in (\Sigma \times T \times \{0,1\})^{\mathbb{N}}$ such that there is no infinite play on some tree such that

- the play stays on the branch $\mathrm{pr}_{\{0,1\}}(x) \in 2^{<\omega}$,
- this branch is labeled by $(\mathrm{pr}_{\Sigma}(x), a) \in \Sigma^{\leq\omega}$,
- Automaton moves as specified by $\mathrm{pr}_T(x)$,
- and Pathfinder wins.

Since K_a is regular, it is recognized by a deterministic automaton. $\mathcal{B}$ simulates all these word automata on all branches. □

Remark 3.3.9. To obtain complementation for nondeterministic automata, it suffices to solve games obtained from nondeterministic automata. Such games are positional for Automaton since every fixed winning strategy can be simulated without memory. But, due to the negation step, we need to show that they are finite memory for Pathfinder. In Proposition 3.5.11 we show that solving such games is as hard as solving general bounded, countably branching, regularly jumping prioritized Borel games.

3.3.4 Monadic logic

The discussed operations of automata yield, using Proposition 3.3.8, the nontrivial direction in the following result.

Corollary 3.3.10. *If Conjecture 3.2.13 holds, then a language is regular if and only if it is definable in the Borel monadic theory of* $(2^{\leq\omega}, \preceq, \leq)$.

All discussed constructions of automata are computable. Thus, we could deduce decidability of the Borel monadic theory of the binary tree [She75, Conjecture 7A] from Conjecture 3.2.13 (this is our only use of the computability requirement in the conjecture) and the following fact.

Proposition 3.3.11. *It is computable whether an automaton recognizes the empty language.*

Proof. A Borel automaton with winning condition $W \subseteq Q \times \Sigma$ recognizes the empty language if and only if the usual tree automaton with winning condition $\mathrm{pr}_Q(W)$ recognizes the empty language. □

A tree $T\colon 2^{<\omega} \to \Sigma$ is **regular** if the language of $x \in (\{0,1\} \times \Sigma)^{<\omega}$ with $T(\mathrm{pr}_{\{0,1\}}(x)) = \mathrm{pr}_\Sigma(x)$ is regular. Rabin's basis theorem states that every nonempty regular language of trees contains a regular tree.

We call a tree $T\colon 2^{\leq\omega} \to \Sigma$ **regular** if $T|_{\{0,1\}^{<\omega}}$ is regular and for each σ the language of $x \in 2^{\mathbb{N}}$ with $T(x) = \sigma$ is regular. We obtain the following analogue of Rabin's basis theorem by the same proof.

Proposition 3.3.12. *Every nonempty regular language of Borel trees contains a regular tree.*

In particular, the branches of a regular tree are labeled by Boolean combinations of F_σ sets. A consequence of Rabin's basis theorem is that the regular subsets of $2^{<\omega}$ form an elementary substructure of all trees in S2S. We obtain the following analogue.

Proposition 3.3.13. *Assuming Conjecture 3.2.13, the regular subsets of $2^{\leq\omega}$ form an elementary substructure of the Borel monadic structure $(2^{\leq\omega}, \preceq, \leq)$.*

Similarly, the Boolean combinations of F_σ sets in $2^{\leq\omega}$ would form an elementary substructure of the Borel sets. This matches our aims expressed in the introduction to section 3.3.

3.4 Particular cases

3.4.1 Ordered branch conditions

Our aim in this subsection is to show the following result.

Proposition 3.4.1. *Suppose Eve wins a prioritized Borel game on priorities Σ such that the payoff set is ordered in the sense that every $p \in \Sigma$ satisfies $W(p) = \bigcap_{q\leq p} W(q)$ or $W(p) = \bigcup_{q\leq p} W(q)$. Then Eve wins without memory.*

Despite the existence of unbounded games requiring infinite memory, this result applies to unbounded games.

Remark 3.4.2. Usual parity games on graphs, interpreted as nullary branching games by Example 3.2.16, satisfy $W(p) = \{()\}$ or $W(p) = \emptyset$, hence the assumptions of Proposition 3.4.1. We thus recover memory-less determinacy of parity games assuming determinacy. Similarly to [EJ91, Section 4.2], our argument guesses histories for a fixed winning strategy σ_0 with memory, but we define a different procedure: While their guessed history is defined in terms of a property of the resulting move of σ_0, our guessed history only depends on σ_0 up to before the current position.

Definition 3.4.3. (a) If a strategy σ_0 is fixed, given a position x, let $\mathcal{H}(x)$ be the set of possible histories for x that are consistent with σ_0.

(b) Let the p-**profile** $p(h)$ of $h \in \mathcal{H}(x)$ be the string of length $|h| - 1$ that is 1 at indices where the outgoing edge of h has priority p and 0 at other indices.

(c) For each priority p, we define two prewellorders on $\mathcal{H}(x)$. The p-**maximizing order** is defined by $h \leq h'$ if $p(h) \geq p(h')$ lexicographically. The p-**minimizing order** is defined by $h \leq h'$ if the index of the last 1 in $p(h)$ is less than or equal the index of the last 1 in $p(h')$.

Proof of Proposition 3.4.1. Let G be a prioritized Borel game such that for all priorities p the payoff map B satisfies either $B(p) = \bigcap_{q\leq p} B(q)$ or $B(p) = \bigcup_{q\leq p} B(q)$. Suppose that σ_0 is a winning strategy of Eve.

Let $\leq_p$ be the p-maximizing order if $B(p) = \bigcap_{q\leq p} B(q)$ and the p-minimizing order if $B(p) = \bigcup_{q\leq p} B(q)$. Define a prewellorder $\leq'$ as the lexicographic order of the $\leq_p$, prioritizing higher priorities. Let $\leq''$ be the total order that extends $\leq'$ and breaks ties lexicographically. Define a strategy σ as follows: In position x, take $h \in \mathcal{H}(x)$ that is $\leq''$-minimal, and play $\sigma_0(h)$. We show that σ is winning.

Consider a play h_∞ compatible with σ. Let h_n be the history chosen by σ in position $h_\infty|_n$. Since $(h_n|_m)_n$ is a decreasing sequence in the wellorder $\leq''$ for fixed m, it is ultimately constant, whence $h'_\infty = \lim_n h_n$ exists. Moreover, h'_∞ has the same public branch $\langle h'_\infty\rangle = \langle h_\infty\rangle$, and is consistent with σ_0, thus won by Eve.

Let p be the highest priority. We distinguish cases:

First suppose both h_∞ and h'_∞ contain p finitely often, say not after position n. Let G' be the game obtained from the residual game of G at $h'_\infty|_n$ by removing all edges of priority p. Then the residual strategy

of σ_0 in G' is still defined on the corresponding suffix of h_∞. Moreover, the definition of σ in G is the same as in G' as all histories that extend $h'_\infty|_n$ without additional occurrences of p are $\leq_p$-equivalent: this follows as $h' <_p h$ but $h'|_n \geq_p h|_n$ implies that h' contains p after position n, which is impossible in G by assumption and impossible in G' by construction. Thus Eve wins by induction on the number of priorities.

Now suppose that h_∞ or h'_∞ contains p infinitely often.

First, suppose that $B(p) = \bigcap_{q \leq p} B(q)$.

We show that if h_∞ contains p infinitely often, so does h'_∞. Extend $\leq_p$ to compare arbitrary positions by padding the shorter profile with zeros at the end. We always have that $h_{n+1} \leq_p h_n$ and whenever p occurs in the actual play, then $h_{n+1} <_p h_n$. Since the limit of profiles $\lim_{n\to\infty} p(h_n|_m)$ exists and equals $p(h_\infty)|_m$, we conclude that h'_∞ contains p infinitely often.

Thus in all cases, h'_∞ contains p infinitely often. Since h'_∞ is won by Eve, $\langle h_\infty \rangle = \langle h'_\infty \rangle \in B(p) = \bigcap_q B(q)$, whence Eve wins h_∞.

Next, suppose that $B(p) = \bigcup_{q \leq p} B(q)$.

We show that if h_∞ contains p finitely often, so does h'_∞. Let n be such that p does not occur beyond $h_\infty|_n$. Then by induction on n for the position x of $h_\infty|_n$ there is $h \in \mathcal{H}(x)$ that contains p not after position n, whence neither h_n does.

Thus in all cases, h_∞ contains p infinitely often. Since h'_∞ is won by Eve, $\langle h_\infty \rangle = \langle h'_\infty \rangle \in \bigcup_q B(q) = B(p)$, whence Eve wins h_∞. □

Remark 3.4.4. For non-jumping Kőnig games, it is not necessary to use $\leq''$. In this case, any strategy that chooses an arbitrary $\leq'$-minimal element in a given position is winning. Then the h_n may be incompatible, whence $\lim_n h_n$ need not exist, but Kőnig's lemma yields a branch having infinitely many h_n as predecessor, which may serve as h'_∞ in the argument.

3.4.2 External games

Definition 3.4.5. Fix a game and let H be a set of histories of residual games. The **external game** for H is a game with the same public positions, proceeding in rounds where Eve and Adam play once. In a position x, the **internal game** is the residual game at x but such that Eve wins if she reaches a history $h \in H$ or she wins the game without reaching such an h. In each round of the external game, starting in position x, Eve picks a winning strategy τ of the internal game at x and Adam responds with an $h \in H$ compatible with τ (a player who cannot move loses immediately). The play proceeds in the final position of h. Finally, an infinite play is won by the player who wins the concatenate of Adam's moves in the internal games.

Proposition 3.4.6. *A player has a winning strategy for the external game if they have a winning strategy for the original game.*

Proof. If Eve wins the original game using a strategy σ, then Eve wins the external game by always playing the residual strategy of σ. If Adam wins the original game using a strategy σ, then, in a round where Eve plays τ, Adam plays the first move of $\tau * \sigma$ in h. □

If the original game is determined, then Proposition 3.4.6 implies its converse. Moreover, for Eve the converse always holds: Eve wins the original game by simulating a run of the external game and following her strategy for the current internal game. However, for $H = \emptyset$ the converse of Proposition 3.4.6 for Adam is equivalent to determinacy.

If the original game is a tree game, a prioritized Borel game, non-jumping, bounded, Kőnig, or countably branching, so is each internal game. If the original game is a tree game, a prioritized Borel game, bounded, Kőnig, or countably branching, then so is the external game - the priority of a round the maximal priority in the finite play of its internal game. However, if we start with a non-jumping game, the external game may be jumping.

Definition 3.4.7. Let p be a priority. The p-external game is the external game for the set of histories containing a priority $q \geq p$.

The p-external game only uses priorities $q \geq p$, while each of its internal games only uses priorities $r < p$. In particular, if p is the highest priority, then the p-external game is memory-less determined by

Proposition 3.4.1, and if p is the lowest priority, then its internal games are memory-less determined by Proposition 3.4.1. This does not enable induction as it is not clear how to combine finite memory strategies for the external and internal games into a finite memory strategy for the original game in general - it might require infinite memory to remember when the last internal game started. Equivalently, one could remember when priority p occurred for the last time. In particular, we obtain by induction that the following infinite memory suffices:

Proposition 3.4.8. *Each prioritized Borel game is determined with memory for each priority p the position of the last occurrence of p.*

Definition 3.4.9. The **outcome** of a finite play of an internal game is the pair of the final position and the highest priority played. A strategy τ **subsumes** a strategy σ if every outcome compatible with τ is compatible with σ.

For a strategy σ of Eve for the internal game, let the **σ-restricted internal game** be obtained from the internal game by the following procedure. First, replace each position by $\#\Sigma$ copies (sharing the same public position), and change the possible transitions such that the game is in the copy corresponding to a priority p if and only if p is the highest priority played so far. Next, declare that Eve wins if she wins the game without reaching $h \in H$ or if she reaches an $h \in H$ with highest priority p such that (h, p) is a possible outcome of σ.

Lemma 3.4.10. *(a) A strategy subsumes a strategy σ if and only if the corresponding strategy for the σ-restricted internal game induced is winning.*

(b) If a strategy for the σ-restricted internal game requires only n memory states, then the corresponding strategy for the internal game can be simulated using $n \cdot \#\Sigma$ memory states.

Proof. Unraveling definitions yields (a). The additional memory for (b) is the highest priority played so far. □

Remark 3.4.11. In many applications, we could instead use a variant of the external game in which rounds of the internal may last longer than the first occurrence of h. This variant would be equivalent to the external game for the set H' of h' such that there is $H \ni h \preceq h'$.

3.4.3 Witnessing priorities

Throughout this section, we fix a non-jumping prioritized Borel game and a winning strategy σ_0 of Eve.

Definition 3.4.12. Let p be a priority. We define by recursion on a node in P whether it **witnesses** p. The root witnesses p. For an inner node v, let w be the last ancestor that witnesses p. Then v witnesses p if in any play compatible with σ_0, there is an occurrence of a priority $q \geq p$ between v and w.

We now study the external game for the witnesses for p, that is, for the set of nonempty histories that end in a witness of p. The main advantage over the p-external game is that the current internal game depends only on public data.

Lemma 3.4.13. *The external game for the witnesses for p is non-jumping.*

Proof. If the original game has public positions in $A^{<\omega}$, then the external game is non-jumping as a game on $(A^{<\omega})^{<\omega}$. Finally, since the current internal game is public, the history in $(A^{<\omega})^{<\omega}$ can be recovered from the current public position in $A^{<\omega}$. □

The dependence on public data also enables the reconstruction of a strategy for the original game with finite memory.

Lemma 3.4.14. *Suppose that Eve wins the external game for the witnesses for p with m memory states and each strategy of Eve for an internal game is subsumed by a strategy using at most n memory states. Then Eve wins the original game using at most $m \cdot n$ memory states.*

Proof. Eve simulates a run of the external game, using the memory for the external game and the current round of the internal game. Using public information, she reconstructs when the last internal game started, using the memory for the external game she reconstructs which strategy τ she played in this round, and using the memory for the internal game she follows a strategy subsuming τ. □

Lemma 3.4.14 reduces solving the original game to solving the external game and all τ-restricted internal games. We investigate particular cases of these games separately.

Generalizing our notation of Proposition 3.4.1, we say that the payoff set is **ordered above** a priority p_0 if every $p \geq p_0$ satisfies $W(p) = \bigcap_{p_0 \leq q \leq p} W(q)$ or $W(p) = \bigcup_{p_0 \leq q \leq p} W(q)$.

Lemma 3.4.15. *If the payoff set is ordered above p_0, then Eve wins the external game for witnesses for p_0 without memory.*

Proof. By the definition of witnesses, σ_0 guarantees a priority $q \geq p_0$ between any two witnesses. The derived strategy for the external game thus guarantees a priority $q \geq p_0$ in any round. Thus the external game is a game on the priorities above p_0, hence ordered. Proposition 3.4.1 finishes the proof. □

Lemma 3.4.16. *Let τ be a strategy of Eve for an internal game for witnesses for p_0. The payoff set W of the τ-restricted internal game satisfies $W(p) \subseteq \bigcup_{q<p_0} W(q)$ for all p.*

Proof. Let $x_\infty \in W(p)$. By definition of the internal game, no prefix of x_∞ is a witness for p_0. Thus, there is a history h_∞ with $\langle h_\infty \rangle = x_\infty$ that only contains priorities $q' < p_0$. If q is the priority of h_∞, then $q < p_0$ and $x_\infty \in W(q)$. □

This finishes our analysis of the external and internal games, and we now derive consequences.

Corollary 3.4.17. *Any non-jumping prioritized Borel game with two priorities can be won using two memory states.*

Proof. Let $\mathfrak{r}_1 < \mathfrak{r}_2$ be the priorities. Since the payoff set is trivially ordered above $\mathfrak{r}_2$, by Lemmas 3.4.14 and 3.4.15 it suffices to show that every strategy for the internal games for witnesses for $\mathfrak{r}_2$ is subsumed by a strategy using two memory states. By Lemma 3.4.10 it suffices to show that each τ-restricted internal game is memory-less. Since $W(\mathfrak{r}_2) \subseteq W(\mathfrak{r}_1)$ in such a game by Lemma 3.4.16, Proposition 3.4.1 applies. □

The strategy obtained by Corollary 3.4.17 can be described more explicitly. We here give a description for Kőnig games, but the general one using lexicographically maximal lifts is similar. In position x, if priority $\mathfrak{r}_2$ occurred since the last witness for $\mathfrak{r}_2$, take $h \in \mathcal{H}(x)$ such that the profile $\mathfrak{r}_2(h)$ is lexicographically maximal, and play $\sigma_0(h)$. Otherwise follow σ_0 assuming any history that does not contain $\mathfrak{r}_2$ since the last witness.

The assumption of being non-jumping in Corollary 3.4.17 is necessary: In section 3.6.2 we present a regularly jumping counterexample.

Remark 3.4.18. There is no analogous approach to three priorities. If priority $\mathfrak{r}_3$ occurs repeatedly without being witnessed, then choosing a history that maximizes $\mathfrak{r}_3$ only guarantees to reach a branch in $W(\mathfrak{r}_1) \cup W(\mathfrak{r}_2)$.

Proposition 3.4.19. *Consider a non-jumping prioritized Borel game with three priorities $\mathfrak{r}_1 < \mathfrak{r}_2 < \mathfrak{r}_3$. If the payoff set W satisfies at least one of the following conditions, then the game can be won using finite memory independent of the game:*

(a) $W(\mathfrak{r}_1) \subseteq W(\mathfrak{r}_2)$

(b) $W(\mathfrak{r}_1) \subseteq W(\mathfrak{r}_3)$

(c) $W(\mathfrak{r}_3) \subseteq W(\mathfrak{r}_2)$

Proof. We always use Lemma 3.4.14 with witnesses for $\mathfrak{r}_2$. The external game is non-jumping by Lemma 3.4.13 and uses only priorities $\mathfrak{r}_2, \mathfrak{r}_3$ and thus finite memory by Corollary 3.4.17. Thus it suffices to show that the internal game is ordered.

The condition $W(\mathfrak{r}_1) \supseteq W(\mathfrak{r}_2)$ always holds by Lemma 3.4.16. Thus it remains to show that $W(\mathfrak{r}_3) \subseteq W(\mathfrak{r}_1) \cap W(\mathfrak{r}_2)$ or $W(\mathfrak{r}_3) \supseteq W(\mathfrak{r}_1) \cup W(\mathfrak{r}_2)$. The first disjunct reduces to (c) and the second one to (b) by Lemma 3.4.16. Finally, (a) implies the first disjunct as $W(\mathfrak{r}_1) \supseteq W(\mathfrak{r}_3)$ by Lemma 3.4.16. □

The only remaining case in which the $W(\mathfrak{r}_i)$ form a chain is $W(\mathfrak{r}_2) \subsetneq W(\mathfrak{r}_3) \subsetneq W(\mathfrak{r}_1)$. In this case, the conclusion of Proposition 3.4.19 fails by Proposition 3.6.2.

3.4.4 Games of bound one

We consider prioritized Borel games of bound one. Equivalently, the map $Q \to P, q \mapsto \langle q \rangle$ is injective. Thus, we identify them as $Q \subseteq P$ and $\langle q \rangle = q$. We will moreover assume that the game is non-jumping. By Corollary 3.5.13, we can even assume that $x \xrightarrow{a} y$ implies $|y| = |x| + 1$. In this model, the priorities played are the only non-public information.

Proposition 3.4.20. *Every non-jumping prioritized Borel game of bound one with d priorities can be won using d memory states.*

Fix a winning strategy σ_0 of Eve. Call a position **branching** if all outgoing edges have the lowest priority. Call a position **picking** if all outgoing edges end in the same position. Splitting edges, we arrange that every position of Eve is branching or picking.

We first sketch the main idea of the proof, focusing on the case where Eve has only branching positions. We use a variant of the construction of witnesses for priorities of Definition 3.4.12 that circumvents the issue of Remark 3.4.18. For each priority p, we will define sets W_p, whose elements are similar to witnesses, and C_p, whose elements code occurrences of p. While occurrences of C_p guarantee that p occurred in a play consistent with the new strategy, occurrences of W_p yield histories consistent with σ_0 in which p occurred. Our construction interpolates between the trivial case in which the moves of σ_0 are completely determined by whether certain priorities occurred since the last witness and the trivial case in which for any possible move of Eve, there are two different moves that do not affect the payoff - in the latter case Eve can use those moves to code whether high priorities occurred. We also depend on the observation that reacting to a priority can be delayed by a constant number of moves of an appropriate finitary type, enabling to react to a fixed priority only in rounds satisfying some congruences on the relevant priority and the number of similar positions so far.

Proof of Proposition 3.4.20. Let $0 < \cdots < d-1$ be the priorities. We recursively define for $p \in \Sigma \cup \{d\}$ sets W_p and C_p of positions such that

$$W_0 \supseteq \cdots \supseteq W_d \text{ and } C_0 \supseteq \cdots \supseteq C_d \text{ and } W_p \subseteq C_p \text{ and } C_d = \emptyset$$

and for each position $x \in W_0$ a history $h_x \in \mathcal{H}(x)$. These data will satisfy the following additional invariants: After the last occurrence of W_p, h_x only contains priorities less than p. If $W_p \ni w \preceq x \in W_p$ is such that W_{p+1} does not occur strictly between w and x, then h_x extends h_w with highest priority p. For any two subsequent occurrences of $C_p \setminus C_{p+1}$, at least one of them lies in W_p.

The root lies in $W_{d-1} \setminus C_d$. Let these data be given up to including x. Call a child y of x p-**consistent** if, for w the maximal prefix of x in W_p, there is a history $h_w \prec h_{y,p} \in \mathcal{H}(y)$ that contains p as highest priority after w. We say that a position x **remembers** a priority p if the maximal prefix c of x in C_p satisfies $c \notin W_p$.

For each child y of x, we will define priorities $c \geq w$ such that y is w-consistent or $c = w = -\infty$. Then let q be the highest priority remembered by x such that y is q-consistent and set $c' = \max\{c, q\}$ and $w' = \max\{w, q\}$. Finally, if $c', w' \neq -\infty$, we declare $y \in (W_{w'} \setminus W_{w'+1}) \cup (C_{c'} \setminus C_{c'+1})$ and let h_y witness that y is w'-consistent. We now define c, w by cases.

If x is a position of Adam, then $c = w$ is the minimal priority on an edge $x \to y$.

Next, suppose that x is a picking position of Eve. Let L be the set of priorities p such that y is p-consistent and there is an edge $x \xrightarrow{p} y$. Let $c = w = \max L$. Then $y \in W_0$ since the move of σ_0 in h_x guarantees $L \neq \emptyset$.

Finally, suppose that x is a branching position of Eve. Let w be minimal such that y is w-consistent. For y_0 the unique 0-consistent child of x, let $c = 0$. Let the **type** $\mathrm{tp}(x) \subseteq \Sigma$ be the set of p such that x has a p-consistent child different from y_0. Define $r \in \{1, \ldots, d-1\}$ by

$$r \equiv \#\{x' \prec x \mid \mathrm{tp}(x') = \mathrm{tp}(x)\} \mod d-1.$$

For any child y different from y_0 that is p-consistent for some p, let w be minimal such that y is w-consistent and $c = \max\{w, r\}$.

We define a strategy σ of Eve with memory m the highest priority that is not remembered and occurred since C_m.

Consider a picking position with child $y \in W_p \setminus W_{p+1}$. If $p \in L$, play p. Otherwise, play any $q \leq p$.

Consider a branching position x. We say that a priority p demands a move to a child y if at least one of the following conditions holds:

- Demands to code: $p = m$ and $y \in C_p \setminus C_{p+1}$.
- Demands to witness: p is remembered and $y \in W_p \setminus C_{p+1}$.

Let p be the maximal priority that demands some move. Play any move demanded by p. If no priority demands a move, play y_0.

We show that σ is winning. Fix a play h_∞ consistent with σ.

Claim A. *If C_p occurs infinitely often in h_∞, so do W_p and some $q \geq p$.*

Proof. By induction, we may assume that C_{p+1} occurs finitely often. By construction, $C_p \setminus C_{p+1}$ occurs at most once strictly between any two occurrences of W_p. Thus, W_p occurs infinitely often. Finally, some $q \geq p$ occurs at least once between an occurrence of W_p followed by an occurrence of C_p. ■

Let p be the highest priority that occurs infinitely often in h_∞. By Claim A C_{p+1} occurs finitely often. If C_p occurs infinitely often, by Claim A so does W_p, whence the h_x converge to a history with highest priority p and Eve wins.

Finally, suppose that C_p occurs finitely often. In particular, $p > 0$ as σ never leaves C_0.

Claim B. *Eve plays priority p only finitely often.*

Proof. Whenever Eve plays p, the next position is in W_p. ■

Claim C. *The last occurrence of C_p lies in W_p.*

Proof. Suppose otherwise. Let x be a position after the last occurrence of C_p in which p is played. Since $p > 0$, x cannot be a branching position of Eve. Since its child does not lie in W_p, x cannot be a picking position of Eve. Thus x is a position of Adam. Extending h_x by the actual move in h_∞ witnesses that the child is p-consistent. Since p is remembered by x, in the definition of the W_j we have that $w' \geq q \geq p$. In other words, the child lies in W_p, which is absurd. ■

Claim D. *The memory is ultimately constant of value at least p.*

Proof. Since a memory state m can only be replaced by a lower state if C_m occurs, memory states $m \geq p$ are only entered finitely often. Thus, otherwise the memory would be ultimately strictly below p. This is impossible: Since p is ultimately not remembered by Claim C, seeing priority p increases the memory to at least p. ■

Let m_0 be the ultimate value of the memory.

Claim E. *For $q \leq m_0$, in branching positions ultimately there is no q-consistent child but possibly y_0.*

Proof. Since C_p occurs finitely often, ultimately only priorities strictly below p can demand moves. In particular, m_0 does ultimately not demand to code, whence there ultimately are no children in $C_{m_0} \setminus C_{m_0+1}$. We conclude by the definition of types. ■

Let x_0 be a position such that

- C_p does not occur after x_0;
- no priority strictly above p occurs after x_0;
- the memory is in state m_0 from x_0 on; and
- for $q \leq m_0$, in branching positions after x_0 there is no q-consistent child but possibly y_0.

We construct a history h'_∞ such that

(a) $h_{x_0} \prec h'_\infty$;

(b) h'_∞ is consistent with σ_0;

(c) h'_∞ contains the same positions as h_∞;

(d) from x_0 on, Eve does not play priorities above p in h'_∞; and

(e) from x_0 on, Adam makes the same moves in h'_∞ and in h_∞.

We first show that the existence of h'_∞ implies that Eve wins. By Claim B Adam plays priority p infinitely often in h_∞, thus by (e) in h'_∞. Hence, by (d) the highest priority of h'_∞ also is p and Eve wins by (b) and (c).

We now construct h'_∞. We first observe that h'_∞ is uniquely determined by (a), (b) and (e). Thus, we have to show by induction that the unique history satisfying (a), (b) and (e) also satisfies (c) and (d) (however, the existence of this unique history requires to simultaneously verify (c) as (e) can be stated only assuming (c)). For this, we verify (c) in branching positions and (d) in picking positions.

In a branching position x, let q be the highest priority that occurs in h'_∞ after x_0. By (d) and (e), $q \leq m_0$. The move made by σ_0 leads to a q-consistent child, thus by Claim E to y_0. We show (c), that is, that h_∞ moves to y_0, by showing that no priority demands a move. Indeed, after the last occurrence of C_p, only priorities strictly below p can demand moves. Since the memory is $m_0 \geq p$, no such priority can demand to code. If a lower priority demands to witness, it leads to a q'-consistent child different from y_0, again contradicting Claim E.

We finish by establishing (d) for picking positions $x \succeq x_0$ of Eve with child y. Indeed, if σ_0 plays a priority $q > p$, then the resulting history at y witnesses that y is q-consistent, whence $y \in L$. Hence $y \in W_q \subseteq C_p$, contradicting the definition of x_0. □

Remark 3.4.21. If all positions of Eve are picking, then the proof simplifies substantially. Namely, $C_p = W_p$ and Eve's strategy is memory-less - even in the general case, the memory does not affect Eve's moves in picking positions.

3.4.5 Playing only priorities

The definition of **picking** positions generalizes to arbitrary prioritized Borel games: A position is picking if all outgoing edges end in the same position.

Proposition 3.4.22. *Suppose Eve wins a non-jumping prioritized Borel game with d priorities in which all of her positions are picking. Then she wins with 2^{d-1} memory states.*

Proof. Fix a winning strategy σ_0 of Eve. For each priority p, we define a prewellorder $\leq_p$ on $\mathcal{H}(x)$ as the lexicographic order of the q-minimizing orders for $q > p$ and the p-maximizing order. We say that a pair of positions $x \preceq y$ improves p if the $\leq_p$-minimal history of y is not given by adding zeros to the $\leq_p$-minimal history of x.

The winning strategy σ remembers the set I of priorities p such that there was a p-improving edge since the last occurrence of p (since I always contains the lowest priority, this requires only 2^{d-1} memory states). Whenever Eve is allowed to pick a priority, she first updates her memory to the state for the successor and then plays the highest possible priority in I.

We show that σ is winning.

Fix a play h_∞ consistent with σ'. Let p be the highest priority that occurs infinitely often in h_∞.

In h_∞, there are infinitely many p-improving edges. Indeed, if Eve plays p infinitely often, then this holds by definition of σ. Otherwise, Adam plays p infinitely often. Every edge along which Adam could play p is p-improving.

Next, we show that the $\leq_p$-class of the finite prefixes of h_∞ converges and its limit contains finitely many occurrences of priorities higher than p.

We show by descending induction on $q > p$ that the limit only contains finitely many occurrences of q. If there are only finitely many q-improving edges, then the q-maximizing order only enables finitely many occurrences, whence the q-minimizing one does. Thus suppose there are infinitely many q-improving edges. Thus, from some point on Eve cannot play priority q as otherwise h_∞ would contain q infinitely often. In a position of Adam, since ultimately no $q > p$ occurs in h_∞, he has a move that plays a low priority, which will always be preferred by the q-minimizing order.

Since there are infinitely many p-improving edges, but the minimizing orders for higher priorities stabilize by above argument, there is a $\leq_p$-convergent sequence of profiles whose limit has priority p. By a lexicographic

ordering argument or Kőnig's lemma as in Proposition 3.4.1, this sequence of profiles is induced by a convergent sequence of histories, whence $\langle h_\infty \rangle \in W(p)$. □

3.5 Equivalent formulations

3.5.1 Universal games

For a bounded prioritized Borel game, its **complexity** is the pair of the number of priorities and the least bound.

Example 3.5.1. Let κ be a cardinal and (d, N) be a complexity. The **universal game** is played as follows: First, Adam publicly plays a prioritized Borel game of branching κ and complexity (d, N) with two players, Alice and Bob. Second, Eve publicly chooses to either be Alice or Bob. Finally, they play this game.

The branching of the universal game equals the number of Borel subsets of $\kappa^{\mathbb{N}}$, which is 2^κ. Eve has a winning strategy in the universal game: Choose the winning player and follow a winning strategy for them.

A local version of the universal game yields the following observation.

Proposition 3.5.2. *Let κ be a cardinal and (d, N) be a complexity. If all prioritized Borel games of branching κ and complexity (d, N) are determined with finite memory depending on the game, then there is $M \in \mathbb{N}$ such that all prioritized Borel games of branching κ and complexity at most (d, N) are determined with M memory states.*

Proof. For $\kappa < 2$, the conclusion holds unconditionally by Example 3.2.16. Thus, using a standard reduction, we may assume κ infinite. If the conclusion is false, for each n choose a game G_n requiring memory at least n. Consider the universal game where Adam's first move is restricted to $\{G_n \mid n\}$. Then Eve still has a winning strategy, which by assumption can be chosen to use some finite memory M. This induces a winning strategy of G_{M+1} using memory M, which is absurd. □

Proposition 3.5.2 suggests that the uniformity assumption in Conjecture 3.2.13 is unnecessary, but it does not formally imply this since the process need not preserve computability. If the assumptions of Proposition 3.5.2 hold, it is not clear if the memory required for the universal game of complexity (d, N) is computable from (d, N). Otherwise, Borel automata could be negated, but not necessarily efficiently.

The same argument works when restricting to non-jumping games in the hypothesis and conclusion. Proposition 3.5.2 has analogues for the upcoming reformulations in Theorems 3.5.3 and 3.5.10 and their corresponding measures of complexity.

3.5.2 Replacing Borel sets by Boolean combinations of F_σ sets

Assuming only the elementary result that closed games are determined, Borel determinacy is equivalent to "for every Borel $B \subseteq A^{\mathbb{N}}$ such that Eve does not win the Gale-Stewart game with payoff set B, there is an open $U \supseteq B$ such that Eve does not win the Gale-Stewart game with payoff set U". We now reformulate Conjecture 3.2.13 similarly.

For a prioritized Borel game won by Eve with payoff set W, we say that $W(p)$ **can be replaced** by B if $B \subseteq W(p)$ and Eve also wins the game with $W(p)$ replaced by B. For example, if a Gale-Stewart game is interpreted as prioritized Borel game as in Example 3.2.3 (a), then the payoff set $W(\star)$ can be replaced by a closed set.

Theorem 3.5.3. *Let (d, N) be a complexity. The following conditions are equivalent:*

(i) All prioritized Borel games up to complexity (d, N) are determined with a bounded amount of memory.

(ii) For all prioritized Borel games up to complexity (d, N) won by Eve, each $W(p)$ can be replaced by a Boolean combination of F_σ sets with a bounded number of Boolean operands.

Moreover, in this case upper bounds for the memory for (a) and for the number of Boolean operands for (b) can be computed from each other.

Proof. By Fact 3.3.1, the implication (ii) $\Longrightarrow$ (i) follows from the Büchi-Landweber theorem. For (i) $\Longrightarrow$ (ii), fix a bounded memory winning strategy σ. Let $B \subseteq W(p)$ be the set of branches x_∞ such that there is a play on x_∞ that is compatible with σ and has highest priority p. Then $W(p)$ can be replaced by B. For showing B a Boolean combination of F_σ sets, we use [Boj+10, Corollary 3.1]. Definability is straightforward, using predicates for the tree to code σ, including its memory update function. (Formally, the statement of the corollary is not sufficient, but we also need to obtain a bound on the complexity of the Boolean combination. This follows from the proof.) □

The following observation is not used in other results, but provides partial progress towards Theorem 3.5.3.

Lemma 3.5.4. *If 0 is the lowest priority in a Kőnig prioritized Borel game, then $W(0)$ can be replaced by an F_σ set.*

Proof. Fix a winning strategy σ. Let B be the set of branches such that there is a play compatible with σ that ultimately only contains 0. Clearly $W(0)$ can be replaced by B. As the projection of an F_σ set along a compact space, B is F_σ. □

Similarly, if there is a single priority $\star$, then $W(\star)$ can be replaced by a closed set, recovering memory-less determinacy in this case.

The following conjecture strengthens the equivalent of Conjecture 3.2.13 obtained by Theorem 3.5.3. It separates the increase in complexity required for more priorities from the increase required for more states.

Conjecture 3.5.5. *There is a computable $f\colon \mathbb{N} \to \mathbb{N}$ such that for all prioritized Borel games of bound N, each $W(p)$ can be replaced by a Boolean combination of at most $f(N)$ F_σ sets.*

The case $N = 1$ of the strengthened conjecture (with $f(1) = 4$) follows from the following lemma.

Lemma 3.5.6. *Let W_p be as in the proof of Proposition 3.4.20. Then the payoff set $W(p)$ can be replaced by the union of the following sets:*

(a) The set of branches that contain W_p infinitely often, but W_{p+1} finitely often.

(b) The set of branches along that the lexicographic maximum of occurrences of p in some history decreases finitely often, but increases infinitely often.

Proof. The limits of the associated histories of $x \in W_p$ witness that (i) is included in $W(p)$. Using lexicographic orderings on the remaining data as in Proposition 3.4.1 or Kőnig's lemma, (ii) is included in $W(p)$.

We finally show that the strategy σ of Proposition 3.4.20 guarantees to end in (i) or (ii) whenever the highest priority is p. If it does not end in (i), then the strategy ultimately only plays y_0 in branching positions. After the last element x of W_p, the lexicographic maximum does not decrease, but increases infinitely often since Eve cannot play p, but Adam can infinitely often. □

3.5.3 Definability in the external game

We use the following folklore result on compositionality of monadic logic on infinite trees.

Lemma 3.5.7. *Let φ be a sentence of monadic logic for T a Σ-labeled binary tree. There are finite sets H, V of **types**, computable from φ, and surjective maps* tp *from trees to H and from pairs of trees and nodes to V such that, where T_x denotes the full subtree of T rooted at x:*

(a) From $\mathrm{tp}(T) \in H$ we can compute whether $T \models \varphi$ holds.

(b) From $\mathrm{tp}(T, x) \in V$ and $\mathrm{tp}(T_x) \in H$ we can compute $\mathrm{tp}(T) \in H$.

(c) If S, T differ only in the subtrees at a node x, then $\mathrm{tp}(S, x) = \mathrm{tp}(T, x) \in V$.

(d) From $\mathrm{tp}(T, x) \in V$, the label $T(x) \in \Sigma$ and the type $\mathrm{tp}(T_{x0}) \in H$ of the subtree at the left child of x, we can compute $\mathrm{tp}(T, x1) \in V$. The analogue holds when swapping left and right.

Proof. Fix an automaton $\mathcal{A}$ for φ. Let $\mathrm{tp}(T) \in H$ be the set of states q of $\mathcal{A}$ such that $\mathcal{A}$ accepts T if the initial state is replaced by q.

Moreover, for each node x, whether $\mathcal{A}$ - possibly with changed initial state - accepts T depends only on T without T_x and $\mathrm{tp}(T_x)$. Thus we set $\mathrm{tp}(T, x)$ as the map sending a state q to the set of $h \in H$ such that $\mathcal{A}$ with initial state q accepts T with T_x replaced by a tree of type h. □

Definition 3.5.8. A **potential** for a prioritized Borel game with highest priority p is a binary relation R on positions satisfying the following conditions:

(a) For all positions x, there is a strategy σ of Eve that guarantees to either win the game seeing p finitely often or reach a position y with xRy seeing p at least once.

(b) Let x_i be positions such that x_0 is initial and $x_0 R x_1 R x_2 R \dots$. Then $\lim_i \langle x_i \rangle \in W(p)$.

Lemma 3.5.9. *Eve wins a game if and only if there is a potential.*

Proof. Given a potential, we define a winning strategy that remembers a position x and follows the strategy of Definition 3.5.8 (a). Initially, x is the root, and whenever the current position y satisfies xRy and the highest priority occurred in between, x is replaced by y.

Conversely, using Proposition 3.4.6, suppose that $x \mapsto \sigma_x$ is a winning strategy for the external game for the highest priority. Declare that xRy holds if there is a play of σ_x that ends in y (and thus contains the highest priority necessarily on the last edge). Then σ_x witnesses (a) and (b) follows as the strategy is winning. □

Theorem 3.5.10. *Let (d, N) be a complexity. The following conditions are equivalent:*

(i) All games up to complexity (d, N) are determined with a bounded amount of memory.

(ii) There is a formula $\varphi(x, y, T)$ in monadic logic such that for all games won by Eve of complexity (d, N) there is an infinite labeled tree T such that $\varphi(-, -, T)$ defines a potential.

Moreover, in this case an upper bound for the memory for (a) and a formula for (c) can be computed from each other.

Proof. For (i) $\implies$ (ii), fix a finite memory strategy σ and consider the induced strategy $x \mapsto \sigma_x$ for the external game. Let $\varphi(x, y, \sigma)$ express that σ_x may end in y; more explicitly, there is a history compatible with σ that ends in y, passes through x and sees the highest priority immediately before y and at no other place between x and y.

For (ii) $\implies$ (i), fix a definable potential R, say with parameter T. We define a winning strategy using as memory $\mathrm{tp}(T, x)$ of Lemma 3.5.7 for x the memory used in Lemma 3.5.9, which can be maintained using (d). By (b) and (d), in position z for any $y \succeq z$ we can compute $\mathrm{tp}(T, x, y)$, whence by (a) the memory determines $\{y \succeq z \,|\, xRy\}$. In particular, it determines the minimal $x' \preceq x$ with $\{y \succeq z \,|\, xRy\} = \{y \succeq z \,|\, x'Ry\}$. Now follow the strategy given in the definition of R for x', which can be assumed to use finite memory by induction on d. □

3.5.4 Automata induce general games

Our aim in this subsection is to show the following result.

Proposition 3.5.11. *Every bounded, non-jumping, countably branching prioritized Borel game is equivalent to the associated game of a non-deterministic Borel automaton from the perspective of Pathfinder.*

A (strengthened) converse of Proposition 3.5.11 holds by definition: Associated games of alternating Borel automata are bounded, non-jumping, and binary branching. Thus Proposition 3.5.11 tells us that Conjecture 3.2.13, apart from the restriction to non-jumping and countably branching games, is only superficially more general than required for our approach to decidability of the Borel monadic theory of the binary tree. (As mentioned in Remark 3.3.9, Automaton never requires memory to win the associated game of a non-deterministic Borel automaton, but we need to show that Pathfinder only requires finite memory.)

There is another sense in which Conjecture 3.2.13 needs to be considered when studying decidability of this theory: For each fixed complexity (d, N) and memory bound n, the statement "all bounded, non-jumping, countably branching prioritized Borel games of complexity at most (d, N) can be solved with n memory states" is expressible as a sentence in the Borel monadic theory. Thus, any decidability proof at least needs to reduce Conjecture 3.2.13 to a semi-decidable statement, while any undecidability proof due to our reduction needs to refute Conjecture 3.2.13.

To establish Proposition 3.5.11, we start by proving a series of lemmas.

Lemma 3.5.12. *Every tree game is equivalent to a tree game in which $x \xrightarrow{a} y$ implies $|\langle y\rangle| > |\langle x\rangle|$.*

Proof. We consider the external game for H the set of histories that strictly increase $|\langle x\rangle|$. Clearly, the external game is a tree game in which $x \xrightarrow{a} y$ implies $|\langle y\rangle| > |\langle x\rangle|$. Thus, it suffices to show that the original game is equivalent to the external game. Using an analogue of Lemma 3.4.14, it suffices to show that the internal game is memory-less determined. This internal game is a usual parity game as in Example 3.2.16. □

Similarly to Lemma 3.4.13, this reduction preserves non-jumping games, whence we deduce:

Corollary 3.5.13. *Every non-jumping tree game is equivalent to a tree game in which $x \xrightarrow{a} y$ implies $|\langle y\rangle| = |\langle x\rangle| + 1$.*

Lemma 3.5.14. *Every bounded prioritized Borel game admits a finite uniform state choice.*

Proof. Since the game is bounded, there is a not necessarily uniform finite state choice $s_0 \colon Q \to S_0$. A transition type is a subset of $S_0 \times \Sigma \times S_0$. For an edge $v \to v'$ in the public positions $P \subseteq A^{<\omega}$, its transition type is the set of (s, p, s') such that there is a transition $q \xrightarrow{p} q'$ with $\langle q\rangle = v, \langle q'\rangle = v'$ and $s_0(q) = s, s_0(q') = s'$.

Replace the public alphabet A by its cartesian product with the set of transition types and let $S = S_0 \times \{\exists, \forall\}$. Define $(i, \exists) \xrightarrow{(a,\tau);p} (j, \forall)$ and $(i, \forall) \xrightarrow{(a,\tau);p} (j, \forall)$ whenever $(i, x, j) \in \tau$. The active player of (i, Q) is Q and the initial state is $s_0(q_0)$ with the corresponding initial player. The payoff map encodes the following rules: If, given a public history with first component $v \in A^{<\omega}$, a player chooses an edge of label (a, τ), then τ equals the transition type of the edge $v \to va$. If both players follow the rules, the winner is determined using the original payoff set and the first component of the history. □

Proof of Proposition 3.5.11. The associated games of non-deterministic Borel automata are precisely the prioritized Borel games satisfying the following conditions: They are binary branching, the players alternate, there is a uniform state choice, Pathfinder always has exactly two moves, which always lead to different public positions y that do not depend on the state and satisfy $|\langle y\rangle| = |\langle x\rangle| + 1$, and Automaton's moves do not change the public position.

It is straightforward to eliminate the assumption that the players alternate. The restriction that Pathfinder has exactly two moves can then be weakened to playing a countably branching game via a folklore reduction - for instance, moves that are natural numbers can be simulated by playing their unary representation, and positions where Pathfinder only has a single possible move can be simulated by making the other move enter a state won by Automaton.

We next show how to simulate a dependency of the public positions available to Pathfinder on the current state of the game and not only its underlying public position. Indeed, add a new state from which Automaton always wins and enable Automaton to enter this state if and only if Pathfinder plays a forbidden move before.

Now, we can eliminate the restriction on Automaton's moves. Whenever Automaton would have the opportunity to play a public letter a in the original game, in the modified game they can move into a state of Pathfinder in which they can only play a.

Thus, it remains to eliminate the restriction is that different moves of Pathfinder lead into different public positions. Then, by Corollary 3.5.13 we may replace the assumption $|\langle y\rangle| = |\langle x\rangle| + 1$ by being non-jumping, and Lemma 3.5.14 yields a uniform state choice.

For this purpose, we present a gadget that enables Pathfinder to choose between two states p, q without emitting public information, which suffices to simulate any other possible move. This gadget consists of the following steps:

- Automaton privately plays an element $i \in \mathbb{Z}/2\mathbb{Z}$.

- Pathfinder publicly plays an element $j \in \mathbb{Z}/2\mathbb{Z}$.
- The play proceeds in state p if $i = j$ and in state q else.

Playing a round in this gadget emits no public information usable for Pathfinder. More formally, if a winning strategy depends on the move j of Pathfinder, then the strategy always assuming that $j = 0$ wins, too. □

Remark 3.5.15. While our reduction introduces many new states, the number of new priorities is a constant independent of the game.

Remark 3.5.16. In Proposition 3.5.11, the assumption of being non-jumping could be weakened to being regularly jumping - by definition, this also requires to assume a priori that a uniform state choice exists. Namely, a transition of the form $s \xrightarrow{L;a} t$ for a regular language L can be replaced by an automaton with a unique initial state s and a unique final state t that recognizes L without entering s or t twice. Such non-deterministic automata easily exist for any regular language if we allow transitions along the empty word. This process requires to introduce up to two additional lowest priorities to ensure that a player who never leaves the automaton corresponding to such a transition loses.

Theorem 3.5.3 leads to a topological reformulation of Conjecture 3.2.13.

Proposition 3.5.17. *Conjecture 3.2.13 is equivalent to the following statement: Let Σ be a finite alphabet, $A_1, \dots, A_n \subseteq (\Sigma \times \{0,1\})^{\mathbb{N}}$ be ω-regular and $B_1, \dots, B_n \subseteq \{0,1\}^{\mathbb{N}}$ be Borel such that Eve wins the Gale-Stewart game on the full tree with payoff set $\bigcup_i (A_i \cap (\Sigma^{\mathbb{N}} \times B_i))$. Then there are $B_i' \subseteq B_i$ that are Boolean combinations of F_σ sets, of complexity depending computably only on $A_1, \dots, A_n$, such that Eve also wins $\bigcup_i (A_i \times (\Sigma^{\mathbb{N}} \times B_i'))$.*

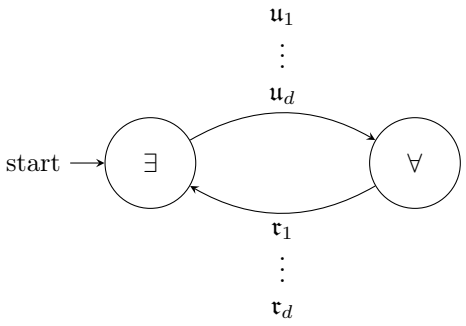
Proof. By Theorem 3.5.3 we only have to show the equivalence of this game model with prioritized Borel games. Any such game yields a prioritized Borel game by Remark 3.2.5. The converse follows from Proposition 3.5.11, using Σ to encode the states of the automaton. □

3.6 Lower bounds

We present some examples of games that require unpleasant amounts of memory. Since the notion of a finite memory strategy for a prioritized Borel game is more flexible than for a game on a finite graph, we give detailed arguments for the lower memory bounds. They yield optimality of many of our previous upper bounds. We also discuss relations to other notions of complexity like topological complexity of payoff sets.

3.6.1 Games with bound one require unbounded memory

The following game, the d-**copying game**, generalizes Example 3.2.8 to d priorities.



Here, Eve wins if the highest i such that $\mathfrak{r}_i$ occurs infinitely often equals the highest j such that $\mathfrak{u}_j$ occurs infinitely often. Again, Eve wins using d memory states by copying Adam's moves, and this memory is optimal. Since the game (or, more precisely, the equivalent game in which Adam's moves $();\mathfrak{r}_i$ are replaced by $\mathfrak{u}_1;\mathfrak{r}_i$) has bound one, this implies that the upper bound of Proposition 3.4.20 is optimal.

Proposition 3.6.1. *No strategy of Eve with $d-1$ memory states wins the d-copying game.*

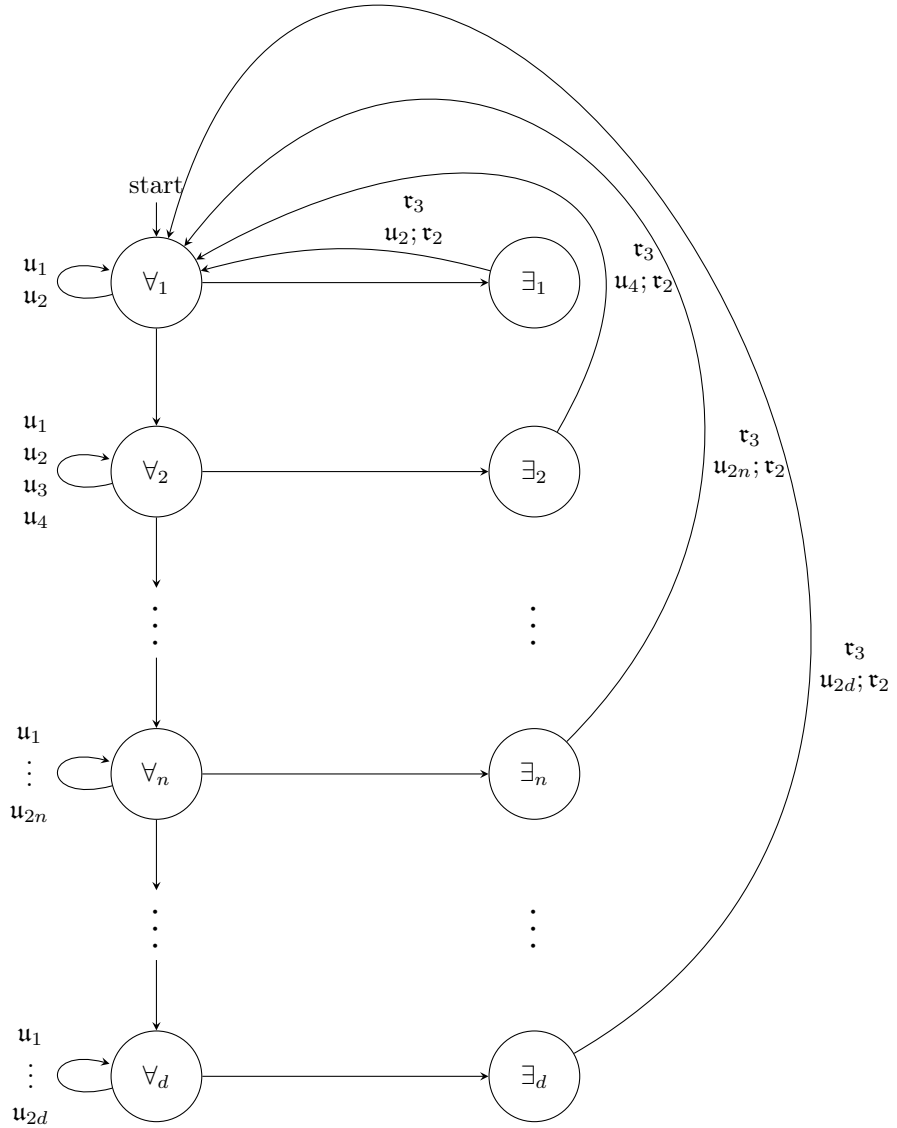
Proof. Let M be the set of memory states, let $\sigma\colon \{\mathfrak{u}_1,\dots,\mathfrak{u}_d\}^{<\omega}\times M\to\{\mathfrak{u}_1,\dots,\mathfrak{u}_d\}$ denote Eve's strategy and let $\tau\colon \{\mathfrak{u}_1,\dots,\mathfrak{u}_d\}^{<\omega}\times M\times\{\mathfrak{r}_1,\dots,\mathfrak{r}_d\}\to M$ denote the memory update function (it clearly suffices for Eve to update her memory after Adam's moves).

We define $x_n\in\{\mathfrak{u}_1,\dots,\mathfrak{u}_d\}^{<\omega}$ and $m_n\in M$ by recursion. Let $x_0=()$ and m_0 be the initial memory state. Given x_n,m_n, since $\#M<d$ there are two distinct r_{n0},r_{n1} such that $\tau(x_n,m_n,r_{n0})=\tau(x_n,m_n,r_{n1})$. Set $m_{n+1}=\tau(x_n,m_n,r_{n0})$ and $x_{n+1}=x_n\sigma(x_n,m_{n+1})$.

For any $y\in 2^{\mathbb{N}}$, the play where Eve plays $\lim_n x_n$ and Adam plays $r_{n,y(n)}$ in his n-th turn is consistent with σ. Since there are two such plays where the highest i such that $\mathfrak{r}_i$ occurs infinitely often differs, (σ,τ) is not winning. □

3.6.2 Three priorities require unbounded memory

The following example, the **d-alternating game**, shows that the memory required to solve a non-jumping prioritized Borel game with three priorities may depend on the number of states.



If the highest priority is $\mathfrak{r}_1$, then Eve always wins; if the highest priority is $\mathfrak{r}_2$, then Adam always wins; if the highest priority is $\mathfrak{r}_3$, then Eve wins if and only if the highest i such that $\mathfrak{u}_i$ occurs infinitely often is even.

We first describe a winning strategy of Eve (using unnecessarily much, but finite memory). If Eve enters $\exists_n$ for the i-th time, she plays $\mathfrak{r}_3$ if i is even and $\mathfrak{u}_{2n};\mathfrak{r}_2$ else.

Proposition 3.6.2. *No strategy of Eve with d memory states wins the d-alternating game.*

Proof. Let M be the set of memory states, let $\sigma\colon \{\mathfrak{u}_1,\dots,\mathfrak{u}_d\}^{<\omega}\times M\times\{\exists_1,\dots,\exists_n\}\to\{\mathfrak{r}_3,(\mathfrak{u},\mathfrak{r}_2)\}$ denote Eve's strategy and let $\tau\colon \{\mathfrak{u}_1,\dots,\mathfrak{u}_d\}^{<\omega}\times M\times\{\forall_1,\dots,\forall_d\}\to M$ denote the memory update function (it clearly suffices for Eve to update her memory before moves of Adam).

We call a map $s\colon \{\mathfrak{u}_1,\dots,\mathfrak{u}_d\}^{<\omega}\to M$ a **generalized memory state**. First, suppose there is a generalized memory state s such that from some point (compatible with σ,τ) on, Eve cannot reach some $(x,\forall_1,s(x))$ without passing through $\exists_d$. This case is impossible for $d=1$ and handled by induction for $d>1$ as it leads to a solution for the $(d-1)$-alternating game not entering s, which can be transformed into a solution with $d-1$ memory states.

Otherwise, for all s,x,m, there is a partial play from $(x,\forall_1,m)$ to some $(x',\forall_1,s(x'))$ that does not pass through $\exists_d$. Let U be the set of x such that for all $m\in M$, after

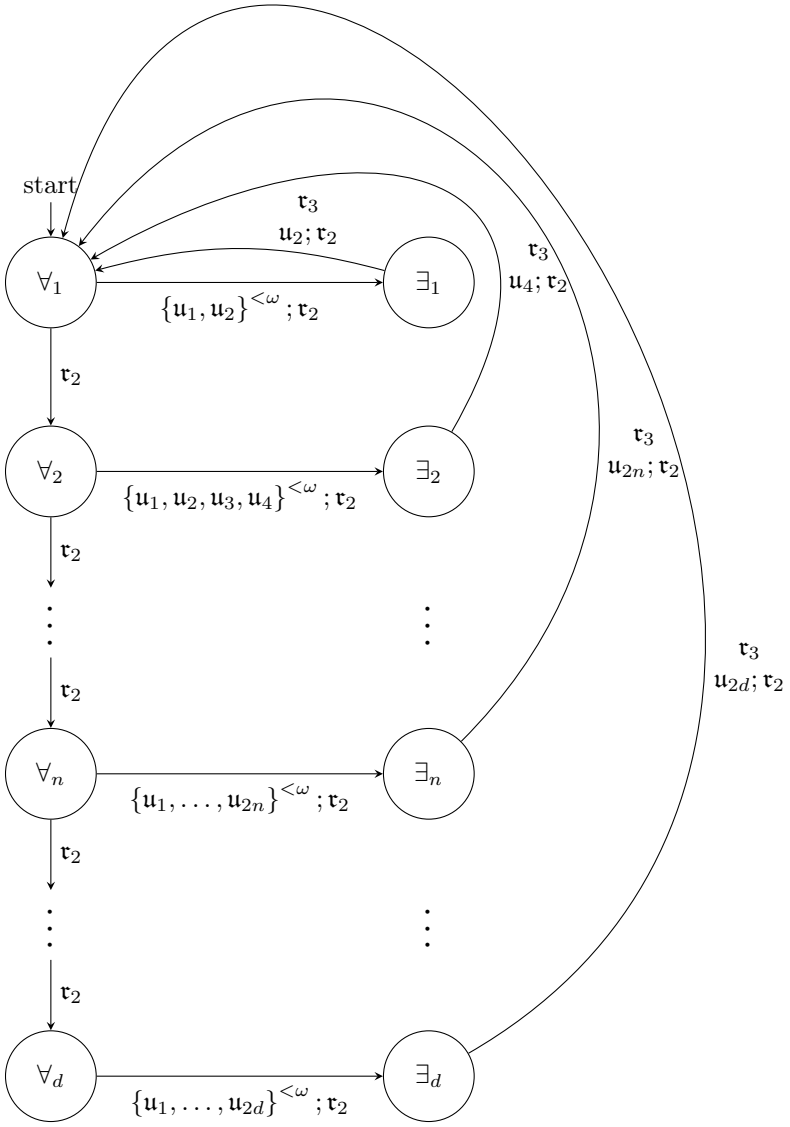
$$(x,\forall_1,m)\to(x,\forall_d)\to(x\mathfrak{u}_{2d-1},\forall_d)\to(x\mathfrak{u}_{2d-1},\exists_d)$$

Eve plays $(\mathfrak{u}_{2d},\mathfrak{r}_2)$. If each x can be extended to some $y\in U$, then Adam can enforce a play with highest priority $\mathfrak{r}_2$ by only allowing Eve to move in this situation, which is absurd. Otherwise, passing to a subtree we may assume that $U=\emptyset$. In other words, there is a generalized memory state s such that $\sigma(x,s(x),\exists_d)=\mathfrak{r}_3$ for all x. Alternating the partial plays obtained from our assumption for this s with the partial plays in the definition of U, Adam can enforce a play with highest priorities $\mathfrak{u}_{2d-1},\mathfrak{r}_3$, which is absurd. □

The d-alternating game shows that the analogue of Proposition 3.4.19 with $W(\mathfrak{r}_2)\subsetneq W(\mathfrak{r}_3)\subsetneq W(\mathfrak{r}_1)$ fails: In this case the required memory may depend on the game. In particular, it shows that the dependence on the bound in Conjecture 3.2.13 is necessary. The analogously defined ω-alternating game shows that unbounded prioritized Borel games may require infinite memory.

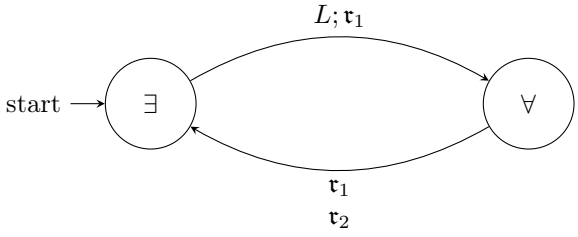
3.6.3 Jumps for two priorities

The d-alternating game can also be expressed as a regularly jumping game with only two priorities, employing regular jumps to omit $\mathfrak{r}_1$:



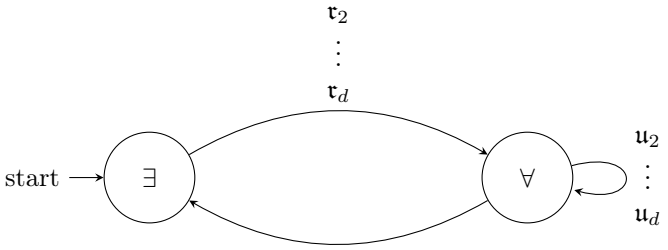
This shows that the non-jumping assumption in Corollary 3.4.17 and the additional priority in Remark 3.5.16 are necessary.

Thus, the problem of finite memory for regularly jumping prioritized Borel games for two priorities is open. Our methods do not solve the following game, parametrized by a regular language $L \subseteq \{\mathfrak{u}_1,\mathfrak{u}_2\}^{<\omega}$ and two Borel subsets $W(\mathfrak{r}_1), W(\mathfrak{r}_2) \subseteq \{\mathfrak{u}_1,\mathfrak{u}_2\}^{\mathbb{N}}$:



3.6.4 Playing only priorities requires memory

So far, in all examples of prioritized Borel games that are not memory-less for Eve, she sometimes played public letters. Thus, one may expect that in every game where Eve requires memory she sometimes has a choice regarding the public letters or states she plays. The following example shows that this is not the case.



If the play ends in $\{\mathfrak{u}_2, \dots, \mathfrak{u}_d\}^{<\omega}$ or the highest priority is $\mathfrak{r}_1$, then Eve always wins. Else Eve wins if and only if the highest i such that $\mathfrak{r}_i$ occurs infinitely often equals the highest j such that $\mathfrak{u}_j$ occurs infinitely often.

A finite memory winning strategy for Eve is playing the highest $\mathfrak{r}_i$ such that $\mathfrak{u}_i$ occurred since her last turn or $\mathfrak{r}_2$ if none did. Similarly to Proposition 3.6.1, Eve cannot win using less than $d-1$ memory states. In particular, Proposition 3.4.22 requires some memory (although the bounds do not match).

3.6.5 Replacing branch conditions

We provide lower bounds for the function f of Conjecture 3.5.5. We will need the following folklore result, comparing the difference hierarchy [Kec95, 22.E] and the Mostowski index of an ω-regular language, that is, the least number of priorities of a deterministic parity automaton.

Proposition 3.6.3. *Let $L \subseteq A^{\mathbb{N}}$ be ω-regular. Then L has Mostowski index with $n+1$ priorities, of which the highest is rejecting, if and only if $L \in D_n(F_\sigma)$.*

This implies analogous results for the dual class $\check{D}_n(F_\sigma)$ and for $D_n(F_\sigma) \cap \check{D}_n(F_\sigma)$.

Proof. Necessity follows from the facts that the parity language itself lies in $D_n(F_\sigma)$ and that the function mapping a word to the sequence of priorities of the run of an automaton provides a Wadge reduction. For sufficiency, suppose that L does not have this Mostowski index. This yields a semantic flower [Del+24] with stem $w \in A^{<\omega}$ and petals $v_1, \dots, v_{n+1} \in A^{<\omega}$. Then mapping a sequence of priorities $p_1 p_2 \dots$ to $w v_{p_1} v_{p_2} \dots$ defines a Wadge reduction from the $\check{D}_n(F_\sigma)$-complete parity language to L, whence $L \notin D_n(F_\sigma)$. □

We first observe a linear lower bound for f. For this, we restrict to games where all positions are controlled by Adam (formally, $Q = Q_\forall$). Such games correspond to non-deterministic parity automata $\mathcal{A}$ on infinite words. Then Eve wins if and only if for each priority p, the set $W(p)$ contains all words on which $\mathcal{A}$ produces p as highest priority. In particular, if $W(p)$ is chosen to be this set, then it cannot be replaced by any proper subset and it suffices to show that $W(p)$ has sufficient topological complexity.

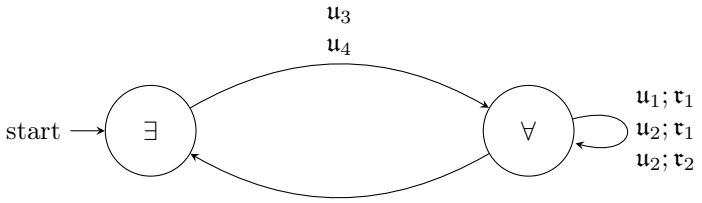
It is straightforward to construct automata such that this procedure yields a linear lower bound. However, this procedure only yields linear lower bounds. Namely, while determinization incurs an exponential increase of the number of states, a language that is recognized by a non-deterministic automaton with n states has Mostowski index $O(n)$ (this follows from [Saf88], noting that complexities of Rabin and priority conditions are linearly related [BKS10, Theorem 7]), thus lies in $D_{O(n)}(F_\sigma)$ by Proposition 3.6.3.

This lower bound persists in the case of non-jumping games on two priorities although the required memory is constant by Corollary 3.4.17.

Finally, we precisely compute the topological complexity of a set to replace $W(\mathfrak{r}_2)$ in a non-jumping prioritized Borel game of bound one and with priorities $\mathfrak{r}_1 < \mathfrak{r}_2$. The case of $W(\mathfrak{r}_1)$ was already solved in Lemma 3.5.4. In contrast to the above example, this requires a game where not only Adam is playing.

Proposition 3.6.4. *Consider a non-jumping prioritized Borel game of bound one and with priorities $\mathfrak{r}_1 < \mathfrak{r}_2$. Then $W(\mathfrak{r}_2)$ can be replaced by a set that is $\check{D}_3(F_\sigma)$, but not in general by a set that is $D_3(F_\sigma)$.*

Proof. The set obtained by Lemma 3.5.6 is $\check{D}_3(F_\sigma)$ since $\mathfrak{r}_2$ is the highest priority (for intermediate priorities it would only be $D_4(F_\sigma)$). For the converse consider the following prioritized Borel game of bound one (again, the bound only holds after passing to an equivalent game with $\forall \xrightarrow{\mathfrak{u}_3} \exists$):



Eve wins if the pair of highest priorities is among $(\mathfrak{r}_2,\mathfrak{u}_4),(\mathfrak{r}_1,\mathfrak{u}_3),(\mathfrak{r}_2,\mathfrak{u}_2),(\mathfrak{r}_1,\mathfrak{u}_2),(\mathfrak{r}_1,\mathfrak{u}_1)$. A winning strategy of Eve is playing $\mathfrak{u}_4$ if $\mathfrak{r}_2$ happened since her last turn.

Suppose for the sake of contradiction that there is $A \subseteq \{\mathfrak{u}_1,\mathfrak{u}_2,\mathfrak{u}_3,\mathfrak{u}_4\}^{\mathbb{N}}$ in $D_3(F_\sigma)$ such that $W(\mathfrak{r}_2)$ can be replaced by A. Iterating Fact 3.3.1 we obtain a labeling $\{\mathfrak{u}_1,\mathfrak{u}_2,\mathfrak{u}_3,\mathfrak{u}_4\}^{<\omega} \to \{\ell_1,\ell_2,\ell_3,\ell_4\}$ such that A consists of the branches where the highest ℓ_i occurring infinitely often is odd. We obtain a contradiction by describing a winning strategy σ of Adam for the game with $W(\mathfrak{r}_2)$ replaced by A:

- if the current position is labeled ℓ_4, play $\mathfrak{u}_2;\mathfrak{r}_2$,
- if the current position is labeled ℓ_3, let Eve play,
- if the current position x is labeled ℓ_2 or ℓ_1 and there is $y \in \{\mathfrak{u}_1,\mathfrak{u}_2\}^{\mathbb{N}}$ containing $\mathfrak{u}_2$ infinitely often such that xy has highest label ℓ_2, then follow y playing either $\mathfrak{u}_2;\mathfrak{r}_2$ or $\mathfrak{u}_1;\mathfrak{r}_1$,
- if the current position is labeled ℓ_2 and there is no such y, play $\mathfrak{u}_2;\mathfrak{r}_1$,
- if the current position is labeled ℓ_1 and there is no such y, play $\mathfrak{u}_1;\mathfrak{r}_1$.

Consider a play h_∞ compatible with σ.

If the highest label is ℓ_4, then $\mathfrak{r}_2$ occurs infinitely often, but $\langle h_\infty\rangle \notin A$, whence Adam wins.

Suppose the label priority is ℓ_3. Since $\mathfrak{r}_2$ occurs only finitely often, it suffices to show that $\mathfrak{u}_4$ occurs infinitely often. If $\mathfrak{u}_4$ occurs finitely often, then the highest priority is $\mathfrak{u}_3$, whence $\langle h_\infty\rangle \notin W(\mathfrak{r}_2) \supseteq A$, which contradicts that ℓ_3 is the highest label.

Suppose the highest label is ℓ_2. Thus the highest priority is $\mathfrak{u}_2$. Moreover, $\mathfrak{r}_2$ occurs infinitely often as $\langle h_\infty\rangle$ plays the role of xy in the definition of σ. Since $\langle h_\infty\rangle \notin A$, this implies that Adam wins.

If the highest label is ℓ_1, then the highest priority is $\mathfrak{u}_1$, whence $\langle h_\infty\rangle \notin W(\mathfrak{r}_2) \supseteq A$, which is absurd. □

Remark 3.6.5. "In the specified game, $W(\mathfrak{r}_2)$ cannot be replaced by a set in $D_3(F_\sigma)$" is expressible as a sentence in the F_σ monadic theory of the binary tree, thus in S2S [Rab69, Theorem 2.6]. It is not clear to the author whether it would be feasible to decide by an actual optimized implementation.

3.7 Generalizations

While we restricted to Borel sets, most of our results generalize assuming determinacy for larger classes of sets. We never use any property of Borel sets beyond determinacy. This makes generalizations of Conjecture 3.2.13 to, say, projective sets, plausible. We rarely use uncountable branching, which can be an obstruction to such generalizations. We also do not depend on the full axiom of choice, and thus could consider the full axiom of determinacy and the unrestricted monadic theory of $2^{\leq\omega}$.

Another possible generalization involves the monadic theory of $2^{<\alpha}$. We tried to generalize S2S, the case $\alpha=\omega$, to the case $\omega+1$. Increasing α causes at least two kinds of additional difficulties. First, the behavior of automata for such a monadic theory along a single branch should be related to automata for words of length α, which become more intricate for greater α [BZ83; Nee08a; Nee08b]. Second, our approaches require determinacy, which beyond $2^{<\omega\cdot 2}$ requires large cardinals even for Borel sets [Agu21] and fails for games of length ω_1 [Myc64, Theorem 7.3].

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